ENGINEERING PROGRAMMES

ZAMBIA QUALIFICATIONS FRAMEWORK (ZQF) - LEVEL 6

SYLLABUS

FOR

DIPLOMA
IN
MAINTENANCE AND MANAGEMENT OF ELECTRICAL AND MECHANICAL EQUIPMENT

Programme No. 412

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ACRONYMS

A.P.S	**ARITHMETIC PROGRESSIONS**
AQC	**ACCEPTANCE QUALITY CONTROL**
CDIO	**CONCEIVING, DESIGNING, IMPLEMENTING & OPERATING**
CLM	**CHINA LUANSHYA COPPER MINE**
CMOS	**COMPLEMENTARY METAL OXIDE SEMICONDUCTORS**
CNC	**COMPUTER NUMERICAL CONTROL**
DC	**DIRECT CURRENT**
DEMUX	**DEMULTIPLEXERS**
ECM	**ELECTRO-CHEMICAL MACHINING**
EDM	**ELECTRO DISCHARGE MACHINING**
EMF	**ELECTROMOTIVE FORCE**
FOLI	**FIRST- OUT LAST- IN**
G.P.S	**GEOMETRIC PROGRESSIONS**
HVTC	**HARBIN VOCATIONAL & TECHNICAL COLLEGE**
I/O	**INPUT/OUTPUT**
LAN	**LOCAL AREA NETWORK**
MBO	**MANAGEMENT BY OBJECTIVES**
MIG	**METAL INERT GAS**
MUX	**MULTIPLEXERS**
NC	**NUMERICAL CONTROL**
P&ID	**PROCESS AND INSTRUMENTATION DRAWING STANDARDS**
PCB	**PRINTED CIRCUIT BOARD**
QHSE	**QUALITY, HEALTH, SAFETY, AND ENVIRONMENT**
SFG	**SIGNAL FLOW GRAPHS**
SMPS	**SWITCH-MODE POWER SUPPLY**
TEVETA	**TECHNICAL EDUCATION VOCATIONAL AND ENTREPRENEURSHIP TRAINING AUTHORITY**
TIG	**TUNGSTEN INERT GAS**
TTL	**TRANSISTOR- TRANSISTOR LOGIC**
UPS	**UNINTERRUPTED POWER SUPPLY**

1.0 ABOUT TEVETA

The Technical Education, Vocational and Entrepreneurship Training Authority (TEVETA) is an institution established under TEVETA Act No. 13 of 1998 read together with the Amendment Act No. 11of 2005. Its functions include, to regulate, coordinate and monitor education, vocational and entrepreneurship training in consultation with stakeholders.

TEVETA executes its regulatory function through the provision of services, among others, the development, review and approval of TEVETA Curricula in consultation with industry, employers, employees and other stakeholders.

2.0 ACKNOWLEDGEMENTS

The Technical Education, Vocational and Entrepreneurship Training Authority (TEVETA) wishes to express sincere appreciation to the following persons who participated in the development of the Diploma in Maintenance and Management of Mechanical and Electrical Equipment

S/N	NAME	POSITION	ORGANISATION
1.	LIANG Chimin	Director	Sino-Zam Vocational College of Science and Technology
2.	Chakupa Sakala	Lecturer	Luanshya Technical and Business College
3.	CHEN Manqian	Teacher	Harbin Vocational and Technical College
4.	Ernest Shamano	Lecturer	Northern Technical College
5.	GUAN Tong	Head of the Teaching and Research Section of Robotic Technology	Harbin Vocational and Technical College
6.	HU Jiangchuan	Head of the Teaching and Research Section of Mechatronics Technology	Harbin Vocational and Technical College
7.	JI Dongjun	Teacher	Harbin Vocational and Technical College
8.	JIANG Tao	Director of Higher Education Research Institute	Harbin Vocational and Technical College
9.	LIU Jianguo	Vice President	Harbin Vocational and Technical College

10. LIU Shen	President	Harbin Vocational and Technical College
11. QIU Zhixin	Teacher	Harbin Vocational and Technical College
12. SONG Kai	Secretary General	National Nonferrous Metals Vocational Education and Teaching Guidance Committee
13. SONG Xiaoli	Teacher	Harbin Vocational and Technical College
14. SUN Fucai	Head of Practical Training Section of Mechanical and Electrical Engineering	Harbin Vocational and Technical College
15. SUN Huimin	Associate Professor	Harbin Vocational and Technical College
16. WU Yuanhui	Teacher	Nanjing Institute of Industry Technology
17. YANG Miaosen	Teacher	Harbin Vocational and Technical College
18. YONG Liying	Dean of School Mechanical and Electrical Engineering	Harbin Vocational and Technical College
19. ZHANG Mingzhu	Teacher	Beijing Polytechnic College
20. ZHAO Dan	Head of General Management of School of Mechanical and Electrical Engineering	Harbin Vocational and Technical College
21. Jorick Yamvwa	Principal	Sino-Zam Vocational College of Science and Technology
22. Noah Monday Banda	Lecturer (Secretary)	Sino-Zam Vocational College of Science and Technology
23. Kennedy Bowa	Manager- Curriculum Development	TEVETA
24. Namuchindo Chikoti	Curriculum Development Specialist	TEVETA

TEVETA would also like to express sincere appreciation to China Luanshya Copper Mine (CLM), for providing the technical and financial support to develop this curriculum.

3.0 RATIONALE

In recent years, the Zambian economy has changed shape, with a growing focus on production and value addition in the manufacturing sector. This growth has been supported by improved and increasing technological change and increased demand for various products by society, both local and foreign. In addition, the Seventh National Development Plan (7NDP) lists this sector as a priority sector for the achievement of the 'Vision 2030'.

Globally the manufacturing sector today is characterised by highly automated mechanical and electrical equipment and tools. It is this automated machinery that has allowed countries such as China to grow at a rapidly fast pace. In Zambia however, the sector still lags behind. One reason for this is the lack of competent manpower to work with these new technologies on the labour market. As a result, there has emerged a growing demand for skilled and competent personnel to support the manufacturing sector.

One of the key areas for skills development is the Electromechanical Maintenance Technologist. These are responsible for installing new systems, maintaining existing systems and troubleshooting electrical and mechanical issues in a modern manufacturing facility. Currently, in Zambia, training in this area is only offered at degree level and this has presented a skills gap.

In view of this, the Diploma in Maintenance and Management of Electrical and Mechanical Equipment has been developed to provide highly qualified electromechanical Maintenance Technologist to meet the demands of the formal and informal manufacturing sector.

4.0 PROGRAMME PURPOSE

The purpose of the program is to equip the trainee with knowledge, skills and appropriate attitudes to effectively and efficiently carry out maintenance on electromechanical equipment and manage electromechanical systems in a manufacturing plant.

5.0 PROGRAMME OUTCOMES

On completion of the programme the trainee will be able to:

5.1 Apply mathematics in solving electromechanical problems
5.2 Interpret engineering drawings, symbols and circuits
5.3 Apply electronic systems in electromechanical engineering
5.4 Apply basic electrical principles
5.5 Communicate effectively
5.6 Apply safe workshop practices in an Electromechanical environment
5.7 Apply computer skills in the workplace
5.8 Run profitable enterprises
5.9 Use engineering materials
5.10 Maintain motor control systems.
5.11 Maintain hydraulic and pneumatic systems
5.12 Apply electrical principles of mechanical operations in Manufacturing
5.13 Maintain electrical and mechanical equipment
5.14 Apply digital electronics in control of electrical machines and equipment
5.15 Carryout welding and Locksmithing
5.16 Maintain production line systems
5.17 Maintain a safe working environment
5.18 Manage a project efficiently and effectively

6.0 PROGRAM DURATION

Three (3) years or 3850 hours inclusive of a minimum of three (3) months or 480 hours of Industrial Attachment.

7.0 PROGRAMME OUTLINE

YEAR 1			
MODULE NO	**TITLE**	**CORE/ NON-CORE**	**DURATION (HOURS)**
412-01-A	Engineering Mathematics I	CORE	150 Hours
412-02-A	Engineering Drawing	CORE	250 Hours
412-03-A	Electronics	CORE	180 Hours
412-04-A	Electrical Technology I	CORE	170 Hours
412-05-A	Communication Skills	NON-CORE	80 Hours
412-06-A	Workshop Practice and Processes	CORE	200 Hours
412-07-A	Introduction to Computers	NON-CORE	80 Hours

412-08-A	Entrepreneurship	NON-CORE	100 Hours
	Subtotal		1200 Hours
YEAR 2			
412-09-B	Electrical Technology II	CORE	150 Hours
412-10-B	Material Science	CORE	120 Hours
412-11-B	Motor Maintenance Technology	CORE	140 Hours
412-12-B	Maintenance of Hydraulic and Pneumatic system	CORE	120 Hours
412-13-B	Engineering Mathematics II	CORE	200 Hours
412-14-B	Mechanical Manufacturing Technology	CORE	100 Hours
412-15-B	Mechanical and Electrical Equipment Maintenance Technology I	CORE	120 Hours
	Industrial Attachment		480 Hours
	Subtotal		**1430**
YEAR 3			
412-16-C	Mechanical and Electrical Equipment Maintenance Technology II	CORE	300 Hours
412-17-C	Production Line Adjustment	CORE	300 Hours
412-18-C	Welding Technology	CORE	300 Hours
412-19-C	Management and Organisation	NON-CORE	120 Hours
412-20-C	Locksmith Operation	CORE	200 Hours
	Subtotal		**1220 Hours**
	Total		3850

8.0 TEACHING/LEARNING STRATEGIES

- Interactive lectures
- Class exercises

- Group discussions
- Role plays
- Simulations
- Video examples
- Field visits
- Field observations
- On-the-job trainings

9.0 PROGRAMME EVALUATION

TEVETA shall evaluate the programme as follows:

9.1 Formative evaluation

9.1.1 Purpose

To determine on an on-going basis whether the programme is being implemented as planned and to provide advice on improvements.

9.1.2 Major areas of Evaluation

Programme aims and objectives, trainees' entry requirements, course content, teaching and learning activities, learning resources, qualifications of trainers, accreditation of training organisations and institutes.

9.1.3 Evaluation Instruments

Questionnaires, structured interviews, observations, checklists, examination/test records, participation and attendance in sessions

9.1.4 Sources of Information

Trainees, trainers, administrators, trainees' records, training organisations and institute.

9.2 Summative Evaluation

9.2.1 Major areas of evaluation

Programme aims, trainee's entry requirements, course content, learning resources, teaching/learning activities, qualification of trainers, assessors, examiners accreditation of training organisations and institutes, and graduate' performance in employment in accordance with qualification descriptions.

9.2.2 Evaluation Instruments

Questionnaires, structured interviews, observations, checklists, records, final integrated examinations.

9.2.3 Sources of information

Trainees, trainers, assessors, examiners, administrators, sponsors, employers, training organisations and institutes, union officials and the public.

10.0 TRAINEE ENTRY REQUIREMENTS:

10.1 Direct entry

10.1.1 Grade 12 School Certificate with a minimum of five (5) credits including English Language, Mathematics, Physics or Science.

10.2 Mature age entry

10.2.1 A minimum of Certificate in Advanced Instrumentation or Electrical / Electronic Engineering or a related field from an accredited institution

10.2.2 Minimum of two (2) years working experience.

10.3 Exemptions

Exemptions will be awarded as per TEVETA guidelines on Exemptions.

11.0 ASSESSMENT

11.1	Continuous assessment	40%
11.2	Final Examinations	60%
11.3	Pass Mark	50%

12.0 ATTENDANCE

The trainee must have an attendance of minimum 85% to be eligible for the final examinations.

13.0 PROGRESSION REQUIREMENTS

13.1 A trainee who passes all modules prescribed for that year will progress to the following academic year or graduate in the case of a trainee in his/her final year.

13.2 Trainees failing in a maximum of three (3) modules shall be allowed to proceed to the following academic year subject to clearing the failed modules within two (2) consecutive examination sessions failure to which they may not be allowed to proceed to the following academic year until all the failed modules are cleared.

13.3 A trainee who has failed a minimum of four (4) modules shall not be permitted to proceed to the following academic year of study. The trainee will have to clear the all the failed modules within two (2) consecutive examination sessions before being allowed to proceed to the following academic year.

14.0 STAFFING

14.1 Minimum of Bachelor's Degree in Electromechanical Engineering and/ or a related field.

14.2 Two years industrial experience.

14.3 A teaching qualification from an accredited and registered institution.

14.4 A TEVETA Accredited trainer.

15.0 CERTIFICATION

Successful candidates will be awarded a Diploma in Maintenance and Management of Mechanical and Electrical Equipment by the Technical Education Vocational and Entrepreneurship Training Authority (TEVETA)

MODULE: 412-01-A ENGINEERING MATHEMATICS I

UNIT A 1.1 SOLVING ALGEBRAIC PROBLEMS ANALYTICALLY GRAPHICALLY

Nominal Duration: 18 Hours Credits: 1.8

General Learning Outcome: SOLVE ALGEBRAIC EQUATIONS ANALYTICALLY & GRAPHICALLY

Specific Learning Outcome: On completion of this unit the trainee will be able to:

1.1.1 Solving indicial and logarithmic problems
1.1.2 Apply the principles of a straight line
1.1.3 Solve linear equations
1.1.4 Solve quadratic equations
1.1.5 Solve linear and quadratic equations graphically
1.1.6 Solve equations using matrices and determinants

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Solving indicial and logarithmic problems
  - The laws of indices
  - Laws of logarithms
  - Change of base
  - Natural logarithms
  - Common logarithms
  - Indicial equations
- Applying the principles of a straight line
  - Degree of straight line
  - The x-y axis plane

    - Independent axis (abscissa)
    - Dependent axis (ordinate)
    - The equation of a straight line
    - Passing through the origin
    - Passing through 2 points
    - Focus of a straight line
    - Gradient form (y=mx+c)
    - General form (ax+by=C=0)
    - Intercept form
    - Distance between points
    - Mid-point
    - Angle between 2 lines
    - Reduction of non- linear lines to linear ones
- Solving algebraic equations analytically & graphically
    - Linear equations in one variable
- Solving linear simultaneous equations by:
    - Elimination method
    - Substitution method
    - A graphical method
    - Quadratic equations
    - Quadratic equations having 2 terms (e.g. x2 – 3=0)
    - Trial & error method
    - Factorisation method
    - Completing the square
    - Use of the formula for quadratic equations
- Solving equations graphically
    - Linear simultaneous equations in 2 unknowns
    - Quadratic equations
    - Simultaneous equations; one linear and the other quadratic
- Solving linear simultaneous equations in using matrices and determinants
    - Order of matrices
    - Types of matrices
        - Identity matrix
        - Square matrix
        - Diagonal matrix
        - Row matrix
        - Column matrix
    - Addition & subtraction of matrices
    - Multiplication of matrices
    - Transpose of a square matrix
    - Inverse of a square matrix
    - Solution of equations having 3 unknowns.
- Solve indicial and logarithmic problems correctly
- Solving algebraic equations analytically & graphically correctly

- Solving equations graphically correctly
- Solving linear simultaneous equations in correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Books:

- Bostock ,L and Chandler,S (Latest edition),Mathematics – The Core Course for A-Level – English
- Society/Stanley Thornes Stroud K. A. (2004),Engineering Mathematics, 4 th Edition – Macmillan

MODULE: 412-01-A ENGINEERING MATHEMATICS I

UNIT A 1.2 APPLYING GEOMETRY

Nominal Duration: 16 Hours Credits: 1.6

General Learning Outcome: APPLY COMPLEX NUMBERS

Specific Learning Outcome: On completion of this unit the trainee will be able to:

1.2.1 Apply the principles of the straight line
1.2.2 Apply the circle theorem
1.2.3 Apply the ellipse
1.2.4 Apply the parabola
1.2.5 Apply the hyperbola

Conditions: The trainee must have access to:

- Text books and materials handbooks
- Materials laboratories
- Catalogues, tutorials and demonstrations
- Library
- Calculators
- Internet (Worldwide Web)

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Applying the principles of a straight line
  - Degree of straight line
  - The x-y axis plane
    - Independent axis (abscissa)
    - Dependent axis (ordinate)
  - The equation of a straight line
    - Passing through the origin
    - Passing through 2 points
  - Focus of a straight line
    - Gradient form (y=mx+c)

    - General form (ax+by=C=0)
    - Intercept form
    - Distance between points
    - Mid point
    - Angle between 2 lines
    - Reduction of non- linear lines to linear ones
- Applying the circle theorem
    - Equation of a circle
        - Centre at the origin
        - Centre at (h.k)
    - The general equation of a circle
        - Radius and circle
    - Circles and Tangents
    - Gradients of perpendicular lines
    - Distance between a point (eg centre of circle) and straight line
- Applying the ellipse
    - Major and minor axis
    - Equation of an ellipse
- Applying the parabola
    - Focus
    - Vertex
    - Directrix
    - Equation of an ellipse
- Applying the hyperbola
    - Focus
    - Vertex
    - Directrix
    - Equation of an hyperbola

Assessment Criteria:

- Apply the principles of the straight line correctly
- Apply the circle/ ellipse theorem correctly
- Apply the parabola correctly
- Apply the hyperbola correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Tests
- Examinations
- Assignments

Recommended Books:

- Stroud ,K.A (2001) ,*Engineering Mathematics 5 th edition Macmillan*
- Backhouse, J.K.E (1997),*Essential Pure Mathematics*

MODULE :412-01-A ENGINEERING MATHEMATICS I

UNIT A1.3 APPLYING TRIGONOMETRY TO SOLVE PROBLEMS

Nominal Duration: 20 Hours Credits: 2.0

General Learning Outcome: APPLY TRIGONOMETRY

Specific Learning Outcome: On completion of this unit the trainee will be able to:

1.3.1 Use radians
1.3.2 Apply trigonometry
1.3.3 Use properties of trigonometric relations
1.3.3 Plot graphs of trigonometric functions

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Using radians
  - Definition of the radian
  - Conversions from degrees to radians & vice-versa
  - Length of an arc
  - Area of a sector
  - Area of a segment
- Using trigonometric ratios
  - Basic trigonometric ratios
  - The four quadrants
  - Trigonometric reciprocal ratios
  - Trigonometric ratios for standard angles
  - Graphs of Sinθ, Cosθ, Tanθ

- Using properties of trigonometric relationships
  - Identities
  - Compound angles
  - Double angles
  - Trigonometric equations
  - Products of Sines and Cosines
  - Forms of RSin(θ ± α) RCos(θ ± α) &RSin (αω ± t)
- Plotting graphs of trigonometric functions

Assessment Criteria:

- Using radians correctly
- Using trigonometric ratios correctly
- Using properties of trigonometric relationships correctly
- Plotting graphs of trigonometric functions correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises / Assignments / Tests

Recommended Books:

- BostockL and Chandler S (Latest edition),*Mathematics – The Core Course for A-Level –English*Stroud K. A. (2004),*Engineering Mathematics, 4 thEdition – Macmillan*

MODULE: 412-01-A ENGINEERING MATHEMATICS I

UNIT A 1.4 USING SERIES TO SOLVE PROBLEMS

Nominal Duration: 18 Hours Credits: 1.8

General Learning Outcome: USE SERIES TO SOLVE PROBLEMS

Specific Learning Outcome: On completion of this unit the trainee will be able to:

1.4.1 Apply arithmetic progressions
1.4.2 Apply geometric progressions
1.4.3 Apply the binomial theorem

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Applying arithmetic progressions (A.P.s)
  - Nth term of an arithmetic progression
  - Sum of n terms of an arithmetic progression Arithmetic mean
- Applying geometric progressions (G.P.s)
  - Nth term of a geometric progression
  - Sum of n terms of a geometric progression
  - Geometric mean
- Applying the binomial theorem
  - Pascal's triangle
  - Binomial theorem
  - Approximations

Assessment Criteria:

- Apply arithmetic progressions applied correctly

- Apply geometric progressions applied correctly
- Apply the binomial theorem applied correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Books

- Bostock L and Chandler S (Latest edition),*Mathematics – The Core Course for A-Level –English*Stroud K. A. (2004), *Engineering Mathematics, 4 thEdition – Macmillan*
- Backhouse Etal (1007), *Essential Pure Mathematics*

MODULE: 412-01-A ENGINEERING MATHEMATICS I

UNIT A 1.5 SOLVING PROBLEMS USING VECTORS

Nominal Duration: 18 Hours Credits: 1.8

General Learning Outcome: SOLVE PROBLEMS USING VECTORS

Specific Learning Outcome: On completion of this unit the trainee will be able to:

1.5.1 Solve vector problems in two dimensions
1.5.2 Add phasors
1.5.3 Solve vectors problems in three dimensions

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Solving vector problems in two dimensions
  - Definition of vector
  - Unit vectors
  - Addition and subtraction of vector
  - Resolution of a vector
- Adding vectors
  - Phasor rotation
  - Phasor diagrams
  - Addition and subtraction of phasors
- Solving vector problems in three dimensions
  - Addition and subtraction
  - Unit vectors
  - Direction cosines
  - Angle between vectors
  - Scalar product

- Work done by a force

Assessment Criteria:

- Solve vector problems in two dimensions correctly
- Add vectors correctly
- Solve vector problems in three dimensions correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Books:

- Morris N. M. (1994), *Mastering Mathematics for Electrical & Electronic Engineering – Macmillan*
- Stroud K. A. (2004),*Engineering Mathematics, 4th Edition – Macmillan*

MODULE: 412-01-A ENGINEERING MATHEMATICS I

UNIT A 1.6 APPLYING COMPLEX NUMBERS

Nominal Duration: 16 Hours Credits: 1.6

General Learning Outcome: APPLY COMPLEX NUMBERS

Specific Learning Outcome: On completion of this unit the trainee will be able to:

1.6.1 Apply the j-operator
1.6.2 Represent complex numbers
1.6.3 Manipulate complex numbers algebraically
1.6.4 Apply complex numbers

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Apply the j-operator
  - The imaginary number, j 2= −1
- Representing complex numbers
  - Rectangular form (Z = x + iy)
  - The Argand diagram
  - Polar form { Z= r (Sinθ + Cosθ) } including abbreviations
  - Exponential form (Z = re jθ, Euler's formula)
- Manipulating complex numbers algebraically
  - Addition and subtraction
  - Multiplication
  - The conjugate of a complex number
  - Division
  - De Moivre's theorem

- Addition, subtraction, multiplication &division
- Roots of complex numbers

- Applying complex numbers
  - RCL circuits

Assessment Criteria:

- Apply the j-operator correctly
- Represent complex numbers correctly
- Manipulate complex numbers algebraically correctly
- Apply complex numbers correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Books:

- Morris N. M. (1994),*Mastering Mathematics for Electrical & Electronic Engineering – Macmillan*
- Stroud K. A. (2004), *Engineering Mathematics,4 th Edition – Macmillan*
- Bostock L and Chandler S (Latestedition), *Mathematics – The Core Course for A-Level –English*

MODULE: 412-01-A ENGINEERING MATHEMATICS I

UNIT A 1.7 APPLYING CALCULUS

Nominal Duration: 24 Hours Credits: 2.4

General Learning Outcome: APPLY CALCULUS

Specific Learning Outcome: On completion of this unit the trainee will be able to:

1.7.1 Obtain differential coefficients for algebraic and trigonometric functions
1.7.2 Apply differentiation
1.7.3 Obtain integrals for algebraic and trigonometric functions
1.7.4 Apply integration

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Obtaining differential coefficients
  - Gradient of a curve from first principles
  - Algebraic functions
  - Trigonometric functions
  - Hyperbolic functions
  - Products and quotients
  - Function of a function
  - Implicit functions
  - Inverse functions
  - Partial differentiation
- Applying differentiation

    - Maximum and minimum values and points of inflexion
    - Rates of change
    - Maclaurin series
- Obtaining integrals for algebraic and trigonometric functions
    - Algebraic functions
    - Trigonometric function
    - Integration by substitution
    - Integration by parts
    - Definite integrals
    - Area under a curve
- Applying integration
    - Mean values
    - Centroids
    - Root Mean Squares

Assessment Criteria:

- Differential coefficients for algebraic and trigonometric functions obtained correctly
- Differentiation solutions correctly applied
- Integrals for algebraic and trigonometric functions correctly obtained

Integration correctly applied

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Books:

- Straud A. K,*Engineering Mathematics 4 th Edition*
- Bird and May (1991) ,*Technician Mathematics 22 nd Edition*

MODULE: 412-01-A ENGINEERING MATHEMATICS I

UNIT A1.8 APPLYING POLYNOMIALS

Nominal Duration: 20 Hours Credits: 2.0

General Learning Outcome: APPLYING POLYNOMIALS

Specific Learning Outcome: On completion of this unit the trainee will be able to:

1.8.1 Apply remainder theorem
1.8.2 Apply factor theorem
1.8.3 Sketch polynomials

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Applying reminder theorem
  - Long division
  - Synthetic division
- Applying factor theorem
  - Factor theorem
  - Roots (Real and Complex)
- Sketching polynomials
  - Y-intercept
  - Roots
  - Stationary points

Assessment Criteria:

- Remainder theorem correctly applied

- Factor theorem correctly applied
- Polynomials correctly sketched

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Books:

- Straud A. K, *Engineering Mathematics 4th Edition*
- Bird and May (1991) ,*Technician Mathematics 22 nd Edition*
- Backhouse et.al (1997), *EssentialPureMathematics, Longman*

MODULE: 412-02-A ENGINEERING DRAWING

UNIT A2.1 USING DRAWING INSTRUMENTS AND EQUIPMENT

Nominal Duration: 30Hours Credits: 3

General Learning Outcome: USE DRAWING INSTRUMENTS AND EQUIPMENT

Learning Outcomes: On completion of this unit, the trainee will be able to:

2.1.1 Use types of instrument
2.1.2 Use types of equipment
2.1.3 Use drawing standards
2.1.4 Dimension the drawing

Conditions: The trainee must have access to:

- Text books and drawing standards – BS308 Parts 1, 2 & 3; ISO 1101, ISO 2595.
- Catalogues, tutorials and demonstrations
- Relevant illustrations and manuals
- Library
- Calculators
- Drafting machine and mini drafter
- Parallel ruling straight edge machine

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Using various types of instruments Drawing board
- Using various types of equipment
- Drafting machine and mini drafter
- Parallel ruling straight edge machine
- Using drawing standards

- Relevant international Standards

Assessment Criteria:

- Various types of instruments used properly
- Various types of equipment used correctly
- Drawing standards used correctly
- Drawing correctly dimensioned

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Drawing tutorials
- Tests
- Examinations
- Assignments

Recommended Textbooks

- Simmons and Maguire, A Manual of Engineering Drawing Practice
- Aggrowal M. L. and Garg R. K, (1997), Engineering Drawing, DhanpatRai and company
- Dhawan R. K. (2000), A text book of MachineDrawing S. Chand and Company Ltd
- Colm H simmons and Denniss Maguire (1993), Progrssive Engineering Drawing for BTEC studies, ELBS.

MODULE :412-02-A ENGINEERING DRAWING

UNIT A2.2 PRODUCING GEOMETRICAL FIGURES

Nominal Duration: 30 Hours Credits: 3

General Learning Outcome: PRODUCE GEOMETRICAL FIGURES

Learning Outcomes: On completion of this unit, the trainee will be able to:

2.2.1 Layout of drawing sheet
2.2.2 Use Conventions
2.2.3 Use Dimension
2.2.4 Use Symbols and Abbreviations
2.2.5 Use Line types
2.2.6 Apply Plane geometrical drawings
2.2.7 Apply types of geometry

Conditions: The trainee must have access to:

- Text books and drawing standards 1, 2 & 3; ISO 1101, ISO 2595.
- Catalogues, tutorials and demonstrations
- Relevant illustrations and manuals
- Library
- Calculators

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities:

- The trainer must ensure that the learning activities for the trainee include the following:
- Layout of drawing sheet
- Using Symbols and Abbreviation
- Mechanical symbol and abbreviation
- Electrical symbols and abbreviations

- Electronic symbols and abbreviations
- Using Line types
- Applying Basics of geometrical drawings

Assessment Criteria:

- Drawing sheet used properly
- Conventions used correctly
- Dimension used properly
- Symbols and Abbreviations used correctly
- Line types used appropriately
- Plane geometrical drawings applied correctly
- Geometry types applied adequately

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Drawing tutorials
- Tests
- Examinations
- Assignments

Recommended Textbooks

- Simmons and Maguire, A Manual of Engineering Drawing Practice
- Aggrowal M. L. and Garg R. K., (1997) ,Engineering Drawing, DhanpatRai and company
- Dhawan R. K. (2000), A text book of Machine Drawing S. Chand and Company Ltd

MODULE: 412-02-A ENGINEERING DRAWING

UNIT A2.3 PRODUCING ENGINEERING DRAWINGS

Nominal Duration: 30HoursCredits: 3

General Learning Outcome: PRODUCE ENGINEERING DRAWINGS

Learning Outcomes: On completion of this unit, the trainee will be able to:

2.3.1Apply plane geometric

2.3.2Carry out free hand sketching

2.3.3Use Solid projections

2.3.4Section the projections

2.3.5Draw Assembly drawings

2.3.6Draw exploded engineering parts

2.3.7Apply development construction

Conditions: The trainee must have access to:

- Text books and drawing standards
- Catalogues, tutorials and demonstrations
- Relevant illustrations and manuals
- Library
- Calculators
- Mechanical and Electrical Components

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Application of Plane Geometrical Drawings
- Carrying out free hand sketching
- Using solid Projection
- Applying Sectioning

- Purpose of sectional views
- Part sectional and sectional views
- Cutting plane lines and identification letters
- Different types of sections i.e. full section, half section, auxiliary sections
- ISO convections in sectioning
- Drawing Assembly drawings
- Machine drawing
- Convectional representation of common engineering parts
- Part identification and part list
- Fittings (Bolt, nut, washers, bracket, and pulleys)
- Assembling in orthographic
- Assembling in Pictorial drawing
- Drawing Exploded Engineering parts
- Principles of exploded drawing
- Using exploded drawing in orthographic project
- Using exploded drawing in isometric projection
- Applying Development
- Parallel development
- Radial development
- Triangulation

Assessment Criteria:

- Plane geometric construction applied correctly
- Free hand sketching carried out properly
- Solid projections drawn accordingly
- Sectioning done correctly
- Assembly drawings drawn correctly
- Exploded engineering parts drawn correctly
- Development construction drawn correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Drawing tutorials
- Tests
- Examinations
- Assignments

Recommended Textbooks:

- Simmons and Maguire, A Manual of Engineering Drawing Practice
- Aggrowal M. L. and Garg R. K(1997), *Engineering Drawing, DhanpatRai and company*
- Dhawan R. K. (2000) ,*A text book of Machine Drawing S. Chand and Company Ltd*
- Colm H simmons and Denniss Maguire (1993), *Progrssive Engineering Drawing for BTEC studies, ELBS.*

MODULE : 412-02-A ENGINEERING DRAWING

UNIT A2.4 PRODUCING ELECTRICAL, ELECTRONICS AND INSTRUMENTATION DRAWINGS

Nominal Duration: 30 Hours Credits: 3

General Learning Outcome: Produce Electrical and Instrumentation Drawings

Learning Outcomes: On completion of this unit, the trainee will be able to:

2.4.1 Electrical and Electronics symbols

2.4.2 Electrical circuits types

2.4.3 Electrical drawing standards and Controls

2.4.4 Apply Instrumentation drawing standards

Conditions: The trainee must have access to:

- Text books and drawing standards
- Catalogues, tutorials and demonstrations
- Relevant illustrations and manuals
- Library
- Calculators

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities:The trainer must ensure that the learning activities for the trainee include the following:

- Using Electrical and Electronic Symbols Abbreviations
- Symbols and signs
- Electrical nomenclature (resistors, capacitors, indicators, transformers)
- Instrumentation nomenclature
- Applying electrical circuit Types
- Principle type of diagram or system Block or system diagram
- Circuit or flow diagram Wiring or piping diagram Installation diagram

- Supplementary diagram
- Apply Electrical drawing standards and control Fuse and sockets
- Amperemeter, voltameter and wattmeter Electroplating Rectifier
- Remote control Relay control
- Apply Operational Electrical Circuits
- Single phase circuits operations Three phase circuit operations
- Direct on line
- Reverse forward
- Start delta etc.
- Apply Instrumentation drawing standards
- Indicators (flow, pressure, temperature and pressure)
- Recorders (flow, pressure, temperature and pressure)
- Signals
- Transducers
- Block diagrams
- Signal flow diagrams

Assessment Criteria:

- Electrical and Electronics symbols used appropriately Electrical circuits applied correctly
- Operational Electrical drawing standards and Controls used correctly
- Instrumentation drawing standards applied properly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Drawing tutorials
- Tests
- Examinations
- Assignments

Recommended Textbooks

- Anwani M L, (2006), Basic Engineering Drawing Mechanical, Electrical and Electronics, DhanpatRai and company
- Simmons and Maguire, A Manual of Engineering Drawing Practice
- Aggrowal M. L. and Garg R. K, (1997) ,Engineering Drawing, DhanpatRai and company
- Dhawan R. K. (2000) ,A text book of Machine Drawing S. Chand and Company Ltd

- Colm H simmons and Denniss Maguire (1993), Progrssive Engineering Drawing for BTEC studies, ELBS.

MODULE : 412-02-A ENGINEERING DRAWING

UNIT A2.5 PRODUCING PNEUMATIC AND HYDRAULIC DRAWINGS

Nominal Duration:30 HoursCredits: 3

General Learning Outcome: PRODUCE PNEUMATIC AND HYDRAULIC DRAWINGS

Learning Outcomes: On completion of this unit, the trainee will be able to:

2.5.1 Use Symbols

2.5.2 Use Conventions

2.5.3 Apply pneumatic drawing standards

2.5.4 Apply hydraulic drawing standards

Conditions: The trainee must have access to:

- Text books and drawing standards
- Catalogues, tutorials and demonstrations
- Relevant illustrations and manuals
- Library
- Calculators

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Using Symbols
- Using Conventions
- Apply pneumatic drawing standards of Air and its properties
- Apply hydraulic drawing standards

Assessment Criteria:

- Symbols used appropriately
- Conventions used correctly

- Pneumatic drawing standards applied properly
- Hydraulic drawing standards applied properly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Drawing tutorials
- Tests
- Examinations
- Assignments

MODULE: 412-02-A ENGINEERING DRAWING

UNIT A2.6 PRODUCING PROCESSAND INSTRUMENTATION DRAWINGS

Nominal Duration: 40 Hours **Credits: 4**

General Learning Outcome: PRODUCE PROCESS AND INSTRUMENTATION DRAWINGS

Learning Outcomes: On completion of this unit, the trainee will be able to:

2.6.1 Describe Process

2.6.2 Use Process and Instrumentation Drawing standards

2.6.3 Draw Process loops

2.6.4 Draw documentation

Conditions: The trainee must have access to:

- Text books and drawing standards – BS308 Parts 1, 2 & 3; ISO 1101, ISO 2595.
- Catalogues, tutorials and demonstrations
- Relevant illustrations and manuals
- Library
- Calculators
-

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities:The trainer must ensure that the learning activities for the trainee include the following:

- Describing Process
- Sketch the process o Use correct symbols o Design the process
- Using Process and Instrumentation Drawing standards (P&ID)
- Different types of Lines o Layout
- Use of correct symbols
- Drawing Process loops o Main

- Compose Drawing documentation

Assessment Criteria:

- Process described as required
- Process and Instrumentation Drawing standards (P&ID) used properly
- Process loops drawn correctly
- Drawing documentation composed as required

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Drawing tutorials
- Tests
- Examinations
- Assignments

Recommended Textbooks

- Ostrowsky, Engineering Drawing with CAD applications, ELBS.
- SolidWorks “Part and Assembly Basic”
- Interactive Programme: Solid Professor I

MODULE :412-02-A ENGINEERING DRAWING

UNIT A 2.7USING 2D SOLID CAD SYSTEMS TO GENERATE DRAWING

Nominal Duration:30Hours Credits:3

General Learning Outcome: USE 2D AND 3D SOLID CAD SYSTEMS TO GENERATE DRAWINGS

Learning Outcomes: On completion of this unit, the trainee will be able to:

2.7.1 AutoCAD system command
2.7.2 Setting template and CAD tools and drawing setup
2.7.3 Applying AutoCAD commands
2.7.4 Applying AutoCAD layers
2.7.5 Applying construction techniques
2.7.6 Using Blocks and drawing input and export
2.7.7 Applying dimensions
2.7.8 Applying cross hatching
2.7.9 Plotting and paper space

Conditions:The trainee must have access to:

- Computer and software for AutoCAD
- Computer Laboratory
- Library
- Manuals and catalogue
- 2D Sketching

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities:The trainer must ensure that the learning activities for the trainee include the following:

- Using Computer Aided Design (CAD)
  - Starting and close AutoCAD

  - Create and manage drawing files
  - Making new drawing
  - Saving changes to drawing files
- Set template and CAD tools and drawing setup
  - Manual drawing
  - Backing up drawing files
  - Absolute coordinates
  - Relative coordinates
  - Polar coordinates
- Applying AutoCAD commands
  - Zoom (window, previous, extents, scale) Object snap overrides
  - Running object snaps and Auto snap
  - System variables (Snap, Grid and limits
  - Creating lines, Arc and circles
  - Creating points
  - Polylines (creating and editing)
- Applying AutoCAD Layer
  - User and world coordinates
  - Inquiry tools for points
  - Drawing setup (using wizard, template files)
  - Managing layers (current layer, on and off, freeze and thaw)
  - Making new layers (Assign color and line layers)
- Applying AutoCAD Construction Techniques Offset
  - Using Blocks and drawing import and export Making block definitions
  - Inserting block definitions and drawing files Cut and paste using the windows clipboard Importing and exporting blocks
- Applying dimensions
  - Dimensions relating to distance
  - Ordinate dimensions
  - Angular dimensions
  - Creating leaders
  - Dimensioning Arcs and circles
  - Dimension styles
- Applying Cross hatching
  - Defining the boundary
  - Hatch parameters
  - Changing existing hatch objects
- Plot and Paper space
  - Output device
  - Configuring a plot
  - Paper space

Assessment Criteria:

- AutoCAD system command used correctly

- Template and CAD tools carried out as required
- AutoCAD commands carried out as required
- AutoCAD layers made according to requirements
- AutoCAD construction techniques used properly
- Blocks and drawing input and export carried out correctly
- Dimensions applied correctly
- Cross hatching used correctly
- Plot and paper space made according to measurements

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Drawing tutorials
- Tests
- Examinations
- Assignments

Recommended Textbooks

- Ostrowsky, Engineering Drawing with CAD applications, ELBS.
- SolidWorks “Part and Assembly Basic”
- Interactive Programme: Solid Professor I

MODULE :412-02-A ENGINEERING DRAWING

UNIT A 2.8USING 2D SOLID CAD SYSTEMS TO GENERATE A DRAWING

Nominal Duration: 30Hours Credits: 3

General Learning Outcome: USE 2D AND 3D SOLID CAD SYSTEMS TO GENERATE DRAWINGS

Learning Outcomes: On completion of this unit, the trainee will be able to:

2.8.1 Inserting parts in drawing module
2.8.2 Applying Dimension
2.8.3 Editing drawing sheet
2.8.4 Applying Symbols
2.8.5 Editing drawing
2.8.6 Using design journal
2.8.7 Using drawing menu

Conditions:The trainee must have access to:

- Computer with AutoCAD software
- Solid work, inventor, solid work software
- Computer Laboratory
- Library

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving
-

Learning activities: The trainer must ensure that the learning activities for the trainee include the following:

- Insert parts in drawing module
  - 3D Concepts
  - Pull down menu
  - Making new drawing

    - Apply orthographic drawing
    - Section
- Apply dimensioning
    - Dimensioning styles
    - Dimensioning Arcs and circles
    - Creating leaders
- Editing drawing sheet
    - Producing drawing sheets)
    - Drawing menu system
    - Erase and OOPs
    - Move and copy
    - Rotate
- Create Bill of Materials
    - Exporting files in other format for other programs
    - Producing drawings as per ISO/BS
    - Editing drawing sheets
- Editing 3D drawing
    - Producing drawing as per ISO/BS
    - Create the polygon, Arch and circles
    - Creating selection sets for edit commands
- Use Design Journals
    - Demonstrating the journals
    - Edit journals
- Use drawing menu
    - Offset, trim, extend, fillet, champer, array, mirrow, sketch

Assessment Criteria:

- Inserting parts in drawing module correctly
- Applying Dimension correctly
- Editing drawing sheet correctly
- Applying Symbols correctly
- Editing drawing correctly
- Using design journal correctly
- Using drawing menu correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Drawing tutorials
- Tests
- Examinations
- Assignments

Recommended Textbooks

- Ostrowsky, Engineering Drawing with CAD applications, ELBS.
- SolidWorks “Part and Assembly Basic”
- Interactive Programme: Solid Professor I
- GrgEyrich and Paul Wallach (1990), *Drafting with AutoCAD, Mitchell McGraw Hill*

MODULE: 412-03-A ELECTRONICS

UNIT A3.1 USING SEMICONDUCTOR THEORY

Nominal Duration: 30 Hours **Credits: 3**

General Learning Outcome : USE SEMICONDUCTOR THEORY

Specific learning outcomes: On completion of this unit the trainee will be able to:

3.1.1 Use Properties of Semiconductor Crystals

3.1.2 Use Doped Semiconductors

3.1.3 Use p-n Junction

Conditions: The students must have access to:

- Library
- Handouts
- Tutorials
- Internet
- Catalogues

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee includes the following:

- Using Properties of Semiconductor Crystals
  - Silicon and Germanium Atoms
  - Valence Electrons
  - Single Crystals, Energy band gap
  - Electrons and Holes
  - Conductivity in the crystals
- Using Doped Semiconductors
  - n-type Semiconductors

- p-type Semiconductors
- n-type Semiconductors Conductivity
- p-type Semiconductors Conductivity

- Using p-n Junction
  - Formation of p-n junction diode
  - Characteristics of the p-n junction
  - Rectification, Clippers and Clampers
  - Zener diode and application

Assessment Criteria:

- Properties of Semiconductor Crystals used correctly
- Doped Semiconductors used correctly
- P-n Junction used correctly

Assessment Methods: To achieve the above assessment the student will be given:

- Exercises
- Assignments
- Tests

Recommended Books:

- Robert L. Boylestad Louids Nashelsky: *Electronic Device and Circuit Theory, Prentice hall of India PL, New Delhi, 2002.*
- Cooper William David: *Solid – state devices: Analysis and Application, Reston publishing company, Virginia USA*
- Malvino: principles of electronics, Longman UK.

MODULE: 412-03-A ELECTRONICS

UNIT A 3.2 APPLYING JUNCTION DIODES, BIPOLAR JUNCTION TRANSISTORS AND UNIPOLAR TRANSISTORS (FET's)

Nominal Duration: 40 Hours **Credits: 4**

General Learning Outcome : APPLY JUNCTION DIODES, BIPOLAR JUNCTION TRANSISTORS AND UNIPOLAR TRANSISTORS (FET'S)

Specific learning outcomes: On completion of this unit the trainee will be able to:

3.2.1 Apply Device Characteristics

3.2.2 Apply Bias arrangement

3.2.3 Apply Alternating Current circuit examples

Conditions: The student must have access to:

- Library
- Handouts
- Tutorials
- Internet
- Catalogues

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee includes the following:

- Applying Device Characteristics
  - Device concepts and construction
  - Natural family Characteristics
- Applying Bias arrangement
  - Biasing methods
  - D.C Circuit analysis
  - Graphical methods
- Applying Alternating Current circuit examples

Assessment Criteria:

- Device Characteristics applied correctly
- Bias arrangement applied correctly
- Alternating Current circuit examples applied correctly

Assessment Methods: To achieve the above assessment the student will be given:

- Exercises
- Assignments
- Tests

Recommended Books:

- Robert L. Boylestad Louids Nashelsky: *Electronic Device and Circuit Theory, Prentice hall of India PL, New Delhi, 2002.*
- Cooper William David: *Solid – state devices: Analysis and Application, Reston publishing company, Virginia USA*
- Malvino: principles of electronics, Longman UK.

MODULE: 412-03-A ELECTRONICS

UNIT A 3.3 USING AMPLIFIERS

Nominal Duration: 40 Hours **Credits: 4**

General Learning Outcome : USE AMPLIFIERS

Specific learning outcomes: On completion of this unit the trainee will be able to:

3.3.1 Use Single and multistage Stage amplifiers

3.3.2 Use Differential and Operational Amplifiers (Op - Amp)

3.3.3 Use Oscillations

Conditions: The student must have access to:

- Library
- Handouts
- Tutorials
- Internet
- Catalogues

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Using Single and multistage Stage amplifiers
  - Coupling methods
  - Classification
- Using Differential and Operational Amplifiers (Op - Amp)
  - Configurations and analysis
- Using Oscillations

Assessment Criteria:

- Single and multistage Stage amplifiers used correctly

- Differential and Operational Amplifiers (Op - Amp) used correctly
- Oscillations used correctly

Assessment Methods: To achieve the above assessment the student will be given:

- Exercises
- Assignments
- Tests

Recommended Books:

- Robert L. Boylestad Louids Nashelsky: *Electronic Device and Circuit Theory, Prentice hall of India PL, New Delhi, 2002.*
- Cooper William David: *Solid – state devices: Analysis and Application, Reston publishing company, Virginia USA*
- Malvino: principles of electronics, Longman UK.

MODULE: 412-03-A ELECTRONICS

UNIT A 3.4 IDENTIFYING SPECIAL SOLID STATE DEVICES

Nominal Duration: 40 Hours **Credits: 4**

General Learning Outcome : IDENTIFY SPECIAL SOLID STATE DEVICES

Specific learning outcomes: On completion of this unit the trainee will be able to:

3.4.1 Identify Light Emitting Diodes (LED's)
3.4.2 Identify Laser Diode
3.4.3 Identify Unijunction Transistors (UJT)
3.4.4 Identify Thyristors

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials
- Internet
- Catalogues

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Identifying Light Emitting Diodes (LED's)
- Identifying Laser Diode
- Identifying Unijunction Transistors (UJT) and
- Identifying Thyristors

Assessment Criteria

- Light Emitting Diodes (LED's) correctly identified

- Laser Diode correctly identified
- Unijunction Transistors (UJT) correctly identified
- Thyristors correctly identified

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests
- Experiments

Recommended Readings/Texts

- Robert L. Boylestad Louids Nashelsky: *Electronic Device and Circuit Theory, Prentice hall of India PL, New Delhi, 2002.*
- Cooper William David: *Solid – state devices: Analysis and Application, Reston publishing company, Virginia USA*
- Malvino: principles of electronics, Longman UK.

MODULE: 412-03-A ELECTRONICS

UNIT A 3.5 USING FILTERS

Nominal Duration: 30 Hours **Credits: 3**

General Learning Outcome : USING FILTERS

Specific learning outcomes: On completion of this unit the trainee will be able to:

3.5.1 Use RC/RL Passive filters

3.5.2 Use RC/RL Active filters

3.5.3 Use Filter transformation

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials
- Internet
- Catalogues

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Using RC/RL Passive filters
  - Low Pass Filter (LPF)
  - High Pass Filter (HPF)
  - Band Pass Filter (BPF)
  - Band Refect or Band Stop Filter (BSF)
- Using RC/RL Active filters
  - Low Pass Filter (LPF)
  - High Pass Filter (HPF)
  - Band Pass Filter (BPF)
  - Band Refect or Band Stop Filter (BSF)

- Using Filter transformation
  - LPF to HPF transformation
  - LPF to BPF transformation
  - LPF to BRF transformation

Assessment Criteria

- RC Passive filters used correctly
- RC Active filters used correctly
- Filter transformation used correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Readings/Texts

- Robert L. Boylestad Louids Nashelsky: *Electronic Device and Circuit Theory, Prentice hall of India PL, New Delhi, 2002.*
- Cooper William David: *Solid – state devices: Analysis and Application, Reston publishing company, Virginia USA*
- Malvino: principles of electronics, Longman UK.

MODULE: 412-04-A ELECTRICAL TECHNOLOGY I

UNIT A 4.1 APPLYING ATOMIC THEORY

Nominal Duration: 30 Hours **Credits: 3**

General Learning Outcome: APPLY ATOMIC THEORY

Specific learning outcomes: On completion of this unit the trainee will be able to:

4.1.1 Draw the atomic structure

4.1.2 Identify types of bonding

4.1.3 Show relationship between current and electrical charge

Conditions: The trainee must have access to:

- Library
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities:

The Trainer must ensure that the learning activities for the trainee include the following:

- Drawing the atomic structure
  - Protons
  - Electrons
  - Neutrons
- Identifying types of bonding
  - Comic bonding
  - Covalent bonding
- Showing the relationship between current and charge
  - The relationship between current and charge
  - Predicting the direction of current flow

Assessment Criteria

- Atomic structure drawn correctly
- Identify types of bonding correctly
- Relationship between current and charge shown correctly
- Direction of current predicted accurately

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Books

- Edward Hughes (1998), *Electrical Technology. Addison Wesley , Longman Ltd*
- Noel Morris: *Electrical and Electronics Engineering Principles*
- J. O Bird: *Higher Electrical Technology*
- B. L. Theraja: *Fundamentals of Electrical and Electronic Engineering*

MODULE: 412-04-A ELECTRICAL TECHNOLOGY I

UNIT A4.2 USING DIRECT CURRENT (DC) CIRCUITS

Nominal Duration: 50 Hours **Credits: 5**

General Learning Outcome : USE DIRECT CURRENT (DC) CIRCUITS

Specific learning outcomes: On completion of this unit the trainee will be able to:

4.2.1 Compute electrical quantities

4.2.2 Interpret electrical laws and theorems

4.2.3 Apply symbols and units

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Computing electrical quantities
  - Current
  - Voltage
  - Resistance

- Interpreting electrical laws and theorems
  - Ohms Law
  - Kirchhoff's law
  - Superposition Theorem
  - Norton's Theorem
  - Thevenin's Theorem

- Applying Symbols and Units
  - Using different types of resistors
  - Carbon composition resistors
  - Wire wound resistors

Assessment Criteria

- Electrical quantities computed accurately
- Electrical laws and theorems interpreted correctly
- Symbols and units applied correctly
- Different types of resistors used correctly
- Use different types of capacitors correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Books

- Edward Hughes (1998), *Electrical Technology. Addison Wesley , Longman Ltd*
- Noel Morris: *Electrical and Electronics Engineering Principles*
- J. O Bird: *Higher Electrical Technology*
- B. L. Theraja: *Fundamentals of Electrical and Electronic Engineering*

MODULE: 412-04-A ELECTRICAL TECHNOLOGY I

UNIT A4.3 CALCULATING POWER AND ENERGY

Nominal Duration: 40 Hours Credits: 4

General Learning Outcome : CALCULATE POWER AND ENERGY

Specific learning outcomes: On completion of this unit the trainee will be able to:

4.3.1 Calculate electrical power, energy, and electromotive force

4.3.2 Connect lamps

4.3.3 Compute maximum power transfer

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Calculate electrical power, energy, electromotive force
  - Calculating electrical power
  - Calculating energy
  - Computing electromotive force
- Connecting lamps
  - Fluorescent tubes
  - Sodium vapour lamps
  - Mercury vapour lamps
  - Halogen lamps

    - Neon lamps
- Compute maximum power transfer
    - Load power
    - Supply power
    - Internal power loss
    - Power factor

Assessment Criteria

- Electrical power calculated correctly
- Electrical energy calculated correctly
- Electromotive force (EMF) computed accurately
- Lamps connected correctly
- Compute maximum power transfer correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests
- projects

Recommended Books

- Edward Hughes (1998), *Electrical Technology. Addison Wesley , Longman Ltd*
- Noel Morris: *Electrical and Electronics Engineering Principles*
- J. O Bird: *Higher Electrical Technology*
- B. L. Theraja: *Fundamentals of Electrical and Electronic Engineering.*

MODULE: 412-04-A ELECTRICAL TECHNOLOGY I

UNIT A4.4 EMPLOYING LAWS AND THEOREMS

Nominal Duration: 30 Hours Credits: 3

General Learning Outcome : EMPLOY LAWS AND THEOREMS

Specific learning outcomes: On completion of this unit the trainee will be able to:

4.4.1 Apply electrical laws

4.4.2 Using theorem to solve problems

Conditions: The trainee must have access to:

- Library
- Tutorial
- Assignment
- Handouts

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Apply electronics laws
  - Applying Ohm's law
  - Solving problems involving Kirchhoff's law
  - Applying Thevenin's theorem
- Using theorem to solve problems
  - Computing Norton's theorem to solve problems
  - Using superposition theorem to solve problems

Assessment Criteria

- Ohm's law applied correctly
- Kirchhoff's law problems solved accurately
- Thevenin's theorem applied accurately
- Norton's theorem problems solved accurately
- Superposition theorem problems solved correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Exercises

Recommended Books

- Edward Hughes (1998), *Electrical Technology. Addison Wesley , Longman Ltd*
- Noel Morris: *Electrical and Electronics Engineering Principles*
- J. O Bird: *Higher Electrical Technology*
- B. L. Theraja: *Fundamentals of Electrical and Electronic Engineering*

MODULE: 412-05-A COMMUNICATION SKILLS

UNIT A5.1 USING A LIBRARY

Nominal Duration: 10 hours **Credits: 1**

General Learning Outcome: USE A LIBRARY

LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

5.1.1 Illustrate filing systems
5.1.2 Identify references and dictionaries
5.1.3 Identify periodicals
5.1.4 Prepare bibliography in any written work
5.1.5 Use the library facilities to obtain required information

Conditions:The trainee must have access to:

- Books
- Library

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Illustrating the filing system
  - Classifying books according to class works
  - Arranging books alphabetically according to the first three letters of the author's name
- Identifying the references and dictionaries
  - Using material within the library
  - Selected materials in this section are either in short supply or are of special nature e.g. dictionaries.
- Identifying periodicals in the library
  - Specialised section where user finds newspapers, magazines and current information
- Preparing bibliography in any written work
- Learning author, subject, description of a book and location in the library sometime this is called catalogue.
- Using the library facilities to obtain required information
  - Look for relevant information from books

- Book classification and location
- Internet (www), library to locate relevant literature.

Assessment Criteria

5.1.1 Illustrating the filling system is properly done
5.1.2 Identify the references and dictionaries correctly done
5.1.3 Identifying periodical correctly written
5.1.4 Preparing bibliography is correctly written
5.1.5The library facilities to obtain required information used correctly

Assessment Methods:To achieve the above assessment the trainee will be given

- Assignment
- Tests
- Practical

MODULE NO: 412-05-ACOMMUNICATION SKILLS

UNIT 5.2 DISCUSSING PROCESS OF COMMUNICATION

Nominal Duration: 10 hours **Credits: 1**

General Learning Outcome : DISCUSS PROCESS OF COMMUNICATION

LEARNING OUTCOMES: On completion of this unit the trainee should be able to:

5.2.1 Recognize the nature and purpose of communication.

5.2.2 Identify the communication process.

5.2.3 Identify barriers of effective communication

5.2.4 Apply strategies of effective communication

5.2.5 Identifying the flow of communication in organization

Conditions:The trainee must have access to:

- Library
- Lecture notes
- Language Laboratory

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Recognizing the nature and purpose of communication.
  - Nature and purposes of communication
  - The Use of English
  - Choices of communication
- Identifying the communication process.
  - Communication circle
  - Channels of communication (types)
    - Verbal
      - Meetings
      - Briefings
    - Non Verbal
      - Paralanguage

        - Kinesics
        - Proximie
        - Colour
        - time
    - Written
        - Journals
        - Notices
        - Memos
        - Reports

- Identifying barriers of effective communication
    - Non-verbal signs
        - Noise
        - Semantic problem
    - Poor listening skills
        - Poor timing
    - Prejudgment
        - I attitude syndrome
    - Emotional response
        - Information overload
- Applying strategies to effective communication
    - Extend language knowledge through reading (Read widely)
    - Listen attentively
    - Be open minded
    - Time your communication
    - Status
    - Maintain High Standards\
    - Obtain feedback
- Identifying the flow of communication in organization
    - Vertical
        - Upward
        - Down ward
    - Horizontal
        - Lateral
    - Diagnosed communication

Assessment Criteria

5.2.1 Recognize the nature and purpose of communication correctly
5.2.2 Identify the communication process as required
5.2.3 Identify barriers of effective communication correctly
5.2.4 Apply strategies of effective communication correctly
5.2.5 The flow of communication in organization correctly identified

Assessment Methods:**To achieve the above assessment the trainee will be given**

- Tests
- Practical tests

- Assignments
- Projects

MODULE: 412-05-ACOMMUNICATION SKILLS

UNIT A5.3 APPLYING STUDY SKILLS

Nominal Duration: 10 hours **Credits: 1**

General Learning Outcome : APPLY STUDY SKILLS

LEARNING OUTCOMES: On completion of this unit the trainee should be able to:

5.3.1 Acquire skills for intensive reading

5.3.2 Develop appropriate writing skills

5.3.3 Develop listening and speaking skills

Conditions: The trainee must have access to:

- Library,
- Videos,
- Role-plays,
- Workshop and work place.

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Acquiring skills for intensive reading
- Developing appropriate writing skills
  - Main points from a text
  - Summarizing paragraphs
  - Technical description and definitions
  - Systematic approach to business writing
- Five key rules to business writing
  - Straight to the point
  - KISS principle
  - Active and not passive voice
  - Use the right tone
  - Use modern language
- General guidelines in composing business letters
  - Chose simple words
  - Use appropriate tone

- Be precise
- Be accurate
- Check consistency
- Use initiative
- Use active and not passive voice
- Write as you speak

- Developing listening and speaking skills
  - Presenting an argument
  - Questioning skills

Assessment Criteria

5.3.1 Acquire skills for intensive reading correctly
5.3.2 Develop appropriate writing skills correctly
5.3.3 Develop listening and speaking skills correctly

Assessment Methods:**To achieve the above assessment the trainee will given**

- Tests
- Assignments
- Projects
- Group work

MODULE: 412-05-A COMMUNICATION SKILLS

UNIT A5.4 APPLYING BUSINESS CORRESPONDENCE SKILLS

Nominal Duration: 10 hours **Credits: 1**

General Learning Outcome : APPLY CORRESPONDENCE SKILLS

LEARNING OUTCOMES: On completion of this unit the trainee should be able to:

5.4.1 Write different types of letters

5.4.2 Respond to memorandum and carry out appropriate action

5.4.3 Respond to circulars and carry appropriate action.

5.4.4 Complete forms and documents

Conditions: The trainee must have access to:

- Library,
- Videos,
- Role-plays
- Workshop and work place

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Writing different types of letters
  - Qualities of a business letter
    - Paper quality
    - Envelope
    - Type face
    - Logo and Corporate Image
    - General Impression
  - Methods of displaying business letters
  - Structure of a business letter
    - Opening paragraphs
    - Main body
    - Closing paragraph
  - Categories of Business letters
    - Letter of conformation
    - Acknowledgement
    - Enquiry
    - Reply to enquiry

- Complaint
- Adjustment
- Collection letter
- Recommendation

- Types of letters
  - Formal
  - Informal

Assessment Criteria

5.4.1 Different types of letters written correctly
5.4.2 Memos and carrying appropriate action responded to correctly.
5.4.3 Appropriate action in response to circulars carried out appropriately.
5.4.4 Forms and documents completed correctly.

Assessment Methods:**To achieve the above assessment the trainee will be given**

- Written assignments
- Group Discussions
- Practical

MODULE 412-05-A COMMUNICATION SKILLS

UNIT A5.5 WRITING REPORTS

Nominal Duration: 10 hours Credits: 1

General Learning Outcome : WRITE REPORTS

LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

5.5.1 Apply research techniques
5.5.2 Write differentiate types of reports
5.5.3 Use report format
5.5.4 Compose different types of report
5.5.5 Write a proposal

Conditions: The trainee must have access to:

- Library
- Handouts
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed

Smarter/Bolder/Driven

- Time conscious
- Problem solving
-

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Applying research techniques
  - Library
  - Internet
- Writing different types of reports
  - Routine or regular report
    - Financial report
    - Equipment maintenance report
    - Staff appraisal
  - Specialised commissioned reports
    - Market research
    - Market forecasting
    - Investigation report

- Using report format
  - Different report formats
- Composing different types of report
  - Informal
  - Formal
- Writing a proposal
  - Components of a proposal

Assessment Criteria

5.5.1 Research techniques applied as per requirement
5.5.2 Differentiate types of reports written correctly
5.5.3 Report format used correctly
5.5.4 Different types of reports composed correctly
5.5.5 A proposal written appropriately

Assessment Methods:To achieve the above assessment the trainee will begiven

- Tests
- Assignments
- Projects
- Group work

MODULE 412-05-A COMMUNICATION SKILLS

UNITA5.6 APPLYING INTERPERSONAL SKILLS IN ANORGANISATION

Nominal Duration: 15 hours **Credits:1.5**

General Learning Outcome : APPLY INTERPERSONAL SKILLS IN ANORGANISATION

Learning Outcomes:On completion of this unit the trainee will be able to:

5.6.1 Consult with colleagues, Supervisors and Organisations
5.6.2 Carry out joint considerations of problems
5.6.3 Receive instructions
5.6.4 Participate in decision making
5.6.5 Handle customers
5.6.6 Get permission

Conditions: The trainee must have access to:

- library
- Handouts
- Lecture notes

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smarter/Bolder/Driven
- Time conscious
- Problem solving

Learning Activities:The Trainer must ensure that the learning activities for the trainee include the following:

- Consulting with colleagues, Supervisors and Organisations
- Carrying out joint considerations of problems
  - Importance of joint consideration
    - Reaching same conclusion
    - Importance of policy or decision
    - Favourable attitudes
    - More better decisions
    - Suggestions available
    - More commitment
- Receiving instructions
- Participating in decision making

- Leadership exercise
- Decisions emerges as the general feeling of the group
- Gives opportunity for all members to have a say

- Handling customers
- Getting/Granting permissions

Assessment Criteria

5.6.1 Consult with colleagues, Supervisors and Organisations correctly
5.6.2 Carry out joint considerations of problems correctly
5.6.3 Receive instructions correctly
5.6.4 Participate in decision making
5.6.5 Handle customers correctly
5.6.6 Get or grant permission correctly

Assessment Methods:To achieve the above assessment the trainee will be given

- Tests
- Assignments
- Group work
- Practical

MODULE 412-05-A COMMUNICATION SKILLS

UNITA5.7 USING TECHNICAL ENGLISH

Nominal Duration: 15 hours **Credits: 1.5**

General Learning Outcome : USE TECHNICAL ENGLISH

LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

5.7.1 Identify technical tools
5.7.2 Apply technical instructions
5.7.3 Use technical descriptions

Conditions:The trainee must have access to:

- Library
- Handouts
- Workshop

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smarter/Bolder/Driven
- Problem solving
- Time conscious

Learning Activities:The Trainer must ensure that the learning activities for the trainee include the following:

- Identifying technical tools
  - Tools
- Applying technical instructions
  - Job cards
  - Operation of tools
  - Grammar used
- Using technical descriptions
  - Mechanisms
  - Processes of tools

Assessment Criteria

5.7.1 Identify technical tools correctly
5.7.2 Apply technical instructions correctly
5.7.3 Use technical descriptions correctly

Assessment Methods:To achieve the above assessment the trainee will be given

- Tests

- Assignments
- Group work
- Practical

References:

- Peter, I. (2004), *Communication in business, Longman, London,*
- *Roy C, Rodger G 2008 Building construction handbook, Butter Heinemann, Oxford*
- Shirley Taylor (2009) *,Effective communication, Bolton ,England*
- Stanton, P.N. (1982), *Communication in Business, DPP ,London*
- Wiley-Huseman, L and Penrose, (2010), *Business communication strategies and skills, London, DPP*

MODULE: 412-06-A WORKSHOP PRACTICE AND PROCESSES

UNIT A 6.1 USING VARIOUS TYPES OF TOOLS

Nominal Duration: 30 Hours **Credits: 3**

General Learning Outcome: USE VARIOUS TYPES OF TOOLS

Learning Outcomes: On completion of this unit, the trainee will be able to:

6.1.1 Use hand tools

6.1.2 Operate Power Tools

6.1.3 Employ Measuring tools

6.1.4 Operate Machine tools

Conditions: The trainee must have access to:

- Lectures/tutorials
- Workshop facilities
- Manuals and reference books
- Library
- IT equipment
- Calculator

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Operating Power Tools
- Hand drill
- Power drills
- Hand grinding machines
- Pedestal grinding machines
- Employing Measuring tools
- Steel rule
- Vernier callipers
- Compasses
- Operating Machine tools
- Lathe
- Grinding machine
- Milling machine

- Hydraulic press

Assessment Criteria:

- Hand tools used correctly
- Power Tools operated properly
- Measuring tools employed correctly
- Machine tools operated as required

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments
- Mini design projects

Recommended Textbooks

Chapman,W A J (1974) *Workshop Technology Part I: An Introductory Course, London: The English Language Book Society & Edward Arnold (Publishers) Ltd.*

Lissaman A.J. and Martin S.J(1987), *Principlesof Engineering Production, 2nd Ed., Hodder and Stoughton*

RadfordJ.D.RichardsonD.B(1983),*Production Engineering Technology, 3rd Edition, Macmillan.*

KempsterM.H.A.(1974), *Introduction to Jig and Tool Design, Hodder and Stroughton, 3rd Edition*

Browne J.W(1965), The Theory of Machine Tools Book 1, Cassell

Browne J.W(1968), The Theory of Machine Tools Book 2, Cassell

GaylerJ.F.W(1977), Shebolt C.R. Metrology for Engineers, Cassell

MODULE :412-06- A WORKSHOP PRACTICE AND PROCESSES

UNIT A 6.2 USING VARIOUS JOINTING AND FASTENING METHODS

Nominal Duration: 50 Hours Credits: 5

General Learning Outcome: USE VARIOUS FASTENING AND JOINTING METHODS

Learning Outcomes: On completion of this unit, the trainee will be able to:

6.2.1Solder
6.2.2Braze
6.2.3Weld
6.2.4Rivet
6.2.5Use Crimping lugs and terminals
6.2.6Use Bolts, nuts and screws
6.2.7Use Push and pull joints
6.2.8Use jointing adhesives

Conditions: The trainee must have access to:

- Lectures/tutorials
- Workshop facilities
- Manuals and reference books
- Library
- IT equipment
- Calculator

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Operating and handling of welding machines
- Welding techniques
- Application
- Riveting
- Types of rivets and their application

- Using Crimping lugs and terminals
- Crimping tools
- Crimping materials
- Application
- Terminals and their applications
- Using Bolts, nuts and screws
- Coarse ISO threaded
- Fine ISO threaded
- Various types of screws and their application
- Using Push and pull joints
- Pneumatic joints
- Hydraulic joints
- Limitations of push/pull joints
- Using jointing adhesives
- Types of adhesives
- Application
- Safety
- Advantages/Limitations

Assessment Criteria:

- Soldering done correctly
- Brazing done properly
- Welding carried out as required
- Riveting performed correctly
- Crimping lugs and terminals used as required
- Bolts, nuts and screws used effectively
- Push and pull joints used effectively
- Jointing adhesives correctly used

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments
- Mini design projects

Recommended Textbooks

Chapman, W A J (1974) Workshop Technology Part I: An Introductory Course, London: The English Language Book Society & Edward Arnold (Publishers) Ltd.

Lissaman A.J. and Martin S.J(1987), Principles of Engineering Production, 2nd Ed., Hodder and Stoughton

Radford J.D.RichardsonD.B(1983),Production Engineering Technology, 3rd Edition, Macmillan.

Kempster M.H.A. (1974), Introduction to Jig and Tool Design, Hodder and Stroughton, 3rd Edition
Browne J.W(1965), The Theory of Machine Tools Book 1, Cassell
Browne J.W(1968), The Theory of Machine Tools Book 2, Cassell
GaylerJ.F.W(1977), SheboltC.R.Metrology for Engineers, Cassell

MODULE :412-06- A WORKSHOP PRACTICE AND PROCESSES

UNIT A 6.3 CARRYING OUT THREADING AND TAPPING

Nominal Duration: 50 Hours Credits: 5

General Learning Outcome: CARRY OUT THREADING AND TAPPING

Learning Outcomes: On completion of this unit, the trainee will be able to:

6.3.1 Differentiate types of threads

6.3.2 Make bolts and nuts

6.3.3 Identify ordinary and self-taping screws

6.3.4 Identify seals

Conditions:The trainee must have access to:

- Lectures/tutorials
- Workshop facilities
- Manuals and reference books
- Library

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities:The trainer must ensure that the learning activities for the trainee include the following:

- Differentiating types of threads
- Tap Types
- Drill-chart
- Threads in a hole
- Tap- handle
- Cutting oil
- Making bolts and nuts
- Die types
- Die-chart
- Threads on Bolts
- Die-Handle
- Cutting oil
- Using ordinary and self-taping screw
- Fasteners Usage
- Mixed

- Machine screws Cap Screws
- Sheet metal screws Set screws
- Wood screws
- Identifyingseals
- Paper
- Rubber
- Polythene
- Hemp
- Oils

Assessment Criteria:

- Differentiate types of threads correctly
- Make bolts and nuts correctly
- Identify ordinary and self-taping science correctly
- Identify seals correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments
- Mini design projects

Recommended Textbooks

OlivoThomas . C. (1990) ,*Fundamentals of Machine Tool technology and manufacturing Processes.New York; C.Thomas Associates*

Zbigniew D. jastrzebski (1987),*The Nature and Properties of Engineering Materials, John wiley and Sons New York*

MODULE :412-06- A WORKSHOP PRACTICE AND PROCESSES

UNIT A 6.4 ASSEMBLING AND DIS-ASEMBLING

Nominal Duration: 50 Hours Credits: 5

General Learning Outcome: ASSEMBLE AND DIS-ASSEMBLE

Learning Outcomes: On completion of this unit, the trainee will be able to:

6.4.1Apply concepts of assembling and dis-assembling

6.4.2Use different modes of connecting sub-assemblies

6.4.3Use different modes of mounting sub-assemblies

Conditions: The trainee must have access to:

- Lectures/tutorials
- Workshop facilities
- Manuals and reference books
- Library
- Calculator
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities:The trainer must ensure that the learning activities for the trainee include the following:

- Applying concepts of assembling and dis-assembling
  - Types of tools
  - First Out Last In (FOLI)
  - Cleaning
  - Test- running
  - Isolation procedure
- Using different modes of connecting sub assemblies
  - Flange type
  - Coupling type
  - Thread type
  - Push and pull types
- Using different modes of mounting

- sub assemblies
- Saddle clamp
- Rawlbolts
- Plug and Screws

Assessment Criteria:

- Apply concepts of assembling and dis-assembling correctly
- Use different modes of connecting sub assemblies correctly
- Use different modes of mounting sub assemblies correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments
- Mini design projects

Recommended Textbooks

Chapman, W A J (1974),*Workshop Technology Part I: An Introductory Course, London: The English Language Book Society & Edward Arnold (Publishers) Ltd.*

Lissaman A.J. and Martin S.J (1987).*Principles of Engineering Production, 2nd Ed., Hodder and Stoughton*

MODULE: 412-06- A WORKSHOP PRACTICE AND PROCESSES

UNIT A 6.5 APPLYING QUALITY CONTROL PRACTICES

Nominal Duration: 20 Hours Credits:2

General Learning Outcome: APPLY QUALITY CONTROL PRACTICES

Learning Outcomes: On completion of this unit, the trainee will be able to:

6.5.1Apply quality assurance practices

6.5.2Apply statistics in quality control

6.5.3Identify technical aspects of quality control

Conditions: The trainee must have access to:

- Internet
- Library
- SHEQ manuals
- Tutorials
- Textbooks/Handbooks
- Industry

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Applying quality assurance practices
- Acceptance quality control
- Set points/parameters Specifications
- Quality control process
- Continuous monitoring
- Corrective measures in process
- Continuous improvement
- Applying statistics in quality control o Frequency distribution
- Averages
- Variability
- Identifying technical aspects of quality control
- Brainstorming
- Cause and effect analysis
- Pareto analysis

- Sampling
- Control charts

Assessment Criteria:

- Apply quality assurance practices correctly
- Apply statistics in quality control correctly
- Identify technical aspects of quality control correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises/practicals
- Assignments
- Tests
- Examinations

Recommended Readings/Texts

BLACK, B J (1997),*Workshop Processes, Practices and Materials, 2nd Ed., London: Arnold.*

KIBBE, R R, NEELY, J E, MEYER, R O and WHITE, W T (2005),*Machine Tool Practices, 6th Ed., San Francisco: Prentice-Hall.*

IRCA (1998),*Health and Safety for Employees, Randburg: International Risk Control America Inc.*

WILLIAMSON, J J (Latest Edition),*General Fire Hazards and Fire Protection, London: Sir Isaac Pitman & Co Ltd.*

MODULE: 412-07-A INTRODUCTION TO COMPUTERS

UNIT A7.1 OPERATING A COMPUTER SYSTEM

Nominal Duration: 13 Hours **Credits:1.3**

General Learning Outcome : OPERATE A COMPUTER SYSTEM

LEARNING OUTCOMES: On completion of this unit the trainee should be able to:

7.1.1 Handle Open, close, login and logout procedure of a personal computer
7.1.2 Identify basic element of a Personal Computer and input/output devices
7.1.3 Handle of Floppies, USB disk and Hard disks.
7.1.4 Maintain a Personal Computer

Conditions: The trainee must have access to:

- Personal computer with keyboard, mouse and printer
- Internet
- Library

Values/ Attitudes:

- Sober minded
- Hard working
- Alert
- Committed
- Honest

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Handling Open, close, login and logout procedure of a personal computer
- Identifying basic element of a Personal Computer
- Inputting /outputting devices
- Handling of Floppies, USB disk and Hard disks.

- Maintaining of a Personal Computer

Assessment Criteria:

- Handle Open, close, login and logout procedure of a
- personal computer correctly
- Identify basic element of a Personal Computer and
- onput/output devices correctly
- Handle of Floppies, USB disk and Hard disks correctly.
- Maintain of a Personal Computer correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- A Practical hands on test
- Written test

MODULE NO: 412-07-A INTRODUCTION TO COMPUTERS

UNIT A7.2USING AN OPERATING SYSTEM

Nominal Duration: 14 Hours **Credits: 1.4**

General Learning Outcome: USE THE OPERATING SYSTEM

LEARNING OUTCOMES: On completion of this unit the trainee should be able to:

7.2.1 Use the file system
7.2.2 Handle libraries
7.2.3 Install software programs
7.2.4 Remove virus on personal Computer system

Conditions:The trainee must have access to:

- Personal computer with keyboard, mouse and Printer
- Internet
- Software programs
- Virus protection software

Values/ Attitudes:

- Sober minded
- Hard-working
- Alert
- Committed
- Honest
- Time conscious
- Smarter/Bolder/Driven
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Using the file system
- Handling libraries
- Installing software programs.
- Removing virus on personal Computer system

Assessment Criteria:

- Use the file system
- Handle libraries
- Install software programs.
- Remove virus on personal Computer system

Assessment Methods: To achieve the above assessment the trainee will be given

- A Practical hands on test
- Written test

MODULE: 412-07-A INTRODUCTION TO COMPUTERS

UNIT A7.3 USE WORD PROCESSING FOR TECHNICAL REPORT

Nominal Duration: 14 Hours **Credits: 1.4**

General Learning Outcome :USE WORD PROCESSING FOR TECHNICAL REPORT

LEARNING OUTCOMES: On completion of this unit the trainee should be able to:

7.3.1 Handle a document
7.3.2 Edit a document
7.3.3 Use fonts
7.3.4 Insert picture, graphics and spreadsheet
7.3.5 Handle tables
7.3.6 Format footer and header
7.3.7 Use Speller and grammar
7.3.8 Index

Conditions: The trainee must have access to:

- Personal computer with keyboard, mouse and printer
- Access to Microsoft word or similar software

Values/ Attitudes:

- Sober minded
- Hard working
- Alert
- Committed
- Honest
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Handling of a document
- Editing a document
- Using fonts
- Insert picture, graphics and spreadsheet.
- Handling tables
- Formatting footer and header
- Using Speller and grammar

- Indexing

Assessment Criteria:

- Handle document correctly
- Edit a document correctly
- Use fonts appropriately
- Insert picture, graphics and spreadsheet correctly
- Handle tables properly
- Format footer and header as per instruction
- Use Speller and grammar correctly
- Index correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Written test
- Assignment of a Practical report

MODULE NO: 412-07-A INTRODUCTION TO COMPUTERS

UNIT A7.4 SENDING AND RECEIVING E-MAIL

Nominal Duration: 11 Hours **Credits: 1.1**

General Learning Outcome: SEND AND RECEIVE E-MAIL

LEARNING OUTCOMES: On completion of this unit, the trainee will be able to:

7.4.1 Compose e-mail
7.4.2 Send e-mail
7.4.3 Receive e-mail
7.4.4 Reply to e-mail
7.4.5 Forward e-mail
7.4.6 Observe etiquette when using e-mail

Conditions: The trainee must have access to:

- Personal computer with Internet access
- Computer user manuals

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- E-mail user name, address and password
  - Internet or Local Area Network (LAN)
  - Opening the e-mail server
  - Creation of address book
- Using e-mail commands when composing e-mail
  - Compose command
  - Use address “To” field
  - Copy field
  - Edit command
  - Document attach command
  - Send command
- Receiving e-mail
  - Open command
  - Saving e-mail in various formats

- Replying to e-mail
  - Use reply command
- Forwarding e-mail
  - Use forwarding command
- Observing etiquette when using e-mail

Assessment Criteria:

7.4.1 Compose e-mail correctly
7.4.2 Send e-mail correctly
7.4.3 Receive e-mail correctly
7.4.4 Reply to e-mail correctly
7.4.5 Forward e-mail correctly
7.4.6 Observe etiquette when using e-mail correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Practical exercise hands on

MODULE NO: 412-07-A INTRODUCTION TO COMPUTERS

UNIT A7.5 USING SPREADSHEET

Nominal Duration: 15 Hours **Credits: 1.5**

General Learning Outcome: USE SPREADSHEET

LEARNING OUTCOMES: On completion of this unit the trainee should be able to:

7.5.1 Manage Spreadsheet
7.5.2 Manage Cells and contents
7.5.3 Handle Rows, columns and content
7.5.4 Handle Sheets and content
7.5.5 Use Functions - Mathematic, logical and statistical
7.5.6 Create database/list of data
7.5.7 Create Diagrams and Charts

Conditions: The trainee must have access to:

- Personal computer with keyboard, mouse and printer
- Access to Microsoft Excel or similar software

Values/ Attitudes:

- Sober minded
- Hard working
- Alert
- Committed
- Honest
- Smarter/Bolder/Driven
- Time conscious
- Problem solving

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- Managing Spreadsheet
- Managing Cells and contents
- Handling Rows, columns and content
- Handling Sheets and content
- Using Functions - Mathematic, logical and statistical
- Creating database/list of data
- Create Diagrams and Charts

Assessment Criteria:

- Manage Spreadsheet properly
- Manage Cells and contents correctly
- Handle Rows, columns and content as per instruction
- Handle Sheets and content correctly

- Use Functions - Mathematic, logical and statistical
- Create database/list of data correctly
- Create Diagrams and Charts correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- A Practical assignment with Excel doing the above exercises from Assessment criteria

MODULE NO: 412-07-A INTRODUCTION TO COMPUTERS

UNIT A7.6 USING THE INTERNET

Nominal Duration: 13 Hours **Credits: 1.3**

General Learning Outcome : USE THE INTERNET

LEARNING OUTCOMES: On completion of this unit, the trainee will be able to:

7.6.1 Enumerate the benefits of using the Internet
7.6.2 Browse the World Wide Web (www)
7.6.3 Save web pages
7.6.4 Download documents from the Web
7.6.5 Observe Rules for using the Internet

Conditions: The trainee must have access to:

- Personal computer with Internet access
- Computer user manuals

Values/ Attitudes:

- Sober minded
- Hard working
- Alert
- Committed
- Honest
- Smarter /Bolder/Driven
- Problem solving
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Enumerating the benefits of using the Internet
- Browsing the World Wide Web (www)
  - Web page addresses
  - Search engines
  - Links
- Saving web pages
  - Various saving formats (html, text, pdf, word, other formats)
- Downloading documents from the Web
  - Selecting required document or part of it
- Observing Rules for using the Internet
  - Malicious software
  - Alteration of third party web pages

Assessment Criteria:

- Enumerate the benefits of using the Internet correctly
- Browse the World Wide Web (www) correctly

- Save web pages correctly
- Download documents from the Web correctly
- Observe Rules for using the Internet correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Practical exercise hands on

Recommended Textbooks

- Personal Computer for beginners

MODULE: 412-08-A ENTREPRENEURSHIP

UNIT A8.1: USING ENTREPRENEURSHIP CONCEPTS

NOMINAL DURATION: 10 Hours CREDITS: 1

General Learning Outcome : Use entrepreneurial concepts

SPECIFIC LEARNING OUTCOMES: On completion of this unit trainees should be able to:

8.1.1 Explain concepts of entrepreneurship
8.1.2 Describe different types of entrepreneurs
8.1.3 Discuss the back ground to emergence of entrepreneurship
8.1.4 Explain the benefits of entrepreneurship

Conditions: **The trainee must have access to:**

- Library facilities
- Hand-outs
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- **Explaining concepts of entrepreneurship**
  - Generic explanation of the terms
    - Enterprise
    - Entrepreneurs
    - Entrepreneurship
    - Enterprise
    - Entrepreneurial competences
  - Define new enterprise creation

- Differentiate between an Entrepreneur and an Intrapreneur

- **Describing the different types of entrepreneurs**
  - Distinguish between pushed and pulled entrepreneurs
  - Explain enterprise, entrepreneurship and the entrepreneur
- **Discussing the back ground to emergence of entrepreneurship**
  - Origin of entrepreneurship
  - Motive of entrepreneurship
  - Importance of entrepreneurship to national development
- **Explaining the benefits of entrepreneurship.**
  - Employment and self-employment
  - Benefits of enterprise development

Assessment Criteria:

- Entrepreneurship concepts well explained
- Types of entrepreneurs correctly described
- Background to emergence of entrepreneurship well discussed
- Benefits of entrepreneurship correctly explained.

Assessment Methods: To achieve the above assessment the trainee will be given;

- Classroom exercise
- Assignments
- Tests
- Case studies

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

Educating the Next Wave of Entrepreneurs-Unlocking entrepreneurial capabilities to meet the global challenges of the 21st Century (2009) World Economic Forum (A Report of the Global Education Initiative)

Encyclopedia of Forms and Precedents, Firth Edition, , Re-issue, Butterworth, Vol. 14, London 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

G A Cole (2003) Management Theory and Practice.

Judy Pakus ,Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry of Commerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

Synthesis of Main Findings from two Case Studies carried out in Ghana and Zambia on Private Technical and Vocational Education and Training (TEVT), International Institute for of Educational Planning, 7-9 rue Eugene-Delacroix, 75116, Paris, UNESCO, March 2003.

Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

Technical Education Vocational and Training TEVT, Statement of Policy and Intent, Government of the Republic of Zambia, 1969.

MODULE: 412-08-A ENTREPRENEURSHIP

UNIT A8.2 BUILDING POSITIVE ATTITUDE TOWARDS ENTREPRENEURSHIP

NOMINAL DURATION: 9 HOURS CREDITS: 0.9

General Learning Outcome : BUILD POSITIVE ATTITUDE TOWARDS ENTREPRENEURSHIP:

SPECIFIC LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

8.2.1 Develop positive thinking
8.2.2 Develop positive ambition
8.2.3 Discuss positive outlook
8.2.4 Develop constructive ambition

Conditions: The trainee must have access to:

- Library
- Personal Entrepreneurial Competences self-rating questionnaire
- Lecture notes
- Internet
- Practical/Field visits

Values/ Attitudes:

- Sober minded
- Honest
- Committed
- Disciplined
- Diligence

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- **Developing positive thinking**

- The mind-set of a positive entrepreneur
- Positive thinking and goal setting

- **Developing positive ambition**
  - The psychological preparedness of an entrepreneur

- **Discussing positive outlook**
  - The entrepreneurs psychological preparedness
- **Developing constructive ambition**
  - Realistic targets
  - Setting realistic targets

Assessment Criteria

- Positive thinking well developed
- Positive ambition well developed
- Positive outlook well discussed
- Constructive ambition appropriately developed.

Assessment Methods: To achieve the above assessment the trainee will be given

- Individual or group assessment
- Case studies
- Tests
- Field visits
- Summary examinations

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

Educating the Next Wave of Entrepreneurs-Unlocking entrepreneurial capabilities to meet the global challenges of the 21st Century (2009) World Economic Forum (A Report of the Global Education Initiative)

Encyclopedia of Forms and Precedents, Firth Edition, , Re-issue, Butterworth, Vol. 14, London 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

G A Cole (2003) Management Theory and Practice

Judy Pakus ,Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry of Commerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

Synthesis of Main Findings from two Case Studies carried out in Ghana and Zambia on Private Technical and Vocational Education and Training (TEVT), International Institute for of Educational Planning, 7-9 rue Eugene-Delacroix, 75116, Paris, UNESCO, March 2003.

Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

Technical Education Vocational and Training TEVT, Statement of Policy and Intent, Government of the Republic of Zambia, 1969.

MODULE: 412-08-A ENTREPRENEURSHIP

UNIT A8.3 DEMONSTRATING PERSONAL ENTREPRENEURIAL COMPETENCES (PECs)

NOMINAL DURATION: 11 HOURS CREDITS: 1.1

General Learning Outcome : DEMONSTRATE PERSONAL ENTREPRENEURIAL COMPETENCES (PECs)

SPECIFIC LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

8.3.1 Identify Personal Entrepreneurial Competences (PECs) relating to achievement cluster

8.3.2 Assess Personal Entrepreneurial Competences relating to achievement to planning cluster

8.3.3 Apply Personal Entrepreneurial Competences relating to achievement to power cluster.

Conditions: The trainee must have access to:

- Library
- Personal Entrepreneurial Competences self-rating questionnaire
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Committed
- Disciplined
- Diligent

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- **Identifying Personal Entrepreneurial Competences relating to Achievement Cluster**
    - Practice Opportunity Seeking and Initiative
    - Practice Risk Taking
    - Practice Demand for Efficiency and Quality
    - Practice Persistence
    - Practice Commitment to the Work Contract

- **Assessing Personal Entrepreneurial Competences relating to Planning Cluster**
    - Practice Information Seeking
    - Practice Goal setting
    - Practice Systematic Planning and Monitoring and Evaluation
    - Assessing the Personal Entrepreneurial Competences
        - Answer a self-rating questionnaire/peer assessment
        - Apply entrepreneurial competences assessment tools (Personal Entrepreneurial Competences self-rating questionnaires)
        - Apply peer assessment
        - Conduct assessment of entrepreneurial competences

- **Applying Personal Entrepreneurial Competences relating to Power Cluster**
    - Practice Persuasion and Networking
    - Practice Independence and self-confidence

Assessment Criteria

- Personal Entrepreneurial Competences correctly identified
- Personal Entrepreneurial Competences correctly assessed

- Personal Entrepreneurial Competences correctly applied

Assessment Methods: To achieve the above assessment the trainee will be given

- Individual exercise
- Group exercises
- Tests
- Assignments
- Examinations

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

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Encyclopedia of Forms and Precedents, Firth Edition, , Re-issue, Butterworth, Vol. 14, London 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

G A Cole (2003) Management Theory and Practice

Judy Pakus ,Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry of Commerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

Synthesis of Main Findings from two Case Studies carried out in Ghana and Zambia on Private Technical and Vocational Education and Training (TEVT), International Institute for of Educational Planning, 7-9 rue Eugene-Delacroix, 75116, Paris, UNESCO, March 2003.

Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

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MODULE: 412-08-A ENTREPRENEURSHIP

UNIT A8.4 ESTABLISHING AN ENTERPRISE

NOMINAL DURATION: 11 HOURS CREDITS: 1.1

General Learning Outcome : ESTABLISH A BUSINESS.

SPECIFIC LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

8.4.1 Identify / generate a business idea
8.4.2 Select the most viable business idea
8.4.3 Develop a business opportunity
8.4.4 Establish an enterprise

Conditions: The trainee must have access to:

- Library
- Lecture notes
- Internet
- Calculator

Values/ Attitudes:

- Sober minded
- Honest
- Committed
- Disciplined
- Diligent

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- **Identifying/ generating a business idea**
  - Identifying sources of business ideas

- Techniques of generating business ideas
- Generate a business idea
- Conduct an environmental scanning for the business idea
- Conduct afeasibility study for the business idea

- **Selecting the most viable business idea**
  - Assessing the most viable business idea
  - Use a micro screening table
    - State solvent demand
    - State availability of qualified personnel
    - State availability of technology/equipment
    - State availability of raw materials
    - State competitors
    - State critical success factors
- **Identifying a business opportunity**
  - Identify sources of business opportunities
  - Outline steps for business opportunity identification
  - Explain considerations in opportunity selection
  - Describe opportunity identification dynamics
    - Discuss the need of the society
    - Discuss capability of the entrepreneur
    - Discuss the resource availability in the environment
- **Establishing an enterprise**
  - Design of the business
    - Generate a business idea
    - Conduct a market research
    - Conduct SWOT analysis

- Develop the business
  - Prepare a business plan
  - Arrange finance (own capital, family, bank)
  - Find location
  - Arrange logistics
- Managing the business
  - Cost and price
  - Market the products
  - Establish record keeping systems
  - Outline time management schedules
  - Apply financial management
  - Conduct stock control
  - Institute business and private guidelines
- Evaluating the business
  - Evaluate financial results
  - Evaluate business results
  - Adjust plans

Assessment Criteria

- A business idea well identified /generated
- The most viable business idea correctly selected
- A business opportunity well developed
- An enterprise well established

Assessment Methods: To achieve the above assessment the trainee will be given

- Individual exercise
- Group exercises
- Tests

- Assignments
- Examinations

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

Educating the Next Wave of Entrepreneurs-Unlocking entrepreneurial capabilities to meet the global challenges of the 21st Century (2009) World Economic Forum (A Report of the Global Education Initiative)

Encyclopedia of Forms and Precedents, Firth Edition, , Re-issue, Butterworth, Vol. 14, London 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

G A Cole (2003) Management Theory and Practice

Judy Pakus ,Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry of Commerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

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Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education Vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

Technical Education Vocational and Training TEVT, Statement of Policy and Intent, Government of the Republic of Zambia, 1969.

MODULE: 412-08-A ENTREPRENEURSHIP

UNIT A8.5 MANAGING AN ENTERPRISE

NOMINAL DURATION: 10 HOURS CREDITS: 1

General Learning Outcome : MANAGE AN ENTERPRISE.

SPECIFIC LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

8.5.1 Apply principles of human resource management
8.5.2 Apply principles of financial management
8.5.3 Manage business operations
8.5.4 Apply principles of marketing management
8.5.5 Use networking management information systems
8.5.6 Adhere to business ethics and values
8.5.7 Fulfil statutory obligations
8.5.8 Undertake strategic planning
8.5.9 Manage credits

Conditions: The trainee must have access to:

- Library
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Committed
- Disciplined
- Diligent

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- **Applying principles of human resource management**

  - Define human resources terms
  - Discuss how to recruit, interview, and retain employees more effectively

- Follow up with new employees in a structured manner
- Advocate for your employees' health and safety
- Provide accurate, actionable feedback to employees
- Act appropriately in situations requiring discipline and termination

- **Applying principles of financial management**

- Using financial terminology
- Understand financial statements
- Identify how to analyze financial statements
- Understand budgets
- Outline how to make budgeting easy
- Understand advanced forecasting techniques
- Understand how to manage the budget
- Identify How to make smart purchasing decisions
- Identify the legal aspects of finance

- **Managing business operations**

- Operations Strategy and managing change
- Operations Strategy and competitiveness
- Project schedule development
- Product design and process selection
- Process analysis
- Total Quality Management
- Supply Chain Management
- Capacity Management
- Just-in-Time and Lean Systems
- Resources Planning Systems
- Aggregate Planning
- Forecasting

- Inventory Control
- Material Requirement Planning
- Operations Scheduling

- **Applying principles of marketing management**

- Define marketing management
- Formulate the marketing mix
- Conduct a market research for products and services
- Evaluate results from the market research
- Introduce SWOT analysis framework

- **Using networking management information systems**

- Define (informal) networking
- Describe Entrepreneur in a SPIDER web environment
- Identify kinds of networks
  - Implicit networks (informal)
  - Explicit networks (Formal)
- State the three basic principles of networking
  - Discuss Trust
  - Discuss Maintenance or active participation
  - Discuss Reciprocity
- Outline networking Rules
- Set Your PEC Barometer- Networking

- **Adhering to business ethics and values**

- Define and understand ethics.
- Understand the benefits of ethics.
- Create strategies to implement ethics at work.

- Recognize social and business responsibility.
- Identify ethical and unethical behavior.
- Learn how to make ethical decisions and lead with integrity.

- **Fulfilling statutory obligations**

- Explain the legal framework for small enterprise
- Discuss forms of business
- Outline advantages and disadvantages of different forms of business
- Discuss registration procedures with PACRA
  - Registration of business name
  - Incorporation of limited companies
  - Registration of co-operatives
- Discuss ZRA procedures
  - Discuss types of taxes
  - Discuss ZRA returns
  - Discuss tax exceptions

- **Undertaking strategic planning**

- Define strategic planning
- Outline the strategic Planning process
- Outline key performance indicators

Assessment Criteria

- Human resource management principles appropriately applied.
- Financial management principles appropriately applied.
- Business operations well executed.
- Marketing management principles well applied.
- Networking management information appropriately used.
- Business ethics and values well adhered to.
- Statutory obligations well complied with.
- Strategic planning well undertaken.

- Credits well managed.

Assessment Methods: To achieve the above assessment the trainee will be given

- Individual exercise
- Group exercises
- Tests
- Assignments
- Examinations

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

Educating the Next Wave of Entrepreneurs-Unlocking entrepreneurial capabilities to meet the global challenges of the 21st Century (2009) World Economic Forum (A Report of the Global Education Initiative)

Encyclopedia of Forms and Precedents, Firth Edition, Re-issue, Butterworth, Vol. 14, London 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

G A Cole (2003) Management Theory and Practice

Judy Pakus Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry of Commerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

Synthesis of Main Findings from two Case Studies carried out in Ghana and Zambia on Private Technical and Vocational Education and Training (TEVT), International Institute for of Educational Planning, 7-9 rue Eugene-Delacroix, 75116, Paris, UNESCO, March 2003.

Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

Technical Education Vocational and Training TEVT, Statement of Policy and Intent, Government of the Republic of Zambia, 1969.

MODULE: 412-08-A ENTREPRENEURSHIP

UNIT A8.6 GROWING AN ENTERPRISE

NOMINAL DURATION: 11 HOURS CREDITS: 1.1

General Learning Outcome : GROW AN ENTERPRISE

SPECIFIC LEARNING OUTCOMES: On completion n of this unit the trainee will be able to:

8.6.1 Identify possible growth avenues
8.6.2 Engage professional staff
8.6.3 Manage business performance
8.6.4 Mobilise finance for growth

Conditions: The trainee must have access to:

- Library
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Committed
- Disciplined
- Diligent

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- **Identifying growth avenues**
    - Explain the business life cycle
    - Explain idea generation
    - Explain business launch
    - Explain business growth and expansion
    - Explain business maturity

- Explain business decline
- Explain business phase out or re-launch
    - Outline types of growth avenues

- **Engaging professional staff**
    - Defining recruitment.
    - Understanding the selection process.
    - Recognizing the GROW model and how to set goals.
    - Preparing for the interview and question process.
    - Identifying and avoiding bias when making offers.
    - Discovering ways to retain talent and measure growth.

- **Managing business performance**
    - Define performance management.
    - Understand how performance management works and the tools to make it work.
    - Learn the three phases of project management and how to assess it.
    - Discuss effective goal-setting.
    - Learn how to give feedback on performance management.
    - Identify Kolb's Learning Cycle.
    - Recognize the importance of motivation.
    - Develop a performance journal and performance plan.

- **Mobilising finance for growth**
    - Explain role of finance in an enterprise
    - Discuss sources of finance for an enterprise
    - State the importance of good financial management

Assessment Criteria

- Possible growth avenues correctly identified.
- Professional staff correctly engaged
- Business performance well managed
- Finance for growth appropriately mobilised

Assessment Methods: To achieve the above assessment the trainee will be given

- Individual exercise
- Group exercises
- Tests
- Assignments
- Examinations

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

Educating the Next Wave of Entrepreneurs-Unlocking entrepreneurial capabilities to meet the global challenges of the 21st Century (2009) World Economic Forum (A Report of the Global Education Initiative)

Encyclopedia of Forms and Precedents, Firth Edition, , Re-issue, Butterworth, Vol. 14, London 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

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Judy Pakus ,Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry of Commerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

Synthesis of Main Findings from two Case Studies carried out in Ghana and Zambia on Private Technical and Vocational Education and Training (TEVT), International Institute for of Educational Planning, 7-9 rue Eugene-Delacroix, 75116, Paris, UNESCO, March 2003.

Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

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MODULE: 412-08-A ENTREPRENEURSHIP

UNIT A8.7: ESTABLISHING BUSINESS NETWORKS

NOMINAL DURATION: 9 HOURS CREDITS: 0.9

General Learning Outcome : ESTABLISH BUSINESS NETWORKS.

SPECIFIC LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

8.7.1 Identify networking agenda

8.7.2 Identify business networks

8.7.3 Join business networks

8.7.4 Attend business forums

8.7.5 Evaluate and control business networks

Conditions: The trainee must have access to:

- Library
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Committed
- Disciplined
- Diligent

LEARNING ACTIVITIES: The trainer must ensure that the learning activities for the trainee include the following:

- **Identifying networking agenda**
  - The importance of networking
  - An enterprise networking web
- **Identifying business networks**
  - Internal networks
  - External networks
- **Joining business networks**
  - Stakeholders and shareholders

  - Terms and conditions
  - Performance Evaluation
- **Attending business forums**
  - Business forum
  - Types of business forums
  - The benefits of business forums
- **Evaluating and controlling business networks**
  - Sustaining business networks
  - Performance of business networks

ASSESSMENT CRITERIA

- Networking agenda correctly identified
- Business networks correctly identified
- Business networks joined
- Business forums appropriately attended.
- Business networks well evaluated and controlled

ASSESSMENT METHODS: To achieve the above assessment the trainee will be given

- Individual exercise
- Group exercises
- Tests
- Assignments
- Examinations

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business.

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

Educating the Next Wave of Entrepreneurs-Unlocking entrepreneurial capabilities to meet the global challenges of the 21st Century (2009) World Economic Forum (A Report of the Global Education Initiative)

Encyclopedia of Forms and Precedents, Firth Edition, Re-issue, Butterworth, Vol. 14, London, 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

G A Cole (2003) Management Theory and Practice

Judy Pakus ,Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry of Commerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

Synthesis of Main Findings from two Case Studies carried out in Ghana and Zambia on Private Technical and Vocational Education and Training (TEVT), International Institute for of Educational Planning, 7-9 rue Eugene-Delacroix, 75116, Paris, UNESCO, March 2003.

Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

Technical Education Vocational and Training TEVT, Statement of Policy and Intent, Government of the Republic of Zambia, 1969.

MODULE: 412-08-A ENTREPRENEURSHIP

UNIT A8.8: EXITING A BUSINESS

NOMINAL DURATION: 7 HOURS CREDITS: 0.7

General Learning Outcome : EXIT A BUSINESS

SPECIFIC LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

8.8.1 Develop an exit strategy

8.8.2 Develop a succession plan

8.8.3 Explain estate management

Conditions: The trainee must have access to:

- Library
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Committed
- Disciplined
- Diligent

LEARNING ACTIVITIES: The trainer must ensure that the learning activities for the trainee include the following:

- **Developing exit strategies**
  - Mergers
  - Acquisitions
  - Debentures
  - Management buy out
- **Developing a succession plan**
  - Business succession planning and its role in your company.
  - Lay the groundwork to develop a succession plan.
  - The importance of mentorship.

- SWOT analysis to set goals.
- Plan, assign roles, and execute the plan.
- Develop support and manage change.
- Obstacles, and evaluate and adapt goals and plans.
- Success of succession plan.

- **Explaining estate management**
  - Estate management
  - Types estate management
  - The enterprise estate management obligations to government

ASSESSMENT CRITERIA

- Exit strategy correctly developed
- Succession plan correctly developed
- Estate correctly managed

ASSESSMENT METHODS: To achieve the above assessment the trainee will be given

- Individual exercise
- Group exercises
- Tests
- Assignments
- Examinations

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

Educating the Next Wave of Entrepreneurs-Unlocking entrepreneurial capabilities to meet the global challenges of the 21st Century (2009) World Economic Forum (A Report of the Global Education Initiative)

Encyclopedia of Forms and Precedents, Firth Edition, , Re-issue, Butterworth, Vol. 14, London, 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

G A Cole (2003) Management Theory and Practice

Judy Pakus ,Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry oCommerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

Synthesis of Main Findings from two Case Studies carried out in Ghana and Zambia on Private Technical and Vocational Education and Training (TEVT), International Institute for of Educational Planning, 7-9 rue Eugene-Delacroix, 75116, Paris, UNESCO, March 2003.

Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

Technical Education Vocational and Training TEVT, Statement of Policy and Intent, Government of the Republic of Zambia, 1969.

MODULE: 412-08-A ENTREPRENEURSHIP

UNIT A8.9: ASSESSSING BUSINESS PERFORMANCE

NOMINAL DURATION: 8 HOURS CREDITS: 0.8

General Learning Outcome : ASSESS BUSINESS PERFORMANCE.

SPECIFIC LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

8.9.1 Set targets

8.9.2 Identify Key Performance Indicators

8.9.3 Structure data collection tools

8.9.4 Outline measuring criteria

8.9.5 Write a report

Conditions: The trainee must have access to:

- Library
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Committed
- Disciplined
- Diligent

LEARNING ACTIVITIES: The trainer must ensure that the learning activities for the trainee include the following:

- **Setting targets**
  - Target setting
  - The target setting process
  - Targets
- **Identifying Key Performance Indicators**
- **Structuring data collection tools**
  - Structured interviews
  - Questionnaire

- Observations
- **Outlining measuring criteria**
- **Writing a report**

ASSESSMENT CRITERIA

- Target well set.
- Key Performance indicators correctly identified.
- Data collection tools well structured.
- Measuring criteria correctly outlined
- Report well written

ASSESSMENT METHODS: To achieve the above assessment the trainee will be given

- Individual exercise
- Group exercises
- Tests
- Assignments
- Examinations

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

Educating the Next Wave of Entrepreneurs-Unlocking entrepreneurial capabilities to meet the global challenges of the 21st Century (2009) World Economic Forum (A Report of the Global Education Initiative)

Encyclopedia of Forms and Precedents, Firth Edition, , Re-issue, Butterworth, Vol. 14, London, 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

G A Cole (2003) Management Theory and Practice

Judy Pakus ,Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry of Commerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

Synthesis of Main Findings from two Case Studies carried out in Ghana and Zambia on Private Technical and Vocational Education and Training (TEVT), International Institute for of Educational Planning, 7-9 rue Eugene-Delacroix, 75116, Paris, UNESCO, March 2003.

Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

Technical Education Vocational and Training TEVT, Statement of Policy and Intent, Government of the Republic of Zambia, 1969.

MODULE: 412-08-A ENTREPRENEURSHIP

UNIT A8.10 MANAGING PROFITS, EXCESS CASH AND RISK

NOMINAL DURATION: 9 HOURS CREDITS: 0.9

General Learning Outcome : Manage profits, excess cash and risk.

SPECIFIC LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

8.10.1 Differentiate between profit and cash

8.10.2 Select alternative investments

8.10.3 Conduct investment appraisal

8.10.4 Carryout risk management

Conditions: The trainee must have access to:

- Library
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Committed
- Disciplined
- Diligent

LEARNING ACTIVITIES: The trainer must ensure that the learning activities for the trainee include the following:

- **Differentiating between profit and cash**
  - Profit
  - Nature of profits
  - Calculating profit
  - Gross profit
  - Net profit
  - Profitability
  - Profitability ratios
- **Explaining cash**

- Money
- The functions of money
- The abilities of money
- Barter
- The deficiencies of barter, rate of exchange indivisibility of goods, double coincident of wants
- The early forms of money
- The measurement of money

- **Selecting alternative investments**
  - Strategic issues in investment
  - The investment process
  - New technology.
- **Conducting investment appraisal**
  - Uncertainty and investment decisions
  - Managers' risk attitudes
- **Carrying out risk management**
  - Risk management
  - Expected NPV and its variability
  - Techniques for assessing project risk
  - The risk-return trade off

ASSESSMENT CRITERIA

- Profit and cash well differentiated
- Alternative investments appropriately found
- Investment appraisal well conducted
- Risk management correctly carried out

ASSESSMENT METHODS: To achieve the above assessment the trainee will be given

- Individual exercise
- Group exercises
- Tests
- Assignments

- Examinations

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

Educating the Next Wave of Entrepreneurs-Unlocking entrepreneurial capabilities to meet the global challenges of the 21st Century (2009) World Economic Forum (A Report of the Global Education Initiative)

Encyclopedia of Forms and Precedents, Firth Edition, Re-issue, Butterworth, Vol. 14, London, 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

G A Cole (2003) Management Theory and Practice

Judy Pakus Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry of Commerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

Synthesis of Main Findings from two Case Studies carried out in Ghana and Zambia on Private Technical and Vocational Education and Training (TEVT), International Institute for of Educational Planning, 7-9 rue Eugene-Delacroix, 75116, Paris, UNESCO, March 2003.

Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

Technical Education Vocational and Training TEVT, Statement of Policy and Intent, Government of the Republic of Zambia, 1969.

MODULE: 412-08-A **ENTREPRENEURSHIP**

UNIT A8.11: **NEGOTIATING AGREEMENTS AND CONTRACTS**

NOMINAL DURATION: **5 HOURS** **CREDITS: 0.5**

General Learning Outcome : Negotiate agreements and contracts.

SPECIFIC LEARNING OUTCOMES: On completion of this unit the trainee will be able to:

8.11.1 Use different approaches of negotiations

8.11.2 Identify qualities of good negotiators

8.11.3 Prepare for negotiations

8.11.4 Manage contracts

Conditions: The trainee must have access to:

- Library
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Committed
- Disciplined
- Diligent

LEARNING ACTIVITIES: The trainer must ensure that the learning activities for the trainee include the following:

- **Using different approaches of negotiations**
  - Basic types of negotiations, the phases of negotiations, and the skills needed for successful negotiating
  - Pre-contract negotiations
  - Contract negotiations
  - Post contract negotiations
  - Basic negotiating concepts
  - WATNA
  - BATNA,

- WAP
- ZOPA

- **Identifying qualities of negotiators**
  - Basic negotiation skills
  - Basic bargaining techniques
  - Strategies for identifying mutual gain
- **Preparing for negotiations**
  - Groundwork for negotiation
  - What information to share and what to keep to yourself
  - Reach consensus and set the terms of agreement
  - Deal with personal attacks and other difficult issues
  - Negotiating process to solve everyday problems
  - Negotiate on behalf of someone else
- **Managing contracts**
  - Contracts
  - Types of tender
    - Open tenders
    - Restricted tenders
  - Tender procedures
  - Bargain for tender

ASSESSMENT CRITERIA

- Approaches of negotiation correctly used
- Qualities of negotiators correctly identified
- Negotiations well prepared for
- Contracts correctly managed

ASSESSMENT METHODS: To achieve the above assessment the trainee will be given

- Individual exercise
- Group exercises
- Tests
- Assignments

- Examinations

REFERENCE MATERIALS

Barrve Hawkins (1993), How to Run a Part-Time Business

Chakonta, D.C., and Nyirenda,N. (2011): Concept Paper –Review of the National Entrepreneurship Curriculum for TEVET.

Contemporary Research in the USA, Germany and Japan on Five Education Issues: Structure of Education Systems, Standards in Education, The Role of School in Adolescents' Lives, Individual Differences amongst Students and Teachers' Lives; Arnold W. Stevenson et al., 1997, Internet.

David Stokes and Nicholas Wilson (2006), Small Business Management and Entrepreneurship (5th Edition), Thomas learning, London

Educating the Next Wave of Entrepreneurs-Unlocking entrepreneurial capabilities to meet the global challenges of the 21st Century (2009) World Economic Forum (A Report of the Global Education Initiative)

Encyclopedia of Forms and Precedents, Firth Edition, Re-issue, Butterworth, Vol. 14, London 1994.

Entrepreneurship and Small Business Management: (4th Edition); Pearson Education, New York.

G A Cole (2003) Management Theory and Practice

Judy Pakus ,Turn Your Ideas into Business Success, London.

Micro, Small and Medium Enterprise Development Policy (2009): Ministry of Commerce, Trade and Industry-Zambia (pp10)

Paul Burns (2001): Entrepreneurship and Small Business, Palgrave Macmillan: New York

Peter F. Drucker (1985); Innovation and Entrepreneurship, Harper and Row, USA,

Questions and Answers on the Dual System of Vocational Training in Germany, Federal Ministry of Education, Science, Research and Technology, D-53170 Bonn, November, 1996.

Robert D. Hisrich, Michael P. Peters, Dean A. Shepherd (2005) Entrepreneurship (Sixth Edition), Tata McGraw – Hill Publishing Company Limited, New York

Robert Graven (2001) Kick-Start Your Business: 100 Days to a Learner, Fitter Organization; Virgin Books Limited. London

Synthesis of Main Findings from two Case Studies carried out in Ghana and Zambia on Private Technical and Vocational Education and Training (TEVT), International Institute for of Educational Planning, 7-9 rue Eugene-Delacroix, 75116, Paris, UNESCO, March 2003.

Technical Education Vocational and Entrepreneurship and Training (TEVT) Policy Paper, Government of the Republic of Zambia, 1996.

Technical Education vocational and Training (TEVET) Statement policy and Intent, GRZ 1999

Technical Education Vocational and Training TEVT, Statement of Policy and Intent, Government of the Republic of Zambia, 1969.

MODULE: 412-09-B ELECTRICAL TECHNOLOGY II

UNIT B 9.1 ANALYSING PRINCIPLES OF MAGNETISM

Nominal Duration: 20 Hours **Credits: 2**

General Learning Outcome: ANALYSE PRINCIPLES OF MAGNETISM

Specific learning outcomes: On completion of the module, trainees will be able to:

9.1.1 Interpret magnetic flux laws

9.1.2 Compute magnetic field strength

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The tutor must ensure that the learning activities for the trainee include the following:

- Interpret magnetic flux laws
  - Permeability of free space
  - Leakage flux
- Compute magnetic field strength
  - Magnetic field strength
  - Reluctance

Assessment Criteria:

- Magnetic flux laws Interpreted correctly
- Magnetic field strength computed accurately

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Tests/Experiments
- Assignments
- Examinations

Recommended Readings/Texts

- JOHNSON, D E, JOHNSON, J R, HILBURN, J L, BELL, D (1991). *Electrical Technology, Simon & Schuster Ltd.*
- GREEN, D C (1997). *Higher Electrical Principles, Addison Wesley Longman Ltd.*
- HUGHES, E (1991). *Electrical Technologies, London: Longman Scientific & Technical.*

MODULE: 412-09-B ELECTRICAL TECHNOLOGY II

UNIT B 9.2 APPLYING PRINCIPLES OF MAGNETISM

Nominal Duration: 20 Hours **Credits: 2**

General Learning Outcome : APPLY PRINCIPLES OF MAGNETISM

Specific learning outcomes: On completion of this unit the trainee will be able to:

9.2.1 Use magnets

9.2.2 Apply induction principles

9.2.3 Apply relays in electrical circuits

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The tutor must ensure that the learning activities for the trainee include the following:

- Using magnets
  - Analysis magnets
  - Calculations magnetic flux
- Applying induction principles
  - Faraday induction principle
  - Flemming lemming induction principle
  - Instantaneous values of EMF
- Applying relays in electrical circuits

- Construction
- Principles of operation

Assessment Criteria:

- Magnets used correctly
- Induction applied correctly
- Relays in electrical circuits applied correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Tests/Experiments
- Examinations

Recommended Readings/Texts

- JOHNSON, D E, JOHNSON, J R, HILBURN, J L, BELL, D (1991). *Electrical Technology, Simon & Schuster Ltd.*
- GREEN, D C (1997), *Higher Electrical Principles, Addison Wesley Longman Ltd.*
- HUGHES, E (1991), *Electrical Technologies, London: Longman Scientific & Technical.*

MODULE: 412-09-B ELECTRICAL TECHNOLOGY II

UNIT B 9.3 APPLYING ELECTROSTATIC THEORY

Nominal Duration: 20 Hours **Credits: 2**

General Learning Outcome : APPLY ELECTROSTATIC THEORY

Specific learning outcomes: On completion of this unit, the trainee will be able to:

9.3.1 Apply laws of attraction and repulsion

9.3.2 Apply capacitors in circuits

9.3.3 Identify dielectrics

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Applying laws of attraction and repulsion
  - Laws of attraction and repulsion
  - Analysis and applying of the laws
- Applying capacitors in circuits
  - Construction
  - Principles of operation
- Identifying dielectrics
  - Principles of operation
  - Characteristic

Assessment Criteria:

- Laws of attraction and repulsion Applied correctly
- Capacitors in circuits Applied correctly
- Dielectrics Identified correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Tests/Experiments
- Examinations

Recommended Readings/Texts

- JOHNSON, D E, JOHNSON, J R, HILBURN, J L, BELL, D (1991). *Electrical Technology, Simon & Schuster Ltd.*
- GREEN, D C (1997). *Higher Electrical Principles, Addison Wesley Longman Ltd.*
- HUGHES, E (1991). *Electrical Technologies, London: Longman Scientific & Technical.*

MODULE: 412-09-B ELECTRICAL TECHNOLOGY II

UNIT B 9.4 EXPLAINING ELECTROMAGNETIC INDUCTION

Nominal Duration: 20 Hours Credits: 2

General Learning Outcome : EXPLAIN ELECTROMAGNETIC INDUCTION

Specific learning outcomes: On completion of this unit, the trainee will be able to:

9.4.1 Define Faraday's and Lenz's Law

9.4.2 Use Flemming's Right Hand Rule

9.4.3 Explain self and mutual inductance

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials
- Laboratories / Demonstrations
- Textbooks / Handbooks

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Explaining Faraday's and Lenz's Law
  - Defining Faraday's Law
  - Defining Lenz's Law
- Using Flemming's Right Hand Rule
  - Flemming's Right Hand Rule
  - Analysis and using of the rule
- Explaining self and mutual inductance
  - Self-inductance
  - Mutual inductance
  - Coefficient of mutual inductance
  - Coefficient of coupling

Assessment Criteria:

- Faraday's Law explained correctly
- Lenz's Law explained correctly

- Flemming's Right Hand Rule used correctly
- Self and mutual inductance explained correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Tests/Experiments
- Examinations

Recommended Readings/Texts

- JOHNSON, D E, JOHNSON, J R, HILBURN, J L, BELL, D (1991). *Electrical Technology, Simon & Schuster Ltd.*
- GREEN, D C (1997), *Higher Electrical Principles, Addison Wesley Longman Ltd.*
- HUGHES, E (1991), *Electrical Technologies, London: Longman Scientific & Technical.*

MODULE: 412-09-B ELECTRICAL TECHNOLOGY II

UNIT B 9.5 ANALYSING SINGLE PHASE CIRCUITS

Nominal Duration: 20 Hours Credits: 2

General Learning Outcome: ANALYSE SINGLE PHASE CIRCUITS

Specific learning outcomes: On completion of this unit, the trainee will be able to:

9.5.1 Explain generation of single phase supply

9.5.2 Explain peak values

9.5.3 Identify average values

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Explaining generation of single phase supply
  - Single phase characteristics
  - Generation of single phase supply
- Explaining peak values
  - Peak values
  - Factors
- Identifying average values
  - Average values
  - Voltage

- Electric current
- Electric power

Assessment Criteria:

- Generation of single phase supply Explained accurately
- Peak values Explained correctly
- Average values Identified correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Tests/Experiments
- Examinations

Recommended Readings/Texts

- JOHNSON, D E, JOHNSON, J R, HILBURN, J L, BELL, D (1991), *Electrical Technology, Simon & Schuster Ltd.*
- GREEN, D C (1997), *Higher Electrical Principles, Addison Wesley Longman Ltd.*
- HUGHES, E (1991), *Electrical Technologies, London: Longman Scientific & Technical.*

MODULE: 412-09-B ELECTRICAL TECHNOLOGY II

UNIT B 9.6 DESCRIBING THREE PHASE SUPPLY

Nominal Duration: 30 Hours **Credits: 3**

General Learning Outcome : Describe three phase supply

Specific learning outcomes: On completion of this unit, the trainee will be able to:

9.6.1 Interpret production of three phase alternating Current supply
9.6.2 Identify line voltage and current
9.6.3 Identify phase voltage and current
9.6.4 Apply power formula

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Interpreting production of three phase alternating current supply
  - Three phases characteristics
  - Production of three phase alternating current
- Identifying phase voltage and current
  - Three phases voltage
  - Three phases current
- Applying power formulae

- Active power formulae
- Reactive power formulae
- Apparent power formulae

Assessment Criteria:

- Production of three phase alternating current supply interpreted correctly
- Line voltage and current identified correctly
- Phase voltage and current identified correctly
- Power formula applied correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Tests/Experiments
- Examinations

Recommended Readings/Texts

- JOHNSON, D E, JOHNSON, J R, HILBURN, J L, BELL, D (1991), *Electrical Technology, Simon & Schuster Ltd.*
- GREEN, D C (1997), *Higher Electrical Principles, Addison Wesley Longman Ltd.*
- HUGHES, E (1991), *Electrical Technologies, London: Longman Scientific & Technical.*

MODULE: 412-09-B ELECTRICAL TECHNOLOGY II

UNIT B 9.7 USING MEASURING INSTRUMENTS

Nominal Duration: 20 Hours Credits: 2

General Learning Outcome : USE MEASURING INSTRUMENTS

Specific learning outcomes: On completion of this unit, the trainee will be able to:

9.7.1 Identify types of measuring instruments

9.7.2 Interpret extending the range of instruments

9.7.3 Identify types of bridges

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Identifying types of measuring instruments
  - Types of measuring instruments
  - Using of instruments
- Interpreting extending the range of instruments
  - Extending the range of instruments
- Identifying types of bridges
  - Types of bridges

Assessment Criteria:

- Types of instruments identified correctly
- Extending the range of instruments interpreted correctly
- Types of bridges identified correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Tests/Experiments
- Examinations

Recommended Readings/Texts

- JOHNSON, D E, JOHNSON, J R, HILBURN, J L, BELL, D (1991), *Electrical Technology, Simon & Schuster Ltd.*
- GREEN, D C (1997), *Higher Electrical Principles, Addison Wesley Longman Ltd.*
- HUGHES, E (1991), *Electrical Technologies, London: Longman Scientific & Technical.*

MODULE: 412-10-B MATERIAL SCIENCE

UNIT B10.1 SELECT MATERIALS

Nominal Duration: 12 Hours **Credits: 1.2**

General Learning Outcome: Select materials

Learning Outcomes: On completion of the module, trainees will be able to:

10.1 Identify economic factors to be considered
10.2 Identify properties of materials
10.3 Specify applications for the materials

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The Tutor must ensure that the learning activities for the trainee include the following:

- Identifying economic factors to be considered
  - Availability
  - Cost
  - Forming methods
- Identifying properties of materials
  - Mechanical properties: tensile strength, modulus of elasticity, hardness and wear resistance, oxidation and corrosion, creep and fatigue characteristics, ductility,
  - Physical properties: density, linear coefficient of expansion, thermal and electrical conductivity, boiling and melting points

- Chemical properties: corrosion resistance, chemical composition, acidity/alkalinity
- Formability: forming methods, ease of manufacture, joining, finishing
- Identifying applications for the materials
- Load conditions: dead load, cyclic load, load magnitude and amplitude,
- Environmental conditions: corrosiveness, temperature,
- Ergonomics: colour, appearance, feel, ease of application

Assessment Criteria:

- Choose economic factors to be considered correctly
- Identify properties of materials correctly
- Specify applications for the materials correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Tests
- Examinations

Recommended Readings/Texts

- ASHBY M F & JONES D R H (1996),*Engineering Materials 1: An Introduction to Their Properties and Applications, 2nd Ed, Oxford: Butterworth-* Heinemann.
- JOHN V B, (1994), *Engineering Materials, London: Macmillan*
- ASHBY M F &JONES D R H (1994),*Engineering Materials 2: An Introduction to Microstructures,* Processing and Design, Oxford: Pergamon.
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*
- FELBECK D K. Introduction to Strengthening Mechanisms, (Latest Ed.), Prentice-Hall.
- ROLLASON E C. Metallurgy for Engineers, (Latest Ed.), London: Arnold

MODULE :412-10-B MATERIAL SCIENCE

UNIT B10.2 IDENTIFYING ATOMIC STRUCTURES

Nominal Duration: 12 Hours **Credits:1.2**

General Learning Objective: IDENTIFYING ATOMIC STRUCTURES

Learning Outcomes:On completion of this unit the trainee will be able to:

10.2.1 Identify Sub-atomic particles
10.2.2 Use concept of Bonding
10.2.3 Describe Crystallography
10.2.4 Describe Amorphous structures

Conditions:The trainee must have access to:

- Library
- Handouts
- Tutorials
- Laboratories / Demonstrations
- Textbooks / Handbooks
- Internet

Learning Activities: The Tutor must ensure that the learning activities for the trainee include the following:

- Identifying sub-atomic particles
  - Electrons
  - Protons
  - Neutrons
  - Atomic number and atomic mass number
  - Isotopes
- Using concept of bonding
  - Ironic bond
  - Covalent bond
  - Metallic bond
  - Vander Waals' bonding
- Describing crystallography
  - Crystal types
  - Crystal structures
  - Crystallographic planes
- Describing amorphous structures
  - Differences with crystalline materials

Assessment Criteria:

- Identify sub-atomic particles correctly
- Use concept of bonding correctly
- Describe crystallography correctly
- Describe amorphous structures correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Tests
- Examinations

Recommended Readings/Texts

- ASHBY M F & JONES D R H (1996),*Engineering Materials 1: An Introduction to Their Properties and Applications, 2nd Ed., Oxford: Butterworth-Heinemann.*
- JOHN V B, (1994), *Engineering Materials, London: Macmillan*
- ASHBY M F & JONES D R H (1994). Engineering Materials 2,*An Introduction to Microstructures, Processing and Design, Oxford: Pergamon.*
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*
- FELBECK D K. Introduction to Strengthening Mechanisms, (Latest Ed.), Prentice-Hall.
- ROLLASON E C. Metallurgy for Engineers, (Latest Ed.), London: Arnold

MODULE :412-10-B MATERIAL SCIENCE

UNIT B10.3 EXPLAINING THE CRYSTAL STRUCTURE OF MATERIALS

Nominal Duration: 12 Hours **Credits:1.2**

General Learning Outcome: Explain the crystal structure of materials

Learning Outcomes: On completion of this unit, the trainee will be able to:

10.3.1 Identify Crystal classes
10.3.2 Explain Miller Notation
10.3.3 Compute measurement in crystals
10.4.4 Discuss defects in crystal lattices
Explain diffusion

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Identify Common Crystal Systems
  - Unit cell
  - Simple cubic
  - Face-centred cubic (fcc)
  - Body-centred cubic (bcc)
  - Hexagonal close packed (hcp, chp)
  - Lattice parameters
  - Crystallographic planes
- Explain the Miller Notation
  - Indices of planes

- Indices of direction
- Compute measurement in crystals
  - Unit cell volume
  - Density
  - Distances between crystallographic planes
  - Angles between crystallographic planes
- Discuss defects in crystal lattices
  - Concept of deformation
  - Slip
  - Twinning
  - Point defects
  - Line defects
  - Other defects
- Explain diffusion
  - Formation of solid solutions
  - Types of solid solutions
  - Changes in solid solutions

Assessment Criteria:

- Identify Common Crystal Systems correctly
- Explain the Miller Notation correctly
- Compute measurement in crystals accurately
- Discuss defects in crystal lattices correctly
- Explain diffusion correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Tests
- Examinations

Recommended Readings/Texts

- ASHBY M F & JONES D R H (1996). *Engineering Materials 1: An Introduction to Their Properties and Applications, 2nd Ed., Oxford: Butterworth- Heinemann.*
- JOHNVB,(1994),*Engineering Materials, London: Macmillan*
- ASHBY M F & JONES D R H (1994). *Engineering Materials 2: An Introduction to Microstructures, Processing and Design, Oxford: Pergamon.*
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*

- FELBECK D K. Introduction to Strengthening Mechanisms, (Latest Ed.), *Prentice-Hall.*
- *ROLLASONEC.Metallurgyfor Engineers, (Latest Ed.), London: Arnold*

MODULE :412-10-B MATERIAL SCIENCE

UNIT B10.4 EXPLAINING THE BOND THEORY OF SOLIDS NOMINAL

Nominal Duration: 12 Hours **Credits:1.2**

General Learning Outcome: Explain the bond theory of solids

Learning Outcomes: On completion of this unit, the trainee will be able to:

10.4.1 Discuss Types of bonds
10.4.2 Describe Metallic conductors
10.4.3 Use Insulators
10.4.4 Use Semi-conductors
10.4.5 Employ Dielectrics

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Discuss Types of bonds
  - Ironic bond
  - Covalent bond
  - Metallic bond
  - Vander Waals' bonding
- Describe Metallic conductors
  - Electrical conductivity
- Use Insulators
  - Behaviour of insulators
- Use Semi-conductors

- Metallic bond
- Covalent bond
- N-type semi-conductor
- P-type semi-conductor
- Employ Dielectrics
- Structure and behaviour
- Dielectric constant

Assessment Criteria:

- Discuss types of bonds correctly
- Describe metallic conductors correctly
- Use Insulators convincingly
- Use Semi-conductors properly
- Employ Dielectrics correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Tests
- Examinations

Recommended Readings/Texts

- ASHBY M F & JONES D R H (1996),*Engineering Materials 1: An Introduction to Their Properties and Applications, 2nd Ed., Oxford: Butterworth-Heinemann.*
- JOHN V B, (1994),*Engineering Materials, London: Macmillan*
- ASHBY M F & JONES D R H (1994). *Engineering Materials 2: An Introduction to Microstructures, Processing and Design, Oxford: Pergamon.*
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*
- FELBECK D K. *Introduction to Strengthening Mechanisms, (Latest Ed.), Prentice-Hall.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*

MODULE :412-10-B MATERIAL SCIENCE

UNIT B10.5 EXPLAINING PHASE DIAGRAMS

Nominal Duration: 12 Hours **Credits:1.2**

General Learning Outcome: Explain phase diagrams

Learning Outcomes: On completion of this unit, the trainee will be able to:

10.5.1 Discuss Introduction to phase diagram
10.5.2 Explain Alloy formation
10.5.3 Draw the Iron-carbon phase diagram
10.5.4 Explain Heat treatment of materials

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities:

The trainer must ensure that the learning activities for the trainee include the following:

- Discuss Introduction to phase diagram
  - Types of equilibrium diagrams–solid solubility, complete and partial solubility
- Explain Alloy formation
  - Eutectics
  - Intermetallic compounds
- Draw the Iron-carbon phase diagram
  - Steels
  - Other alloys and alloying elements
- Explain Heat treatment of materials

- Annealing
- Normalising
- Hardening
- Tempering
- Case hardening

Assessment Criteria:

- Discuss Introduction to phase diagram correctly
- Explain Alloy formation correctly
- Draw Iron-carbon phase diagram properly
- Explain Heat treatment of materials correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments

Recommended Textbooks

- ASHBY M F & JONES D R H (1996),*Engineering Materials 1: An Introduction to Their Properties and Applications, 2nd Ed., Oxford: Butterworth-Heinemann.*
- JOHN V B, (1994), *Engineering Materials, London: Macmillan*
- ASHBY M F & JONES D R H (1994),*Engineering Materials 2: An Introduction to Microstructures, Processing and Design, Oxford: Pergamon.*
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*
- FELBECK D K. *Introduction to Strengthening Mechanisms, (Latest Ed.), Prentice-Hall.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*

MODULE :412-10-B MATERIAL SCIENCE

UNIT B10.6 USING MECHANICAL PROPERTIES OF MATERIALS

Nominal Duration: 10 Hours **Credits:1**

General Learning Objective: Use mechanical properties of materials

Learning Outcomes: On completion of this unit, the trainee will be able to:

10.6.1 Discuss Strength
10.6.2 Compute Hardness
10.6.3 Explain Toughness
10.6.4 Discuss Fatigue failure
10.6.5 Discuss Creep failure
10.6.6 Conduct material tests

Conditions:The trainee must have access to:

- Text books and materials handbooks
- Materials laboratories
- Catalogues, tutorials and demonstrations
- Library
- Calculators
- Internet (Worldwide Web)

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Discuss Strength
  - Tensile strength
  - Compressive strength
  - Stress-strain curve
  - Elasticity
  - Plasticity
- Compute Hardness
  - Brinnel hardness
  - Vickers hardness
  - Rockwell hardness
  - Polymeric materials
- Explain Toughness
  - Impact resistance
- Discuss Fatigue failure
  - Alternating loads

- Fluctuating loads
- Pulsating loads
  - Discuss Creep
  - Temperature effects
  - Conduct material tests
- Tensile tests
- Hardness tests
- Torsional tests
- Bending tests
- Fatigue tests
- Creep tests

Assessment Criteria:

- Discuss Strength correctly
- Compute Hardness correctly
- Explain Toughness correctly
- Discuss Fatigue correctly
- Discuss Creep correctly
- Conduct Material tests correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments

Recommended Textbooks

- ASHBY M F & JONES D R H (1996),*Engineering Materials 1: An Introduction to Their Properties and Applications, 2nd Ed., Oxford: Butterworth-Heinemann.*
- JOHN V B, (1994), *Engineering Materials, London: Macmillan*
- ASHBY M F & JONES D R H (1994),*Engineering Materials 2: An Introduction to Microstructures, Processing and Design, Oxford: Pergamon.*
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*
- FELBECK D K. *Introduction to Strengthening Mechanisms, (Latest Ed.), Prentice-Hall.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*

- JASTRZEBSKI, Z D (1987). *The Nature and Properties of Engineering Materials, 3rd Ed, New York: John Wiley and Sons.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*
- TIMINGS, R L (1998). *Engineering Materials, Volume 1, Harlow: Addison Wesley Longman Limited.*

MODULE :412-10-B MATERIAL SCIENCE

UNIT B10.7 USING STEEL AND NON-FERROUS METALS

Nominal Duration: 10 Hours **Credits:1**

General Learning Objective: Relate Heat Treatment ofSteel

Learning Outcomes: On completion of this unit, the trainee will be able to:

10.7.1 Plain carbon steels
10.7.2 Heat Treatment of steel
10.7.3 Non-Ferrous Metals and Alloys
10.7.4 Jointing Processes

Conditions:The trainee must have access to:

- Text books and materials handbooks
- Materials laboratories
- Catalogues, tutorials and demonstrations
- Library
- Calculators
- Internet (Worldwide Web)

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Describe plain carbon steels
  - Effect of carbon on properties of steel
  - Low, medium and high carbon steels
- Explain Heat Treatment
  - Recrystallisation
  - Cold working and Hot working
  - Annealing
  - Normalising
  - Hardening
  - Tempering
  - Quench hardening, Case hardening
  - Carburising, Induction hardening, Nitriding
- State Non-Ferrous Metals and Alloys
  - Copper-copper alloys, copper-nickel alloys, nickel- nickel alloys
  - Aluminium, magnesium and their alloys
  - Titanium and its alloys
- Explain Jointing processes

- Welding processes
- Soldering
- Brazing
- Bolting

Assessment Criteria:

- Describe plain carbon steels correctly
- Explain Heat Treatment correctly
- State Non-Ferrous Metals and Alloys correctly
- Explain Jointing processes correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments

Recommended Textbooks

- ASHBY M F & JONES D R H (1996),*Engineering Materials 1: An Introduction to Their Properties and Applications, 2nd Ed., Oxford: Butterworth-Heinemann.*
- JOHN V B, (1994), *Engineering Materials, London: Macmillan*
- ASHBY M F & JONES D R H (1994),*Engineering Materials 2: An Introduction to Microstructures, Processing and Design, Oxford: Pergamon.*
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*
- FELBECK D K. *Introduction to Strengthening Mechanisms, (Latest Ed.), Prentice-Hall.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*
- JASTRZEBSKI, Z D (1987). *The Nature and Properties of Engineering Materials, 3rd Ed, New York: John Wiley and Sons.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*
- TIMINGS, R L (1998). *Engineering Materials, Volume 1, Harlow: Addison Wesley Longman Limited.*

MODULE :412-10-B MATERIAL SCIENCE

UNIT B10.8 EMPLOYING POLYMERS

Nominal Duration: 10 Hours Credits:1

General Learning Objective: Employ polymers

Learning Outcomes: On completion of this unit, the trainee will be able to:

10.8.1 Classify types of polymers
10.8.2 Carryout Polymeric reactions
10.8.3 Identify Elastomers
10.8.4 Deformation and fracture of polymers

Conditions:The trainee must have access to:

- Text books and materials handbooks
- Materials laboratories
- Catalogues, tutorials and demonstrations
- Library
- Calculators
- Internet (Worldwide Web)

Learning Activities:The trainer must ensure that the learning activities for the trainee include the following:

- Describe Classification of polymers
  - Types of polymers
  - Crystallinity in polymers
  - Properties of polymers
  - Applications of polymers
- Discuss Polymeric reactions
  - Plasticizers
  - Stabilisers
  - Colorants
- Describe Elastomers
  - Types of elastomers
  - Properties of elastomers
  - Applications of elastomers
- Discuss Deformation and fracture of polymers
  - Temperature effects
  - Fatigue failure

Assessment Criteria:

- Describe Classification of polymers correctly
- Discuss Polymeric reactions correctly
- Describe Elastomers correctly
- Discuss Deformation and Fracture of polymers correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments

Recommended Textbooks

- ASHBY M F & JONES D R H (1996),*Engineering Materials 1: An Introduction to Their Properties and Applications, 2nd Ed., Oxford: Butterworth-Heinemann.*
- JOHN V B, (1994), *Engineering Materials, London: Macmillan*
- ASHBY M F & JONES D R H (1994),*Engineering Materials 2: An Introduction to Microstructures, Processing and Design, Oxford: Pergamon.*
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*
- FELBECK D K. *Introduction to Strengthening Mechanisms, (Latest Ed.), Prentice-Hall.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*
- JASTRZEBSKI, Z D (1987). *The Nature and Properties of Engineering Materials, 3rd Ed, New York: John Wiley and Sons.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*
- TIMINGS, R L (1998). *Engineering Materials, Volume 1, Harlow: Addison Wesley Longman Limited.*

MODULE :412-10-B MATERIAL SCIENCE

UNIT B10.9 EMPLOYING CERAMICS GLASSES AND COMPOSITES

Nominal Duration: 10 Hours **Credits:1**

General Learning Objective: Employ ceramics glasses and composites

Learning Outcomes: On completion of this unit, the trainee will be able to:

10.9.1 Classify types of ceramics
10.9.2 Explain Structure of glasses
10.9.3 Identify Types of composite materials
10.9.4 Explain Dispersion-hardened materials

Conditions:The trainee must have access to:

- Text books and materials handbooks
- Materials laboratories
- Catalogues, tutorials and demonstrations
- Library
- Calculators
- Internet (Worldwide Web)

Learning Activities:The trainer must ensure that the learning activities for the trainee include the following:

- Discuss Classification of ceramics
  - Types of ceramics
  - Structure of Ceramics
  - Properties of ceramics
  - Applications of ceramics
- Discuss glasses
  - Types of glasses
  - Transition temperature
  - Properties of glasses
  - Applications of glasses
- Explain types of composite materials
  - Types of composites including concrete and mortar
  - Composite matrices
  - Reinforcement materials
  - Additives and pigmentation
  - Properties of composites
  - Applications of composites

- Discuss Dispersion-hardened materials
  - Cermets
  - Whiskers
  - Properties of dispersion hardened materials
  - Applications of dispersion hardened materials

Assessment Criteria:

- Discuss Classification of ceramics correctly
- Discuss glasses correctly
- Explain Types of composite materials correctly
- Discuss Dispersion-hardened materials correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments

Recommended Textbooks

- ASHBY M F & JONES D R H (1996),*Engineering Materials 1: An Introduction to Their Properties and Applications, 2nd Ed., Oxford: Butterworth-Heinemann.*
- JOHN V B, (1994), *Engineering Materials, London: Macmillan*
- ASHBY M F & JONES D R H (1994),*Engineering Materials 2: An Introduction to Microstructures, Processing and Design, Oxford: Pergamon.*
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*
- FELBECK D K. *Introduction to Strengthening Mechanisms, (Latest Ed.), Prentice-Hall.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*
- JASTRZEBSKI, Z D (1987). *The Nature and Properties of Engineering Materials, 3rd Ed, New York: John Wiley and Sons.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*
- TIMINGS, R L (1998). *Engineering Materials, Volume 1, Harlow: Addison Wesley Longman Limited.*

MODULE :412-10-B MATERIAL SCIENCE

UNIT B10.10 EXPLAINING ENVIRONMENTAL EFFECTS

Nominal Duration: 10 Hours **Credits:1**

General Learning Objective: Explain environmental effects

Learning Outcomes: On completion of this unit, the trainee will be able to:

10.10.1 Explain Corrosion of metals
10.10.2 Describe Weathering of polymers
10.10.3 Describe Weathering of elastomers
10.10.4 Discuss Pollution

Conditions:The trainee must have access to:

- Text books and materials handbooks
- Materials laboratories
- Catalogues, tutorials and demonstrations
- Library
- Calculators
- Internet (Worldwide Web)

Learning Activities:The trainer must ensure that the learning activities for the trainee include the following:

- Explain Corrosion of metals
  - Types of corrosion – atmospheric, galvanic
  - Factors affecting corrosion
  - Materials that resist corrosion
  - Prevention
- Describe Weathering of polymers
  - Ageing and weather effects in polymers
- Describe Weathering of elastomers
  - Weather effects on elastomers
  - Discuss Pollution
  - Materials that pollute the environment

Assessment Criteria:

- Corrosion of metals explained correctly
- Weathering of polymers described correctly
- Weathering of elastomers described properly

- Pollution discussed correctly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments

Recommended Textbooks

- ASHBY M F & JONES D R H (1996),*Engineering Materials 1: An Introduction to Their Properties and Applications, 2nd Ed., Oxford: Butterworth-Heinemann.*
- JOHN V B, (1994), *Engineering Materials, London: Macmillan*
- ASHBY M F & JONES D R H (1994),*Engineering Materials 2: An Introduction to Microstructures, Processing and Design, Oxford: Pergamon.*
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*
- FELBECK D K. *Introduction to Strengthening Mechanisms, (Latest Ed.), Prentice-Hall.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*
- JASTRZEBSKI, Z D (1987). *The Nature and Properties of Engineering Materials, 3rd Ed, New York: John Wiley and Sons.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*
- TIMINGS, R L (1998). *Engineering Materials, Volume 1, Harlow: Addison Wesley Longman Limited.*

MODULE :412-10-B MATERIAL SCIENCE

UNIT B10.11 BLANK FORMING METHOD

Nominal Duration: 10 Hours Credits:1

General Learning Objective: Blank Forming Method

Learning Outcomes: On completion of this unit, the trainee will be able to:

10.11.1 Understand blank forming technology
10.11.2 Interpret casting
10.11.3 Interpret forging
10.11.4 Interpret welding

Conditions:The trainee must have access to:

- Handouts
- Text books and materials handbooks
- Materials laboratories
- Catalogues, tutorials and demonstrations
- Library
- Calculators
- Internet (Worldwide Web)

Learning Activities:The trainer must ensure that the learning activities for the trainee include the following:

- Describe blank forming technology
  - Explain casting
  - Types of casting
  - Application of casting
- Explain forging
  - Types of forging
  - Application of forging
- Explain welding
  - Types of welding
  - Application of welding

Assessment Criteria:

- Blank forming technology described correctly
- Types and application of casting described correctly
- Types and application of forging described properly

- Types and application of welding described properly

Assessment Methods: To achieve the above assessment, the trainee will be given:

- Tests
- Examinations
- Assignments

Recommended Textbooks

- Zhou Xiulin and Wu Yuanhui, *Machinery manufacturing foundation, Machinery Industry Press*
- ASHBY M F & JONES D R H (1994),*Engineering Materials 2: An Introduction to Microstructures, Processing and Design, Oxford: Pergamon.*
- PASCOE K J (1982),*An Introduction to the Properties of Engineering Materials, Wokingham: van Nostrand Reinhold*
- FELBECK D K. *Introduction to Strengthening Mechanisms, (Latest Ed.), Prentice-Hall.*
- JASTRZEBSKI, Z D (1987). *The Nature and Properties of Engineering Materials, 3rd Ed, New York: John Wiley and Sons.*
- ROLLASON E C. *Metallurgy for Engineers, (Latest Ed.), London: Arnold*
- TIMINGS, R L (1998). *Engineering Materials, Volume 1, Harlow: Addison Wesley Longman Limited.*

MODULE: 412-11-B MOTOR MAINTENANCE TECHNOLOGY

UNIT B11.1 APPLYING PRINCIPLES OF MOTOR MAINTENANCE

Nominal Duration: 30 Hours **Credits: 3**

General Learning Outcome : APPLY PRINCIPLES OF MOTOR MAINTENANCE

Specific learning outcomes: On completion of this unit the trainee will be able to:

11.1.1 Apply principles of motor maintenance
11.1.2 Analyse DC motor
11.1.3 Analyse three-phase induction motor
11.1.4 Analyse control motor

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Applying basic principles of motor maintenance
  - Electromagnetic induction and electromagnetic laws of logarithms
  - Types and applications of motors
  - Safety knowledge of motor maintenance
- Analysing DC motor
  - Dc motor structure and parameters
  - Working principle of DC motor
  - Mechanical characteristics and application of DC motor
- Analysing three-phase induction motor

    - Structure and classification of three-phase asynchronous motor
    - Working principle of three-phase asynchronous motor and nameplate
    - Mechanical characteristics and application of three-phase asynchronous motor
- Analysing control motor
    - Common types and characteristics of control motors
    - Common applications of control motors

Assessment Criteria:

- principles of motor maintenance applied correctly
- DC motor analysed correctly
- three-phase induction motor analysed correctly
- control motor analysed correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Theory test
- Practice test

Recommended Books:

Xiang Cao，*Starting from scratch to study motor maintenance and control circuits*，

MODULE: 412-11-B MOTOR MAINTENANCE TECHNOLOGY

UNIT B 11.2 DISASSEMBLING AND ASSEMBLING ASYNCHRONOUS MOTORS

Nominal Duration: 20 Hours **Credits: 2**

General Learning Outcome : DISASSEMBLE AND ASSEMBLE ASYNCHRONOUS MOTOR

Specific learning outcomes: On completion of this unit the trainee will be able to:

11.2.1 Carryout disassembly of three-phase asynchronous motor
11.2.2 Analyse faults in three-phase asynchronous motor
11.2.3 Carryout maintenance of three-phase asynchronous motor

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts
- Workshops

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Carrying out disassembly of three-phase asynchronous motor
  - Tools and materials needed for motor maintenance
  - Components of motor
  - Dismounting process method of the motor
- Analysing faults in three-phase asynchronous motor
  - Common fault and reason of motor
  - commonly used detection methods of Motor faults or failures

  - Mechanical troubleshooting of three-phase asynchronous motor
  - Electrical fault maintenance of three-phase asynchronous motor
- Carrying out maintenance of three-phase asynchronous motor
  - maintenance of main parts of motor
  - inspection of motor

Assessment Criteria:

- disassembly of three-phase asynchronous motor carried out correctly
- faults in three-phase asynchronous motor analysed correctly
- maintenance of three-phase asynchronous motor carried out correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Theory test
- Practice test

Recommended Books:

Xiang Cao，*Starting from scratch to study motor maintenance and control circuits*，

MODULE: 412-11-B MOTOR MAINTENANCE TECHNOLOGY

UNIT B 11.3 MAINTAINING MOTOR WINDING

Nominal Duration: 20 Hours **Credits: 2**

General Learning Outcome : MAINTAIN MOTOR WINDING

Specific learning outcome: On completion of this unit the trainee will be able to:

11.3.1 Analyse winding structure for motor
11.3.2 Identify tool and material for motor winding embedded technology
11.3.3 Apply embedded technology of winding of three-phase asynchronous motor

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts
- Workshops

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Analysing winding structure for motor
  - Stator winding related terminology
  - Development diagram of three-phase single-layer stator winding
- Identifying tools and materials for motor embedded technology
  - Usage of special tool for motor inlay line
  - Select special material for motor inlay wire
- Apply embedded technology of winding of three-phase asynchronous motor

- Preparation of motor windings before inlaying wire
- Motor winding embedded line technology
- Motor winding embedded line operation
- Dip and dry the motor windings

Assessment Criteria:

- Winding structure for motor analysed correctly
- Tools and materials for motor embedded technology identified correctly
- Embedded technology of winding of three-phase asynchronous motor applied correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Theory test
- Practice test

Recommended Books:

Xiang Cao，*Starting from scratch to study motor maintenance and control circuits*，

MODULE: 412-11-B MOTOR MAINTENANCE TECHNOLOGY

UNIT B 11.4 COMMISSIONING LOW-VOLTAGE ELECTRICAL APPARATUS

Nominal Duration: 20 Hours Credits: 2

General Learning Outcome : COMMISSION LOW VOLTAGE ELECTRICAL APPARATUS

Specific learning outcome: On completion of this unit the trainee will be able to:

11.4.1 Analyse Low voltage fuse

11.4.2 Analyse Low voltage switches

11.4.3 Classify main orders appliances

11.4.4 Analyse Contactors

11.4.5 Analyse Relays

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Analysing Low voltage fuse
  - Function, symbol and parameters of the fuse
  - Structure and working principle of fuse
  - Identification and selection of fuse
  - Fuse installation and testing
- Analysing Low voltage switches
  - Classification of low voltage switches

  - Function, symbol and parameters of commonly used low voltage switch
  - Structure and working principle of commonly used low voltage switch
  - Identification and selection of commonly used low voltage switches
  - Installation and testing of commonly low voltage switch
- Classify main orders appliances
  - Identification and selection of main order appliances commonly used
  - The functions, symbols and parameters of main order appliances commonly used
  - Structure and working principle of main order electric apparatus commonly used
  - Installation and testing of main order appliances
- Analysing Contactors
  - Contactor classification
  - The functions, symbols and parameters of the contactor
  - Structure and working principle of ac contactor
  - Identification and selection of ac contactor
  - Installation and testing of ac contactor
- Analysing Relays
  - Classification of relays
  - Function, symbol and parameter of relay
  - Structure and working principle of relay
  - Identification and selection of relay
  - Relay installation and detection

Assessment Criteria:

- Low voltage fuse analysed correctly
- Low voltage switches analysed correctly
- Main orders appliances classified correctly
- Contactors analysed correctly
- Relays analysed correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Theory test
- Practice test

Recommended Books:

Xiang Cao，*Starting from scratch to study motor maintenance and control circuits*

MODULE: 412-11-B MOTOR MAINTENANCE TECHNOLOGY

UNIT 11.5 INSTALLING ELECTRICAL CONTROL CIRCUIT FOR THREE-PHASE ASYNCHRONOUS MOTOR

Nominal Duration: 30 Hours **Credits: 3**

General Learning Outcome : INSTALL ELECTRICAL CONTROL CIRCUIT FOR THREE-PHASE ASYNCHRONOUS MOTOR

Specific learning outcome: On completion of this unit the trainee will be able to:

11.5.1 Install motor control line

11.5.2 Test positive rotation control circuit of three-phase asynchronous motor

11.5.3 Test positive and negative control circuit of three-phase induction motor

11.5.4 Test sequence control circuit of three-phase asynchronous motor

11.5.5 Design Y—△ voltage control circuit of three-phase asynchronous motor

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Installing motor control line
  - The method of drawing and reading circuit diagrams
  - Principles for drawing and reading layout and wiring diagrams
  - Tools and materials for installing and testing circuits

  - Process requirements for making and installing circuits
- Testing positive rotation control circuit of three-phase asynchronous motor
  - The structure, working principle, installation and testing of the positive-turn control circuit
  - Construction, working principle, installation and testing of continuous forward control circuit
- Testing positive and negative control circuit of three-phase induction motor
  - The structure, working principle, installation and testing of contactor interlock positive and negative control circuit
  - Installation and detection of contactor and button double interlock positive and negative control circuit
- Testing sequence control circuit of three-phase asynchronous motor
  - Installation and testing of control circuits for sequential starting of two motors
  - Installation and detection of two controls forward turn control circuits
  - Control circuit of voltage reducing start of three-phase asynchronous motor
  - Installation and detection of voltage dropping control line of autotransformer
- Designing Y—△ voltage control circuit of three-phase asynchronous motor
  - The installation and detection of Y—△ voltage reducing starting control line
  - Testing of Y—△ voltage reducing starting control line

Assessment Criteria:

- Motor control line Installed correctly
- Positive rotation control circuit of three-phase asynchronous motor tested correctly
- Positive and negative control circuit of three-phase induction motor tested correctly
- Sequence control circuit of three-phase asynchronous motor tested correctly
- Y—△ voltage control circuit of three-phase asynchronous motor designed correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Theory test
- Practice test

Recommended Books:

Xiang Cao，*Starting from scratch to study motor maintenance and control circuits*

MODULE: 412-11-B MOTOR MAINTENANCE TECHNOLOGY

UNIT B11.6 MAINTAINING ELECTRICAL CONTROL CIRCUITS FOR MACHINE TOOLS

Nominal Duration: 20 Hours **Credits: 2**

General Learning Outcome : MAINTAIN ELECTRICAL CONTROL CIRCUITS FOR MACHINE TOOLS

Specific learning outcome: On completion of this unit the trainee will be able to:

11.6.1 Analyse electrical control circuit design basis

11.6.2 Maintain electrical control circuit of the Lathe Machine

11.6.3 Maintain Radial Drilling Machine electrical control line

11.6.4 Maintain Universal Miller electrical control line

Conditions: The trainee must have access to:

- Library
- Laboratory
- Lectures
- Demonstration
- Handouts

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Analyse Electrical control circuit design basis
  - Basic principles for designing electrical control circuits
  - Design the steps of electrical control circuit
- Maintain electrical control circuit of the Lathe Machine
  - The main structure of the lathe Machine
  - Analysis of electrical control circuit of the lathe Machine
  - common electrical fault and maintenance of the Lathe Machine
- Maintain Radial Drilling Machine electrical control line

  - The main structure of the Radial Drilling Machine
  - Analysis of electrical control circuit of Radial Drilling Machine
  - Radial Drilling Machine common electrical failures and maintenance
- Maintain Universal Miller electrical control line
  - The main structure of Universal Milling machine
  - Analysis of electrical control circuit of Universal Milling Machine
  - Universal Milling Machine common electrical failures and maintenance

Assessment Criteria:

- Electrical control circuit design basis analysed correctly
- Electrical control circuit of the Lathe Machine maintained correctly
- Radial Drilling Machine electrical control line maintained correctly
- Universal Miller electrical control line maintained correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Test

Recommended Books:

Xiang Cao，*Starting from scratch to study motor maintenance and control circuits*

MODULE: 412-12-B MAINTENANCE OF HYDRAULIC AND PNEUMATIC SYSTEM

UNIT B12.1 DESCRIBING DESIGN AND CONSTRUCTION OF HYDRAULIC SYSTEM OF GRINDING MACHINE TABLE

Nominal Duration: 25 Hours **Credits: 2.5**

General Learning Outcome ：Describe design and construction of hydraulic system of grinding machine table

Specific Learning Outcome：On completion of this unit the trainee will be able to：

12.1.1 Describe hydraulic grinder system

12.1.2 Examine hydraulic oil

12.1.3 Explain theory of hydraulic transmission

12.1.4 Apply principles of the gear pump

12.1.5 Describing hydraulic cylinders

12.1.6 Describing construction of hydraulic system of grinding table

12.1.7 Use hydraulic system FluidSIM-H simulation

Conditions：The trainee must have access to：

- Education resources:
  - Industrial application video for hydraulic technology
  - Hydraulic Components Gallery
  - Hydraulic circuit FLASH animation
  - FluidSIM-H simulation software
  - Ref://www.yeyanet.com/" and other industry websites
- Equipment and tools:
  - Equipment: hydraulic training device, hydraulic pump station, hydraulic cylinder, manual reversing valve, relief valve, flow valve components, computer, etc.
  - Tools: connecting tubing, pressure relief tools, etc.

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious
- Problem solving

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Describing hydraulic grinder system
  - Hydraulic and pneumatic drive overview
  - Grinding table worktable motion control requirements
  - Working principle of hydraulic transmission
- Examining hydraulic oil
  - Main properties of hydraulic oil
  - Types and characteristics of hydraulic oil
  - Selection of commonly used hydraulic oil
- Explaining theory of hydraulic transmission
  - Hydrostatic property
  - Fluid dynamics
- Applying principles of the gear pump
  - Hydraulic pump working principle and performance parameters
  - Gear pump characteristics, working principle and application
- Describing hydraulic cylinders
  - Hydraulic cylinder type and characteristics
  - Three connection methods for hydraulic cylinders
  - Hydraulic cylinder structure and composition
- Using hydraulic system FluidSIM-H simulation
  - FluidSIM-H simulation software use
  - Grinding table workbench hydraulic system FluidSIM-H simulation
    - Components of hydraulic system
  - Piston pump working principle and characteristics
  - Hydraulic motor working principle and characteristics
  - Design and construction of hoist hydraulic system
  - Working principle and application of pressure reducing valve

- Describing construction of hydraulic system of grinding table
  - Construction of hydraulic pumping station
  - Hydraulic control valve selection
  - Grinding table hydraulic system construction
  - Operation and commissioning of the hydraulic system of the grinding table
    - Principles of control in grinder table hydraulic system
  - Directional valve cognition

- Manual reversing valve working process
- Median function of the reversing valve
- The role and classification of pressure valves
- Working principle and application of overflow valve
- The role and classification of flow control valves
- Working principle and application of flow control valve

▪ Schematic design of grinding table work platform

- Directional control loop design
- Pressure control loop design
- Speed control loop design
- Workbench hydraulic system schematic design

Assessment Criteria:

- Hydraulic grinder system described correctly
- Hydraulic oil examined correctly
- Theory of hydraulic transmission explained correctly
- Principles of the gear pump applied correctly
- Hydraulic cylinders described correctly
- Principles of control in grinder table hydraulic system applied correctly
- Schematic design of grinding table work platform described correctly
- Hydraulic system fluidSIM-H simulation used correctly
- Construction of hydraulic system of grinding table described correctly
- Components of hydraulic system identified correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Assignments
- Project process assessment
- CDIO project report
- Test
- Project summary report requirements
- Phased test

Recommended Books：

Zhang Qin, XuGangtao (2015.3),*"Hydraulic and pneumatic transmission technology" (Second Edition) "Twelfth Five-Year" vocational education national planning textbook - Chinese version*,
Jin Yingji, FengHaiming (2013.2),*"Hydraulic and Pneumatic Technology" "Twelfth Five-Year" Vocational Education National Planning Textbook - Chinese Edition*
Industrial Hydraulic Technology Student Operation Manual
Reference books such as "220 Cases of Modern Hydraulic Technology Application"

MODULE: 412-12-BMAINTENANCE OF HYDRAULIC AND PNEUMATIC SYSTEM

UNIT B12.2 DESCRIBING DESIGN AND CONSTRUCTION OF HYDRAULIC SYSTEM FOR COMBINED MACHINE TOOL POWER SLIDE

Nominal Duration: 25 Hours **Credits: 2.5**

General Learning Outcome ：Describe design and construction of hydraulic system for combined machine tool power slide

Specific Learning Outcome: On completion of this unit the trainee will be able to：

12.2.1 Identify Hydraulic Auxiliary Components

12.2.2 Identify Hydraulic Basic Circuit

12.2.3 Identify Servo Valves and Cartridge Valves

12.2.4 Determine the pressure valve and hydraulic pump required for the hydraulic system of the power slide

12.2.5 Design Electric circuit of power slide hydraulic system

12.2.6 Describe construction and operation of hydraulic system of power slide

12.2.7 Analyse the hydraulic system of the combined machine tool power slide

Conditions：The trainee must have access to：

- Education resources:
  - Industrial application video for hydraulic technology
  - Hydraulic Components Gallery
  - Hydraulic circuit FLASH animation
  - FluidSIM-H simulation software
- Equipment and tools:
  - Equipment: hydraulic training device, hydraulic pump station, hydraulic cylinder, electromagnetic reversing valve, relief valve, button switch, relay, proximity switch, computer, etc.
  - Tools: Connecting tubing, connecting cables, pressure relief tools, etc.

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold

- Driven
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Identifying Hydraulic Auxiliary Components
  - filter
  - Accumulator
  - Fuel tank and its accessories
  - Tubing and pipe joint
  - Sealing means
- Identifying Hydraulic Basic Circuit
  - Unloading circuit
  - Pressure maintaining circuit
  - Balance loop
  - Locking circuit
- Identifying Servo Valves and Cartridge Valves
  - Electrohydraulic servo valve
  - Cartridge valve
- Determining the pressure valve and hydraulic pump required for the hydraulic system of the power slide
  - Working principle and application of sequence valve
  - Working principle and application of pressure relay
  - Blade pump working principle and characteristics
- Designing Electric circuit of power slide hydraulic system
  - Common electrical components in hydraulic systems
  - Design of electrical control loop
    - Directional control loop design
    - Pressure control loop design
    - Speed control loop design
    - Hydraulic control circuit design
  - Fluid Slider Hydraulic System FluidSIM-H Simulation
- Describing construction and operation of hydraulic system of power slide
  - System component selection
  - Construction of hydraulic slide hydraulic system
  - Power slide electrical control loop connection
  - Operation and commissioning of the hydraulic system of the power slide

- Analysing the hydraulic system of the combined machine tool power slide
  - Combined machine tool power slide motion control requirements
  - Working principle and application of electromagnetic reversing valve
  - Working principle and application of electro-hydraulic reversing valve
  - Working principle and application of check valve
  - Online test

Assessment Criteria:

- Hydraulic system of the combined machine tool power slide analysed correctly
- Pressure valve and hydraulic pump required for the hydraulic system of the power slide determined correctly
- Electric circuit of power slide hydraulic system designed correctly
- construction and operation of hydraulic system of power slide described correctly
- Hydraulic Auxiliary Components identified correctly
- Hydraulic Basic Circuit identified correctly
- Servo Valves and Cartridge Valves identified correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Test
- Assignments
- Project process assessment
- CDIO project report

Recommended Books：

Zhang Qin, XuGangtao (2015.3),"Hydraulic and pneumatic transmission technology" (Second Edition) "Twelfth Five-Year" vocational education national planning textbook - Chinese version,

Jin Yingji, FengHaiming (2013.2),"Hydraulic and Pneumatic Technology" "Twelfth Five-Year" Vocational Education National Planning Textbook - Chinese Edition

Yang, H.Y., 2014. Recent fluid power research activities in China. 9th International Fluid Power Conference, Aa-chen, Germany.

Liu, R., Alleyne, A., 2000. Nonlinear force/pressure tracking of an electro-hydraulic actuator. Journal of Dynamic Systems, Measurement, and Control, 122 (1):232-236

MODULE: 412-12-B MAINTENANCE OF HYDRAULIC AND PNEUMATIC SYSTEM

UNIT B12.3 DESCRIBING DESIGN AND CONSTRUCTION OF PNEUMATIC TOOL CHANGE SYSTEM IN MACHINING CENTER

Nominal Duration: 25 Hours **Credits: 2.5**

General Learning Outcome：Describe design and construction of pneumatic tool change system in machining centre

Specific Learning Outcome：On completion of this unit the trainee will be able to：

12.3.1 Describe pneumatic tool change system of CNC machining centre

12.3.2 Identify pneumatic components required for a pneumatic tool change system

12.3.3 Analyse Pneumatic control loop

12.3.4 Describe Pneumatic control circuit design of pneumatic tool change system

12.3.5 Describe Construction and operation of pneumatic tool change system in CNC machining centre

12.3.6 Identify Special Type Cylinders and Vacuum Components

Conditions：The trainee must have access to：

- Education resources:
  - "Hydraulic and pneumatic technology" textbook
  - Industrial application video of pneumatic technology
  - FESTO pneumatic training guide
  - Pneumatic Transmission Student Operation Manual
  - Pneumatic Components Gallery
  - Pneumatic back FLASH animation
  - FluidSIM-P simulation software
  - "http://www.yeyanet.com/" and other industry websites
- Equipment and tools:
  - Equipment: pneumatic training device, air compressor (mute), cylinder, pneumatic reversing valve, one-way throttle valve pneumatic two-piece, computer, etc.
  - Tools: Connecting the air pipe, etc.

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Describing pneumatic tool change system of CNC machining centre
  - Pneumatic System Cognition - Bus Door Pneumatic Control System
  - CNC machining centre tool change system control requirements
  - How pneumatic systems work
  - Pneumatic component installation and system connection
- Identifying pneumatic components required for a pneumatic tool change system
  - Air compressor
  - Use of pneumatic triplex
  - Types and applications of ordinary cylinders
  - Electromagnetic reversing valve (pneumatic valve)
  - Directional control valve
  - Silencer
- Analysing Pneumatic control loop
  - Pneumatic reversing circuit
  - Pneumatic speed control loop
- Describing Pneumatic control circuit design of pneumatic tool change system
  - Design of spindle positioning control loop
  - Design of tool clamping cylinder control loop
  - Tool changer control circuit design
  - Pneumatic circuit design of tool change system in machining centre
  - Design and circuit simulation of electrical control loop for pneumatic tool change system
  - Design of electrical control loop for tool change system in machining centre
  - FluidSIM-P simulation of pneumatic tool change system in machining centre

- Pneumatic Common Circuit
  - Hand operated safety circuit
  - Single reciprocating circuit
  - Online test

- Describing Construction and operation of pneumatic tool change system in CNC machining centre
  - System component selection
  - Construction of pneumatic system of tool change system in CNC machining centre
  - CNC machining centre tool changer control circuit connection
  - Operation and commissioning of tool change system in CNC machining centre
- Identifying Special Type Cylinders and Vacuum Components
  - Special type cylinder
  - Vacuum element

Assessment Criteria:

- Pneumatic tool change system of CNC machining centre described correctly
- Pneumatic components required for a pneumatic tool change system identified correctly
- Pneumatic control loop analysed correctly
- Pneumatic control circuit design of pneumatic tool change system described correctly
- Construction and operation of pneumatic tool change system in CNC machining centre described correctly
- Special Type Cylinders and Vacuum Components
- Pneumatic Common Circuit described correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- test
- Assignments
- Project process assessment
- CDIO project report

Recommended Books：

Zhang Qin, XuGangtao (2015.3),"Hydraulic and pneumatic transmission technology" (Second Edition) "Twelfth Five-Year" vocational education national planning textbook - Chinese version,

Jin Yingji, FengHaiming (2013.2),"Hydraulic and Pneumatic Technology" "Twelfth Five-Year" Vocational Education National Planning Textbook - Chinese Edition

MODULE: 412-12-B MAINTENANCE OF HYDRAULIC AND PNEUMATIC SYSTEM

UNIT B12.4 DESCRIBING ELECTRICAL CONTROL TECHNOLOGY FOR HYDRAULIC AND PNEUMATIC SYSTEMS

Nominal Duration: 25 Hours Credits: 2.5

General Learning Outcome ：DESCRIBE ELECTRICAL CONTROL TECHNOLOGY FOR HYDRAULIC AND PNEUMATIC SYSTEMS

Specific Learning Outcome：On completion of this unit the trainee will be able to：

12.4.1 Describing design and construction of single electric control basic hydraulic circuit
12.4.2 Describing design and construction of double electronically controlled hydraulic circuit
12.4.3 Describing design and construction of single electronically controlled complex hydraulic circuit

Conditions：The trainee must have access to：

- Education resources:
  - Industrial application video for hydraulic technology
  - Hydraulic Components Gallery
  - Hydraulic circuit FLASH animation
  - "http://www.yeyanet.com/" and other industry websites
- Site conditions:
  - The hydraulic control technology training room for students to complete the training tasks, which can accommodate more than 40 people, can be used for hydraulic system design, construction, measurement and drawing.
- Equipment conditions:
  - Bosch Rexroth DS4 electro-hydraulic proportional control training device, including industrial hydraulic and electrical control components, hydraulic components using quick connector connection, electrical components using plug-in cable, student industrial computer.
  - Tool conditions:
    Each set of common multimeters, measurement tools, drawing software, etc.

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Describing Design and construction of single electric control basic hydraulic circuit
  - Jog control hydraulic system design and construction
  - Design and construction of single reciprocating motion control hydraulic system
  - Continuous reciprocating motion control hydraulic system design and construction
- Describing Design and construction of double electronically controlled hydraulic circuit
  - Design and construction of electrical interlock hydraulic system
  - Design and construction of hydraulic system for fast-forward-work automatic conversion
- Describing Design and construction of single electronically controlled complex hydraulic circuit
  - Design and construction of hydraulic system for automatic return of pressure relay control
  - Design and construction of hydraulic system with time delay and single reciprocating control
  - Design and construction of feed motion hydraulic system with time delay and intermediate stop

Assessment Criteria:

- Design and construction of single electric control basic hydraulic circuit described correctly
- Design and construction of double electronically controlled hydraulic circuit described correctly
- Design and construction of single electronically controlled complex hydraulic circuit described correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Test
- Assignment
- Project process assessment
- Project Report

Recommended Books：

Rexroth Electro-Hydraulic Technology Student Handbook

"Safety Procedures for Hydraulic Work Posts"

General technical conditions for hydraulic systems

"General rules and safety requirements for hydraulic transmission systems and their components

Hydraulic and pneumatic technology" textbook

Industrial Hydraulic Technology Student Operation Manual

Electrical-hydraulic student operation manual

Reference books such as "220 Cases of Modern Hydraulic Technology Application"

MODULE: 412-12-B MAINTENANCE OF HYDRAULIC AND PNEUMATIC SYSTEM

UNIT B12.5 MAINTAINING HYDRAULIC AND PNEUMATIC SYSTEM

Nominal Duration: 20 Hours Credits: 2

General Learning Outcome：MAINTAIN HYDRAULIC AND PNEUMATIC SYSTEM MAINTENANCE

Specific Learning Outcome：On completion of this unit the trainee will be able to：

12.5.1 Troubleshooting common faults in hydraulic pumps
12.5.2 Troubleshooting common faults in the relief valve
12.5.3 Troubleshooting common faults of pressure reducing valves
12.5.4 Troubleshooting flow control valve common faults
12.5.5 Troubleshooting common faults in directional control valves
12.5.6 Troubleshooting common faults in hydraulic cylinders and hydraulic motors
12.5.7 Solve the phenomenon of temperature rise
12.5.8 Troubleshooting pressure abnormality
12.5.9 Troubleshooting abnormal traffic
12.5.10 Solve the vibration and noise phenomenon
12.5.11 Solve the problem of hydraulic oil pollution

Conditions：The trainee must have access to：

- Library
- Laboratories
- Handouts
- Tutorials
- Tools

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Troubleshooting common faults in hydraulic pumps
  - Hydraulic pump oil-free discharge phenomenon and treatment method
  - Hydraulic pump noise phenomenon and treatment method
  - Insufficient oil pumping quantity and treatment method
  - Seal oil leakage phenomenon and treatment method
- Troubleshooting common faults in the relief valve
  - Overflow valve pressure is too high (or too low) phenomenon and treatment method
  - Pressure relief valve pressure phenomenon and treatment method
  - The pressure of the relief valve has a slight vibration phenomenon and the treatment method
- Troubleshooting common faults of pressure reducing valves
  - Pressure reducing valve pressure is too high (or too low) phenomenon and treatment
  - Pressure reducing valve pressure indefinite phenomenon and treatment method
- Troubleshooting flow control valve common faults
  - Throttle failure phenomenon and treatment method
  - Movement speed instability phenomenon and treatment method
- Troubleshooting common faults in directional control valves
  - Manual operation of valve stem, oil seal oil leakage phenomenon and treatment method
  - Mechanical operation spool cannot move phenomenon and treatment method
  - Solenoid valve coil burning phenomenon and treatment method
  - Hydraulic control valve does not move phenomenon and treatment method
- Troubleshooting common faults in hydraulic cylinders and hydraulic motors
  - Hydraulic cylinder crawling phenomenon and treatment method
  - Hydraulic cylinder impact phenomenon and treatment method
  - Insufficient thrust or gradual decline in working speed or even stop phenomenon and treatment
- Solve the phenomenon of temperature rise
  - High oil temperature phenomenon and treatment method

  - Overflow valve overheating phenomenon and treatment method
  - Hydraulic pump, motor, hydraulic cylinder overheating phenomenon and treatment method
- Troubleshooting pressure abnormality
  - No pressure phenomenon and treatment method
  - Pressure instability phenomenon and treatment method
  - Low pressure phenomenon and treatment method
  - Excessive pressure phenomenon and treatment method
- Troubleshooting abnormal traffic
  - No flow phenomenon and treatment method
  - Flow rate is too small and processing method
  - Excessive flow and processing methods
- Solve the vibration and noise phenomenon
  - Pump, motor vibration phenomenon and treatment method
  - Hydraulic cylinder vibration phenomenon and treatment method
  - Pipeline vibration phenomenon and treatment method
  - Fuel tank resonance phenomenon and treatment method
  - Oil flow noise phenomenon and treatment method
  - Valve switching noise phenomenon and processing method
- Solve the problem of hydraulic oil pollution
  - Air source failure
    - Air compressor failure
    - Pressure reducing valve failure
    - Pipeline failure
    - Triple piece failure
  - Pneumatic actuator failure
    - Internal and external leakage failure of the cylinder
    - Insufficient output force of the cylinder and unstable operation
    - Bad cylinder buffer effect
    - Cylinder piston rod and cylinder head damage failure
  - Pneumatic reversing valve failure
    - The reversing valve cannot be reversed or the reversing action is slow.
    - Gas leak failure
    - Electromagnetic pilot valve failure
  - Pneumatic auxiliary component failure
    - Oil mister failure
    - Automatic drainer failure

- Muffler failure

Assessment Criteria:

- Troubleshooting common faults in hydraulic pumps
- Troubleshooting common faults in the relief valve
- Troubleshooting common faults of pressure reducing valves
- Troubleshooting flow control valve common faults
- Troubleshooting common faults in directional control valves
- Troubleshooting common faults in hydraulic cylinders and hydraulic
- Solve the phenomenon of temperature rise
- Troubleshooting pressure abnormality
- Troubleshooting abnormal traffic
- Solve the vibration and noise phenomenon
- Solve the problem of hydraulic oil pollution

Assessment Methods: To achieve the above assessment the trainee will be given

- Tests
- Exercise
- Assignments
- Practice test
- Project

Recommended Books：

Shi Weichao, Wang Lijie (2016),*"Hydraulic and Pneumatic Systems and Maintenance" (Electrical and Mechanical Equipment Maintenance and Management, Higher Vocational Education, National Machinery Industry, Vocational Education, Quality Planning Textbook) - Chinese version*

Cui Xuehong (2015.1),*Chinese version of Hydraulic and Pneumatic Systems and Maintenance*

MODULE: 412-13-B ENGINEERING MATHEMATICS II

Nominal Duration: 20 Hours Credits: 2

UNIT B13.1 APPLYING BOOLEAN ALGEBRA

General Learning Outcomes: Solve problems involving logic gates

Specific Learning Outcome: On completion of this unit the trainee will be able to:

13.1.1 Apply the operation of logic gates
13.1.2 Manipulate Boolean expressions
13.1.3 Design logic gates using Karnaugh maps

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials

Values/Attitudes:

- Sober minded
- Hard working
- Alert
- Committed
- Honest
- Smarter/Bolder/Driven
- Time conscious
- Problem solving

Learning Activities:

The trainer must ensure that the learning activities for the trainee include the following:

- **Applying the operation of logic gates**
  - Logic signal levels
  - Logic gates
  - Truth table
- **Manipulating Boolean algebra**
  - Laws of logic
  - DeMorganlogicBoole
  - Simplification of Boolean algebra
- **Designing logic gates**
  - Venn diagrams
  - Karnaugh maps (2 and 3 variable inputs)
  - Minimisation of functions using Karnaugh maps

Assessment Criteria:

- The operation of logic gates applied correctly
- Boolean algebra manipulated correctly
- Logic gates designed correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises

- Assignments
- Tests

Recommended Books:

Morris N. M. (1994): *Mastering Mathematics*
Electrical & Electronic Engineering – Macmillan
Stroud K. A. ((2004) 4thEdition *Engineering Mathematics* – Macmillan

MODULE: 412-13-B ENGINEERING MATHEMATICS II

UNIT B13.2 SOLVING PROBLEMS USING BINARY NUMBERS

Nominal Duration: 20 Hours Credits: 2

General Learning Outcomes: Solve problems using binary numbers

Specific Learning Outcome: On completion of this unit the trainee will be able to:

13.2.1 Convert from one system to another
13.2.2 Add and subtract binary numbers
13.2.3 Multiply and divide binary numbers

Conditions: The trainee must have access to:

- Library
- Handouts
- Tutorials

Values/Attitudes:

- Sober minded
- Hard working
- Alert
- Committed
- Honest
- Smarter/Bolder/Driven
- Time conscious
- Problem solving

Learning Activities:

The trainer must ensure that the learning activities for the trainee include the following:

- **Converting from one system to another**
  - Terminology (binary, octal, decimal, hexadecimal)
  - Converting a decimal number to another base
  - Converting from any base to decimal
  - Converting a binary number to octal
  - Converting a binary number to hexadecimal
- **Adding and subtracting binary numbers**
  - Addition of binary numbers
  - Ordinary subtraction of binary numbers
  - Subtraction of binary numbers by complementation
- **Multiplying and dividing binary numbers**

Assessment Criteria:

- Converting from one system to another done correctly
- Adding and subtracting binary numbers done correctly
- Multiplying and dividing binary numbers done correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises

- Assignments
- Tests

Recommended Books:

Morris N. M. (1994): *Mastering Mathematics for Electrical & Electronic Engineering* – Macmillan
Stroud K. A. ((2004) 4th Edition *Engineering Mathematics*, – Macmillan
Bird and May (latest edition) *Technician mathematics 4 and 5.* Longman

MODULE: 412-13-B ENGINEERING MATHEMATICS II
UNIT B13.3 APPLYING STATISTICS TO SOLVE PROBLEMS

Nominal Duration: 30 Hours Credits: 3

General Learning Outcomes: Apply statistics to solve problems

Specific Learning Outcome: On completion of this unit the trainee will be able to:
13.3.1 Identify types of data
13.3.2 Present data in various forms
13.3.3 Determine measures of central tendency
13.3.4 Determine measures of dispersion

Conditions: The trainee must have access to:
- Library
- Handouts
- Tutorials
- Manuals

Values/Attitudes:

- Sober minded
- Hard working
- Alert
- Committed
- Honest
- Smarter/Bolder/Driven
- Time conscious
- Problem solving

Learning Activities:
The trainer must ensure that the learning activities for the trainee include the following:
- **Identifying types of data**
  - Discrete data
  - Continuous data
- **Presenting data**
  - Arrays
  - Tables
  - Frequency distribution
  - Chart
  - Bar charts
  - Pictograms
- **Determining measures of central tendency**
  - Arithmetic mean
  - Median
  - Mode
- **Determining measures of dispersion**
  - Range
  - Interquartile range
  - Mean absolute deviation
  - Standard deviation

- Coefficient of variation
- Graphs
- Line graphs
- Histograms
- Frequency polygons
- Cumulative frequency curves (ogives)

Assessment Criteria:

- Types of data correctly identified correctly
- Data presented correctly
- Measures of central tendency determined correctly
- Measures of dispersion determined correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Books:

Straud A. K. 4th *Edition, Engineering Mathematics*

Bird and May (1991) 22nd Edition: *Technician Mathematics*

MODULE: 412-13-B ENGINEERING MATHEMATICS II

UNIT B13.4 APPLYING PROBABILITY TO SOLVE PROBLEMS

Nominal Duration: 25 Hours Credits: 2.5

General Learning Outcomes: Apply complex numbers

Specific Learning Outcome: On completion of this unit the trainee will be able to:

13.4.1 Calculate simple probability
13.4.2 Apply probability distribution
13.4.3 Apply significance testing

Conditions: The trainee must have access to:

- Text books and materials handbooks
- Materials laboratories
- Catalogues, tutorials and demonstrations
- Library
- Calculators
- Internet (Worldwide Web)

Values/Attitudes:

- Sober minded
- Hard working
- Alert
- Committed
- Honest
- Smarter/Bolder/Driven
- Time conscious
- Problem solving

Learning Activities:

The trainer must ensure that the learning activities for the trainee include the following:

- **Calculating simple probability**
  - Probability terminologies
  - Simple probability
  - Independent events
  - Dependent events
  - Mutually exclusive events
  - Additional Laws
  - Multiplication laws
  - Probability tree diagrams
- **Applying probability distribution**
  - Binomial probability Distribution
  - Poisson probability Distribution
  - Normal probability Distribution
- **Applying significance testing**
  - Distribution sample mean
  - Confidence limits
  - Statistical decisions based on samples

- Hypothesis
- Null hypothesis

- **Alternative hypothesis**
  - Type I and type II errors
  - Level of significance
  - One and two-tail tests
  - Tests for difference between the two means

Assessment Criteria:

- Simple probability calculated correctly
- Probability distribution applied correctly
- Significance testing applied correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- Tests

Recommended Books:

Stroud ,K.A (2001) 5th edition *Engineering Mathematics Macmillan*
Greer A. (latest edition) *Statistics for Engineers* ELBS/Stanley Thornes

MODULE: 412-13-B ENGINEERING MATHEMATICS II

UNIT B13.5 APPLYING REGRESSION AND CORRELATION TO ANALYSE DATA

Nominal Duration: 25 Hours Credits: 2.5

General Learning Outcomes: Apply regression and correlation to analyse data

Learning Outcome: On completion of this unit the trainee will be able to:

13.5.1 Plot scatter diagrams
13.5.2 Apply regression analysis
13.5.3 Calculate correlation coefficient of a distribution

Conditions: The trainee must have access to:

- Text books and materials handbooks
- Materials laboratories
- Catalogues, tutorials and demonstrations
- Library
- Calculators
- Internet (Worldwide Web)

Values/Attitudes:

- Sober minded
- Hard working
- Alert
- Committed
- Honest
- Smarter/Bolder/Driven
- Time conscious
- Problem solving

Learning Activities:

The trainer must ensure that the learning activities for the trainee include the following:

- **Plotting scatter diagrams**
  - Independent Variables
  - Dependent variables
- **Applying regression analysis**
  - Methods of least squares
  - Regression line
  - Regression coefficients (Y- intercept and gradient)
  - Interpolation and extrapolation
- **Calculating the correlation coefficient of a distribution**
  - Types of correlation
  - Positive, negative, Zero
  - Linear correlation
  - Correlation coefficient
  - Spearmanion coefficient

- Rand correlation coefficient
- Significance of correlation coefficient

Assessment Criteria:

- Plot scatter diagrams done correctly
- Regression analysis applied correctly
- Correlation coefficient of a distribution calculated correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Tests
- Examinations
- Assignments

Recommended Books:

Stroud ,K.A (2001) 5th edition *Engineering Mathematics* Macmillan

Greer A. (latest edition) *Statistics for Engineers* ELBS/Stanley Thornes

MODULE: 412-14-B MECHANICAL MANUFACTURING TECHNOLOGY

UNIT B 14.1 EXPLAINING MECHANICAL PROPERTIES AND HEAT TREATMENT OF METAL MATERIALS

Nominal Duration: 10 Hours **Credits: 1**

General Learning Outcome：EXPLAIN MECHANICAL PROPERTIES AND HEAT TREATMENT CONCEPTS OF METAL MATERIALS

Learning Outcome：On completion of this unit the trainee will be able to：

14.1.1 Explain mechanical properties

14.1.2 Describe heat treatment of metals

14.1.3 Apply mechanical properties to heat treatment of metals

Conditions：The trainee must have access to：

- Library
- Handout
- Course

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- **Explaining mechanical properties of metal materials**
  - Strength
  - Plastic
  - Hardness
  - Fatigue strength
- **Carrying out Heat treatment of metal materials**
  - Tempering
  - Annealing

- Normalizing
- Quenching
- **Applying mechanical properties to heat treatment of metals**

Assessment Criteria :

- Mechanical properties of metal materials explained correctly
- Heat treatment carried out correctly
- Mechanical properties in heat treatment of metals applied correctly

Assessment Methods : To achieve the above assessment the trainee will be given :

- Exercises
- Task
- Test

Recommended Books :

- QiaoShimin, et al. (2003),*Machinery manufacturing foundation (latest edition) - "12th Five-Year" Vocational Education National Planning textbook Chinese version*

MODULE: 412-14-B MECHANICAL MANUFACTURING TECHNOLOGY

UNIT B 14.2 SELECTING MATERIALS

Nominal Duration: 20 Hours **Credits: 2.0**

General Learning Outcome：SELECT MATERIALS

Learning Outcome：On completion of this unit the trainee will be able to：

14.2.1 Classify common alloy steel

14.2.2 Identify common nonferrous metals

14.2.3 Select non-metal materials

Conditions：The trainee must have access to：

- Library
- Handout
- Course

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- **Classifying common alloy steel**
  - Number of alloy steel
  - Alloy structural steel
  - Alloy tool steel
  - Special properties steel
- **Identifying Nonferrous Metals**
  - Aluminium and aluminium alloy
  - Copper and copper alloy
  - Titanium and titanium alloy
  - Sliding bearing alloy
- **Selecting non-metal materials**

- Powder metallurgy materials
- Ceramic materials
- Plastic, rubber material
- Composite material

Assessment Criteria：

- Common alloy steels classified correctly
- Nonferrous identified correctly
- Non-metallic materials selected correctly

Assessment Methods：To achieve the above assessment the trainee will be given：

- Exercises
- Task
- Test

Recommended Books：

- QiaoShimin, et al. (2003),*Machinery manufacturing foundation"(latest edition) - "12th Five-Year" Vocational Education National Planning textbook Chinese version*

MODULE: 412-14-B MECHANICAL MANUFACTURING TECHNOLOGY

UNIT B 14.3 APPLYING MANUFACTURING METHODS OF BLANKING

Nominal Duration: 10 Hours **Credits: 1**

General Learning Outcome ： APPLY MANUFACTURING METHODS OF BLANKING

Learning Outcome：On completion of this unit the trainee will be able to：

14.3.1 Apply methods of casting of blank

14.3.2 Apply methods of forging

14.3.3 Carryout of sheet metal stamping process

Conditions：The trainee must have access to：

- Library
- Handout
- Course

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- **Applying methods of casting of blank**
  - Sand mould casting
  - Special casting
  - Technology of casting structure
  - The requirements for castability to the structure of castings
  - The requirement of casting process to the structure of castings
- **Applying methods of forging**
  - The characteristics of forging production
  - The scope of use of forged production
  - The foundation of forging process
  - The plastic deformation of metal

- The Forgability of metal
- Free forging
- Free forging equipment
- Free forging process
- The classification and forging process of free forging
- Die forging
- Hammer forging
- Die forging
- Die forging on press
- **Carrying out of sheet metal stamping process**
  - Working procedure of sheet metal stamping

Assessment Criteria:

- Apply methods of casting of blank applied correctly
- Apply methods of forging applied correctly
- Carryout of sheet metal stamping process carried out correctly

Assessment Method: To achieve the above assessment the trainee will be given

- Exercises
- Task
- Test

Recommended Books：

- QiaoShimin, et al. (2003),*Machinery manufacturing foundation" (latest edition) - "12th Five-Year" Vocational Education National Planning textbook Chinese version*

MODULE: 412-14-B MECHANICAL MANUFACTURING TECHNOLOGY

UNIT B 14.4 CARRYING OUT METAL CUTTING IN MANUFACTURING

Nominal Duration: 20 Hours **Credits: 2**

General Learning Outcome : CARRYOUT METAL CUTTING IN MANUFACTURING

Specific learning outcomes：On completion of this unit the trainee will be able to：

14.4.1 Carryout quality processing
14.4.2 Set cutting movement
14.4.3 Use geometric angle
14.4.4 Carry out metal cutting
14.4.5 Assess tool wear and tool life
14.4.6 Assess machinability of work piece
14.4.7 Assess metal cutting conditions

Conditions：The trainee must have access to：

- Library
- Handout
- Course

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- **Carryout quality processing**
  - Machining precision
  - Surface quality
- **Setting cutting movement**
  - Cutting motion
  - The surface of the workpiece
  - Cutting amount
- **Using tool geometric angle**

- The composition of the lathe tool
- Tool geometric angle reference system
- Tool standard angle
- Tool working angle
- The parameters of the cutting layer

▪ **Tool material**

- The performance of the tool material
- High speed steel
- Hard alloy

- **Carrying out metal cutting**
  - The formation process of chip

▪ Metal cutting Procedure

▪ Cutting force

- The source, force and force of the cutting force
- The main factors affecting the cutting force

▪ Cutting heat and cutting temperature

- The generation and transmission of the cutting heat
- The distribution of the cutting temperature
- The main factors affecting the cutting temperature

- **Assessing tool wear and tool life**
  - The wear form
  - Wear process of the tool
  - Causes of tool wear
  - Tool life
- **Assessing machinability of workpiece**
  - Relative machinability
  - Classification of workpiece materials
- **Assessing metal cutting conditions**
  - Selection of geometric parameters of cutting tools
  - The choice of tool life
  - The choice of cutting amount
  - The choice of cutting fluid

Assessment Criteria:

- Quality processing carried out correctly
- Cutting movement set correctly
- Geometric angle used correctly
- Metal cutting carried out correctly
- Tool wear and tool life assessed correctly
- Machinability of workpiece assessed correctly
- Metal cutting conditions assessed correctly

Assessment Method: To achieve the above assessment the trainee will be given：

- Exercises
- Task
- Test

Recommended Books：

- QiaoShimin, et al. (2003),*Machinery manufacturing foundation (latest edition) - "12th Five-Year" Vocational Education National Planning textbook Chinese version*

MODULE: 412-14-B MECHANICAL MANUFACTURING TECHNOLOGY

UNIT B 14.5 CARRYING OUT SURFACE MACHINING OF MECHANICAL PARTS

Nominal Duration: 20 Hours **Credits: 2**

General Learning Outcome ： CARRYOUT SURFACE MACHINING OF MECHANICAL PARTS

Specific learning outcomes：On completion of this unit the trainee will be able to：

14.5.1 Apply concepts of metal cutting

14.5.2 Select appropriate gear tooth

14.5.3 Carryout machining of outer surface of mechanical parts

14.5.4 Carryout machining of inner circle surface of mechanical parts

14.5.5 Carryout plane processing for mechanical parts

Conditions：The trainee must have access to：

- Library
- Handout
- Course

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- **Applying concepts of metal cutting**
  - Classification of machine tools
  - The method of making the model of metal cutting machine
  - Movement of machine tools
  - **The basic composition of machine tool transmission**

- **Selecting gear tooth**
  - The machining method of cylindrical gear tooth shape
  - Hobbing
  - Inserting tooth
  - Other machining methods of tooth shape
- **Carrying out machining of outer surface of mechanical parts**
  - **The processing method of the outer surface**
  - Turning on the surface of the outer circle
  - **Grinding the surface of the outer circle**
- **Carrying out machining of inner circle surface of mechanical parts**
  - The machining method of the inner circle surface
  - Drilling
  - Boring
  - Grinding process
- **Carrying out plane processing for mechanical parts**
  - Plane processing method
  - Planing
  - Milling
  - Plane grinding

Assessment Criteria:

- Basic concepts of metal cutting applied correctly
- Machining of outer surface of mechanical parts carried out correctly
- Machining of inner circle surface of mechanical parts carried out correctly
- Plane processing for mechanical parts carried out correctly
- Gear tooth selected correctly

Assessment Method: To achieve the above assessment the trainee will be given：

- Exercises
- Task
- Test

Recommended Books：

- QiaoShimin, et al. (2003),*Machinery manufacturing foundation"(latest edition) - "12th Five-Year" Vocational Education National Planning textbook Chinese version*

MODULE: 412-14-B MECHANICAL MANUFACTURING TECHNOLOGY

UNIT B 14.6 USING MANUFACTURING TECHNOLOGY

Nominal Duration: 20 Hours **Credits: 2.0**

General Learning Outcome：USE MANUFACTURING TECHNOLOGY

Specific learning outcomes：On completion of this unit the trainee will be able to：

14.6.1 Apply special processing methods

14.6.2 Use CNC Machining Technology

14.6.3 Use advanced manufacturing techniques

Conditions：The trainee must have access to：

- Library
- Handout
- Course

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- **Applying special processing methods**
  - EDM
  - ECM
  - Laser processing
  - Ion beam processing
- **Using NC Machining Technology**
  - The concept of digital control and CNC machine tools
  - The difference between CNC machining and non-numerical control machining
  - The composition and classification of CNC machine tools
  - The application range of CNC machine tools

- **Using advanced manufacturing techniques Other advanced manufacturing technologies**
  - Rapid prototyping technology
  - Industrial robot technology
  - Green manufacturing technology

Assessment Criteria:

- Special processing methods applied correctly
- CNC machining technology used correctly
- Advanced manufacturing techniques used correctly

Assessment Method: To achieve the above assessment the trainee will be given：

- Exercises
- Task
- Test

Recommended Books：

- QiaoShimin, et al. (2003),Machinery manufacturing foundation(latest edition) - "12th Five-Year" Vocational Education National Planning textbook Chinese version

MODULE: 412-15-B MECHANICAL AND ELECTRICAL EQUIPMENT MAINTENANCE TECHNOLOGY I

UNIT B15.1 PLANNING MECHANICAL AND ELECTRICAL EQUIPMENT MAINTENANCE PROCESS

Nominal Duration: 60 Hours **Credits: 6**

General Learning Outcome : PLAN MECHANICAL AND ELECTRICAL EQUIPMENT MAINTENANCE PROCESS

Specific learning outcome: On completion of this unit the trainee will be able to:

15.1.1 Plan for repair of mechanical and electrical equipment
15.1.2 Identify types of faults
15.1.3 Identify types of failure and their causes
15.1.4 Outline the general process for parts disassembly and cleaning
15.1.5 Select parts for repair process
15.1.6 Test equipment

Conditions: The trainee must have access to:

- Library
- Training room
- Handout
- Course
- Tool
- Consumable

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Planning for repair of mechanical and electrical equipment
    - Common technical documents before equipment repair
    - Use of technical documents before repair
    - Implementation of technical preparation
    - Matters needing attention in parts surveying and mapping
- Identifying types of faults

    - Part fault type
    - Common faults of power system
    - Common troubles in lubrication system
    - Common faults of transmission system
- Identifying types of failure and their causes
    - Basic concepts of mechanical failure
    - Common failures of mechanical fasteners
    - Failure modes of mechanical parts
    - Failure causes of mechanical parts
        - Deformation failure
        - Fracture failure
        - Wear failure
        - Failure analysis

- Outlining the general process for parts disassembly and cleaning
    - Disassembly of mechanical parts
    - Cleaning of mechanical parts
    - Disassembly of gear pair
    - Removal of locating parts on axle
    - Disassembly of threaded connection
    - Disassembly of interference parts
    - Inspection of parts
    - Preparation before assembly
    - Key points of assembly process
    - Adjustment after assembly of parts
    - Some technological problems during assembly
- Selecting parts for repair process
    - Selection of parts repair process
    - Classification of repair processes
    - Considerations for process selection
    - Establishment of repair plan
- Testing equipment
    - Trial equipment preparation
    - Equipment inspection
    - Equipment maintenance data management
    - Acceptance procedures and contents
    - Fill in the acceptance card.

Assessment criteria:

- Repair of mechanical and electrical equipment planned correctly
- Types of faults identified correctly
- Types of failure and their causes identified correctly
- General process for parts disassembly and cleaning outlined correctly
- Parts for repair process selected correctly
- Equipment tested correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Exercises
- Task
- Theoretical test
- Practice test

Reference books:

- XieJinzhu and Wang Wanyou,*Fault Diagnosis and Maintenance of Electromechanical Equipment*

MODULE: 412-15-B MECHANICAL AND ELECTRICAL EQUIPMENT MAINTENANCE TECHNOLOGY I

UNIT B 15.2 DIAGNOSING FAULTS OF A2-2 GENERAL PARTS

Nominal Duration: 60 Hours **Credits: 6**

General Learning Outcome : DIAGNOSE FAULTS OF A2-2 GENERAL PARTS

Specific Learning outcome: On completion of this unit the trainee will be able to:

15.2.1 Apply methods for repair of mechanical parts
15.2.2 Use lubricating materials
15.2.3 Diagnose faults on gear transmission
15.2.4 Carryout repairs on gear transmission
15.2.5 Diagnose fault on rolling bearings
15.2.6 Carryout repairs on rolling bearings
15.2.7 Diagnose fault on shaft parts
15.2.8 Carryout repairs on shaft
15.2.9 Diagnose fault on sliding bearings
15.2.10 Carryout repairs on sliding bearings
15.2.11 Diagnose fault on Rotating parts and counterweight balancing operation

Conditions: The trainee must have access to:

- Library
- training room
- Handout
- Consumables
- Workshop
- Hydraulic Fluids

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Applying methods for repair of mechanical parts
  - Welding repair of parts
  - Electroplating of parts
  - Spray repair of parts

    - Bonding repair of parts
- Using lubricating materials
    - Types and functions of lubricants
    - Quality index of lubricating oil
    - Selection appropriate lubricants
    - Configuration of special oil number
- Diagnosing faults on gear transmission
    - Functional analysis
    - Type and application of gear transmission
    - Failure modes and preventive measures of gear transmission
    - Fault diagnosis of gear transmission
    - Factors affecting the accuracy of gear operation
- Carrying out repairs on gear transmission
    - gear tooth surface wear
    - gear teeth
    - wheel rim and hub of gear
    - Gear materials
- Diagnosing fault on rolling bearings
    - Functional analysis
    - Types and applications of rolling bearings
    - Matching selection of rolling bearings
- Carrying out repairs on rolling bearings
    - Dismantling
    - Cleaning
    - Assembly of rolling bearings
    - Adjustment of rolling bearing assembly clearance

- Diagnosing fault on shaft parts
    - Material selection of shaft
    - New shaft configuration
    - Disassembly method for shaft parts
    - Cleaning and inspection after shaft dismantling
- Carrying out repairs on shaft
    - shaft failure
    - Shaft assembly
    - Wear roundness of measuring shaft
    - Measuring the parallelism, perpendicularity and coaxiality of the shaft
    - Workpiece magnetization method

- Diagnosing fault on sliding bearings

    - Types of sliding bearings
    - Determination and adjustment of radial clearance of sliding bearing
    - Determination and adjustment of axial clearance of sliding bearing
- Carrying out repairs on sliding bearings
    - Failure repair of sliding bearing
    - Assembly of sliding bearings
    - Measurement of clearance of sliding bearing
- Diagnosing fault on Rotating parts and counterweight balancing operation
    - Unbalanced parts of rotary parts
    - Static balance test principle
    - Static balance checking device
    - Obviously unbalanced inspection and counterweight operation method

Assessment criteria:

- Methods for repair of mechanical parts applied correctly
- Lubricating materials used correctly
- Faults on gear transmission diagnosed correctly
- Repairs on gear transmission carried out correctly
- Fault on rolling bearings diagnosed correctly
- Repairs on rolling bearings carried out correctly
- Fault on shaft parts diagnosed correctly
- Repairs on shaft carried out correctly
- Fault on sliding bearings diagnosed correctly
- Repairs on sliding bearings carried out correctly
- Fault on rotating parts and counterweight balancing operation diagnosed correctly

Assessment methods: To achieve the above assessment the trainee will be given

- Exercises
- Task
- Theoretical test
- Practice test

Reference books:

- XieJinzhu and Wang Wanyou,*Fault Diagnosis and Maintenance of Electromechanical Equipment*

MODULE: 412-16-C MECHANICAL AND ELECTRICAL EQUIPMENT MAINTENANCE TECHNOLOGY II

UNIT C 16.1 MAINTAINING HYDRAULIC TRANSMISSION EQUIPMENT

Nominal Duration: 80 Hours **Credits: 8**

General Learning Outcome: MAINTAIN HYDRAULIC TRANSMISSION EQUIPMENT

Specific Learning outcome: On completion of this unit the trainee will be able to:

16.1.1 Identify common faults in hydraulic system
16.1.2 Rectify faults in hydraulic pump
16.1.3 Rectify faults in hydraulic cylinder
16.1.4 Rectify faults in hydraulic control valves

Conditions: The trainee must have access to:

- Library
- Training room
- Handout
- Course
- Tool
- Consumables

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Identifying common faults in hydraulic system
    - Summary of hydraulic transmission system
    - General steps of fault diagnosis for hydraulic system
    - Method of locating fault
    - Common faults and characteristics of hydraulic system
- Rectifying faults in hydraulic pump
    - Fault analysis of hydraulic pump
    - Common fault diagnosis and repair of gear pump
    - Fault diagnosis and repair of quantitative vane pump

- Rectifying faults in hydraulic cylinder
    - Dismantling and assembling of hydraulic cylinders
    - Common failures and troubleshooting methods of hydraulic cylinders
    - Cause and harm of leakage of hydraulic cylinder
- Rectifying faults in hydraulic control valves
    - Fault diagnosis and repair of directional control valve
    - Fault diagnosis and repair of pressure control valve
    - Removal of throttle valve
    - Fault diagnosis and repair of throttle valve
    - Fault diagnosis and repair of hydraulic auxiliary parts
    - Matters needing attention in use and maintenance of hydraulic system

Assessment criteria:

- Common faults in hydraulic system identified correctly
- Faults in hydraulic pump rectified correctly
- Faults in hydraulic cylinder rectified correctly
- Faults in hydraulic control valves rectified correctly

Assessment methods:

- Exercises
- Task
- Theoretical test
- Practice test

Reference books:

- XieJinzhu and Wang Wanyou,*Fault Diagnosis and Maintenance of Electromechanical Equipment*

MODULE: 412-16-C MECHANICAL AND ELECTRICAL EQUIPMENT MAINTENANCE TECHNOLOGY II

UNIT C 16.2 RECTIFYING FAULT IN ELECTRICAL EQUIPMENT

Nominal Duration: 80 Hours Credits: 8

General Learning Outcome : RECTIFY FAULTS IN ELECTRICAL EQUIPMENT

Specific Learning outcome: On completion of this unit the trainee will be able to:

16.2.1 Rectify faults in electrical equipment
16.2.2 Rectify faults in transformers
16.2.3 Maintain motors in electrical equipment

Conditions: The trainee must have access to:

- Library
- Training room
- Handout
- Course
- Tool
- Consumables

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Rectifying faults in electrical equipment
  - Causes of electrical failure
  - Maintenance of electrical equipment
  - Repair of electrical equipment
- Rectifying faults in transformers
  - Basic working principle of transformer
  - Structure of power transformer
  - Power transformer fault diagnosis
    - Inspection in transformer operation
    - Abnormal phenomena in transformer operation
- Maintaining motors in electrical equipment

- Analysis of the working principle of three direction asynchronous motor
- Fault diagnosis and repair of motor position control part
- Disassembly and assembly of motors
- Rectification of faults in motor

Assessment criteria:

- Faults in electrical equipment rectified correctly
- Faults in transformers rectified correctly
- Motors maintained correctly

Assessment methods:

- Exercises
- Task
- Theoretical test
- Practice test

Reference books:

- XieJinzhu and Wang, *Fault Diagnosis and Maintenance of Electromechanical Equipment*

MODULE: 412-16-C MECHANICAL AND ELECTRICAL EQUIPMENT MAINTENANCE TECHNOLOGY II

UNIT C 16.3 RECTIFYING FAULT IN C20-1 LARGE EQUIPMENT

Nominal Duration: 80 Hours **Credits: 8**

General Learning Outcome : RECTIFY FAULT IN C20-1 LARGE EQUIPMENT

Specific Learning outcome: On completion of this unit the trainee will be able to:

16.3.1 Rectify fault in axial fan
16.3.2 Rectify fault in air compressor
16.3.3 Rectify fault in drainage pump
16.3.4 Maintain C20-1 Large equipment

Conditions: The trainee must have access to:

- Library
- Training room
- Handout
- Course
- Tool
- Consumables

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- **Rectifying fault in axial fan**
  - Types of ventilators
  - Composition of fans
  - Fan form
  - Disassembly and maintenance of axial fan
- **Rectifying fault in air compressor**
  - Common faults in the operation of air compressors
  - Troubleshooting of crankshaft
  - Failure repair of connecting rod

- Troubleshooting of crosshead
- Cylinder failure repair
- Troubleshooting of air valve

- **Rectifying fault in drainage pump**
  - Dismantling of centrifugal drainage pump
  - Troubleshooting of centrifugal pumps
  - Measurement of multistage centrifugal pump parameters
- **Maintaining C20-1 Large equipment**
  - Dismantling multistage centrifugal pumps
  - Troubleshooting multistage centrifugal pumps
  - Maintenance of thrust balance device for multistage centrifugal pumps

Assessment criteria:

- Fault in axial fan rectified correctly
- Fault in air compressor rectified correctly
- Fault in drainage pump diagnosed correctly
- C20-1 Large equipment maintained correctly

Assessment methods:

- Exercises
- Task
- Theoretical test
- Practice test

Reference books:

- XieJinzhu and Wang Wanyou,*Fault Diagnosis and Maintenance of Electromechanical Equipment*

MODULE: 412-16-C MECHANICAL AND ELECTRICAL EQUIPMENT MAINTENANCE TECHNOLOGY II

UNIT C 16.4 RECTIFYING FAULTS IN C20-2 CNC MACHINE TOOL

Nominal Duration: 60 Hours Credits: 6

General Learning Outcome: RECTIFY FAULTS IN C20-2 CNC MACHINE TOOL

Specific Learning outcome: On completion of this unit the trainee will be able to:

16.4.1 Analyse CNC machine
16.4.2 Carryout routine maintenance of CNC machine tools
16.4.3 Rectify fault of main drive system of CNC machine tool

Conditions: The trainee must have access to:

- Library
- Training room
- Handout
- Course
- Tool
- Consumables

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The trainer must ensure that the learning activities for the trainee include the following:

- Analysing CNC machine
  - Composition of CNC machine tools
  - Working principle of CNC machine tools
- Carrying out routine maintenance of CNC machine tools
  - Fault diagnosis
  - Troubleshooting mechanical devices of CNC machine tools
- Rectifying fault of main drive system of CNC machine tool
  - Fault diagnosis and elimination of feed drive system
  - Fault diagnosis and elimination of tool carrier, tool magazine and tool changer
  - Troubleshooting of CNC system

- Zero return, fault diagnosis and elimination of CNC machine tools
- Common fault diagnosis and elimination in CNC system operation
- Common spindle drive system
- Common feed drive system

Assessment criteria:

- CNC machine analysed correctly
- Routine maintenance of CNC machine tools carried out correctly
- Fault of main drive system of CNC machine tool rectified correctly

Assessment methods:

- Exercises
- Task
- Theoretical test
- Practice test

Reference books:

- Xie Jinzhu and Wang Wanyou, *Fault Diagnosis and Maintenance of Electromechanical Equipment*

MODULE: 412-17-C PRODUCTION LINE ADJUSTMENT AND MAINTENANCE

UNIT C 17.1 MAINTAINING AUTOMATIC SORTING SYSTEM

Nominal Duration: 150 Hours **Credits: 15**

General Learning Outcome：MAINTAINING AUTOMATIC SORTING SYSTEM

Specific learning outcomes：On completion of this unit the trainee will be able to：

17.1.1 Explain the use of tools
17.1.2 Describe order of components
17.1.3 Select sensor installation
17.1.4 Install mechanical and electrical components
17.1.5 Analyse Pneumatic control circuit design
17.1.6 Analyse Electrical control circuit design and connection

Conditions：The trainee must have access to：

- Library
- Training rooms
- Handouts
- Tutorials
- Tool
- Consumables

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- Explaining the use of tools
  - Mechanical tools
  - Surveying tools
  - Mapping tools
  - Electrical instrumentation tools
- Describing order of components
  - Sensor
  - Puncher

    - Air compressor
- Selecting sensor installation
    - Photoelectric sensor
    - Rotary encoder
    - Inductive sensor
    - Magnetic switch
- Installing mechanical and electrical components
    - The electromagnetic directional valve
    - Double-acting cylinder
    - Terminal row
    - Control button box
    - The programmable controller
    - Three-phase ac motor slowdown
    - The inverter
- Analysing Pneumatic control circuit design
    - The host of the pneumatic system requirements
    - Pneumatic circuit design
    - Execute component specification
    - Adjust the control valve specification
    - Auxiliary components and protection components
    - Air compressor specification
- Analysing Electrical control circuit design and connection
    - The PLC power supply wiring diagram
    - The distribution of lateral port signal terminal
    - Sensors, solenoid valve, power terminal device such as a lead to the connection between the lateral connection port
    - The power supply connection
    - The I/O point connection

Assessment Criteria：

- Use of tools explained correctly
- Order of components described correctly
- Sensor installation selected correctly
- Mechanical and electrical components installed correctly
- Pneumatic control circuit design analysed correctly
- Electrical control circuit design and connection analysed correctly

Assessment Methods：To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- The theory test

- The practice test

Recommended Books：

Li-ping du, *Automatic production line installation and debugging*

MODULE: 412-17-C PRODUCTION LINE ADJUSTMENT AND MAINTENANCE

UNIT C 17.2 MAINTAINING ASSEMBLY PRODUCTION LINE

Nominal Duration: 150 Hours **Credits: 15**

General Learning Outcome ： MAINTAIN ASSEMBLY PRODUCTION LINE

Specific learning outcomes：On completion of this unit the trainee will be able to：

17.2.1 Select sensor installation

17.2.2 Select transmission system

17.2.3 Install mechanical and electrical components

17.2.4 Install servo drive systems

17.2.5 Analyse Pneumatic control circuit design and installation

17.2.6 Analyse Electrical control circuit design and the connection

Conditions：The trainee must have access to：

- Library
- Training rooms
- Handouts
- Tutorials
- Tools
- Consumables

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- Selecting sensor installation
  - Photoelectric sensor
  - Rotary encoder
  - Magnetic switch

- Selecting transmission system
  - Torque converter

- Hydraulic controls
- Planetary gear trains

- Installing mechanical and electrical components
  - Single electric solenoid directional control valve
  - Double electric solenoid directional control valve
  - Double-acting cylinder
  - Terminal row
  - Control button box
  - The programmable controller
  - Touch screen
- Installing servo drive systems
  - Amplifier
  - Drive compatibility
  - Electrical isolation
  - Protection

- Analysing pneumatic control circuit design and installation
  - The host of the pneumatic system requirements
  - Pneumatic circuit design
  - Execute component specification
  - Adjust the control valve specification
  - Auxiliary components and protection components
  - Air compressor specification
- Analysing electrical control circuit design and the connection
  - The PLC power supply wiring diagram
  - The distribution of lateral port signal terminal
  - Sensors, solenoid valve, power terminal device such as a lead to the connection between the lateral connection port
  - The power supply connection
  - The I/O point connection

Assessment Criteria:

- Sensor installation selected correctly
- Transmission system selected correctly
- Mechanical and electrical components installed correctly
- Servo drive systems installed correctly

- Pneumatic control circuit design and installation analysed correctly
- Electrical control circuit design and the connection analysed correctly

Assessment Methods：To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- The theory test
- The practice test

Recommended Books：

Li-ping Du, *Automatic production line installation and debugging*

MODULE: 412-18-C WELDING TECHNOLOGY

Unit C 18.1 CARRYING OUT ELECTRODE ARC WELDING

Nominal Duration: 60 Hours **Credits: 6**

General Learning Outcome: CARRYOUT ELECTRODE ARC WELDING

Specific Learning Outcomes: On completion of this unit the trainee will be able to:

18.1.1 Explain electrode arc welding
18.1.2 Carryout material cutting
18.1.3 Carryout electrode arc welding
18.1.4 Use arc welding technology
18.1.5 Carryout vertical welding of electrode
18.1.6 Carryout transverse welding of electrode

Conditions: The trainee must have access to:

- Library
- Handouts
- Course
- Welding machine
- Welding rod
- Steel plate
- Material cutting equipment
- Workshop

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- **Explaining electrode arc welding**
  - Melting of electrode and base metal
  - Welding joints and welds
  - Use of welding machines and tools
- **Carrying out material cutting**
  - Assembly of plasma cutting machine
  - Use of plasma cutting machine

    - Use of electric tools
- **Carrying out electrode arc welding**
    - Arc ignition
    - Flat welding
- **Using welding technology**
    - Fillet welding
    - Flat side butt welding
- **Carrying out vertical welding of electrode**
    - Vertical butt welding, one side welding and two side forming
    - T joint vertical fillet welding
- **Carrying out transverse welding of electrode**
    - Transverse butt welding with one side welding
    - One side welding and double face forming of butt joint

Assessment Criteria:

- Electrode arc welding explained correctly
- Material cutting carried out correctly
- Electrode arc welding carried out correctly
- Electrode arc welding technology used correctly
- Vertical welding of electrode carried out correctly
- Transverse welding of electrode carried out correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Task
- Test

Recommended Readings/Texts

Yong Hou et al. (2018).*Welding rod arc welding*. Machinery Industry Press

Yang, H.Y., 2014. Recent fluid power research activities in China. 9th International Fluid Power Conference, Aa-chen, Germany.

Liu, R., Alleyne, A., 2000. Nonlinear force/pressure tracking of an electro-hydraulic actuator. Journal of Dynamic Systems, Measurement, and Control, 122 (1):232-236

MODULE: 412-18-C WELDING TECHNOLOGY

Unit C 18.2 CARRYING OUT ARGON ARC WELDING

Nominal Duration: 60 Hours **Credits: 6**

General Learning Outcome: CARRYOUT ARGON ARC WELDING

Specific Learning Outcomes: On completion of this unit the trainee will be able to:

18.2.1 Explain argon arc welding
18.2.2 Use argon arc welding technology
18.2.3 Carry out argon Arc welding

Conditions: The trainee must have access to:

- Library
- Handouts
- Course
- Welding machine
- Tungsten pole
- Steel plate
- Material cutting equipment
- Workshop

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- **Explaining argon arc welding**
  - Principle of TIG welding
  - Polarity selection of tungsten electrode and electrode
  - Use of welding machines and tools
- **Carrying out argon welding**
  - Wire feeding
  - Flat welding
- **Using argon arc welding technology**
  - Fillet welding

- Flat side butt welding

Assessment Criteria:

- Argon arc welding explained correctly
- Argon arc welding carried out correctly
- Argon arc welding technology used correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Task
- Test

Recommended Readings/Texts

Wu Yejun et al. (2018).Machinery Industry Press.No. 9787111593799

Yang, H.Y., 2014. Recent fluid power research activities in China. 9th International Fluid Power Conference, Aa-chen, Germany.

Liu, R., Alleyne, A., 2000. Nonlinear force/pressure tracking of an electro-hydraulic actuator. Journal of Dynamic Systems, Measurement, and Control, 122 (1):232-236

MODULE: 412-18-C WELDING TECHNOLOGY

Unit C 18.3 CARRYING OUT CO_2GAS SHIELDED ARC WELDING

Nominal Duration: 60 Hours Credits: 6

General Learning Outcome: CARRYOUT CO_2 GAS SHIELDED ARC WELDING

Specific Learning Outcomes: On completion of this unit the trainee will be able to:

18.3.1 Explain CO_2 gas shielded arc welding
18.3.2 Using welding machines and tools
18.3.3 Carrying out Basic CO_2 gas shielded arc welding

Conditions: The trainee must have access to:

- Library
- Handouts
- Course
- Welding machine
- Steel plate
- Material cutting equipment
- Workshop

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- **Explaining CO_2 gas shielded arc welding**
  - Principle of CO_2 gas shielded arc welding
  - Droplet transfer form and parameter adjustment for CO_2 gas shielded arc welding
- **Using welding machines and tools**
- **Carrying out CO_2 gas shielded arc welding**
  - Equipment installation and adjustment
  - Fillet welding
  - Flat side butt welding

Assessment Criteria:

- CO_2 gas shielded arc welding explained correctly
- Welding machines and tools used correctly
- CO_2 gas shielded arc welding carried out correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Task
- Test

Recommended Readings/Texts

Jiang Zedong et al. (2018).*Melting Pole Gas Shielded Welding.*Machinery Industry Press

Yang, H.Y., 2014. Recent fluid power research activities in China. 9th International Fluid Power Conference, Aa-chen, Germany.

Liu, R., Alleyne, A., 2000. Nonlinear force/pressure tracking of an electro-hydraulic actuator. Journal of Dynamic Systems, Measurement, and Control, 122 (1):232-236

MODULE: 412-18-C WELDING TECHNOLOGY

Unit C 18.4 CARRYING OUT TIG WELDING

Nominal Duration: 60 Hours Credits: 6

General Learning Outcome: CARRYOUT TIG WELDING

Specific Learning Outcomes: On completion of this unit the trainee will be able to:

18.4.1 Explain TIG welding

18.4.2 Use TIG welding technology

18.4.3 Carry out TIG welding

Conditions: The trainee must have access to:

- Library
- Handouts
- Course
- Welding machine
- Tungsten pole
- Steel plate
- Material cutting equipment
- Workshop

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- **Explaining TIG welding**
  - Principle of TIG welding
  - Polarity selection of tungsten electrode
  - Use of welding machines and tools
- **Using TIG welding technology**
  - Fillet welding
  - Flat side butt welding
- **Carrying out TIG welding**

- Wire feeding
- Flat welding

Assessment Criteria:

- TIG welding explained correctly
- TIG welding carried out correctly
- TIG welding technology used correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Task
- Test

Recommended Readings/Texts

Wu Yejun et al. (2018).Machinery Industry Press.No. 9787111593799

Yang, H.Y., 2014. Recent fluid power research activities in China. 9th International Fluid Power Conference, Aa-chen, Germany.
Liu, R., Alleyne, A., 2000. Nonlinear force/pressure tracking of an electro-hydraulic actuator. Journal of Dynamic Systems, Measurement, and Control, 122 (1):232-236

MODULE: 412-18-C WELDING TECHNOLOGY

Unit C 18.5 CARRYING OUT MIG WELDING

Nominal Duration: 60 Hours **Credits: 6**

General Learning Outcome: CARRYOUT MIG WELDING

Specific Learning Outcomes: On completion of this unit the trainee will be able to:

18.5.1 Explain MIG welding

18.5.2 Use MIG welding technology

18.5.3 Carry out MIG welding

Conditions: The trainee must have access to:

- Library
- Handouts
- Course
- Welding machine
- Tungsten pole
- Steel plate
- Material cutting equipment
- Workshop

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- **Explaining MIG welding**
  - Principle of MIG welding
  - Polarity selection of Metal electrode
  - Use of welding machines and tools
- **Carrying out MIG welding**
  - Wire feeding
  - Flat welding

- **Using MIG welding technology**
  - Fillet welding
  - Flat side butt welding

Assessment Criteria:

- MIG welding explained correctly
- MIG welding carried out correctly
- MIG welding technology used correctly

Assessment Methods: To achieve the above assessment the trainee will be given:

- Exercises
- Task
- Test

Recommended Readings/Texts

Wu Yejun et al. (2018).Machinery Industry Press.No. 9787111593799
Yang, H.Y., 2014. Recent fluid power research activities in China. 9th International Fluid Power Conference, Aa-chen, Germany.
Liu, R., Alleyne, A., 2000. Nonlinear force/pressure tracking of an electro-hydraulic actuator. Journal of Dynamic Systems, Measurement, and Control, 122 (1):232-236

MODULE: 412-19-C MANAGEMENT AND ORGANISATION
UNIT C19.1 APPLYING MANAGEMENT CONCEPTS
Nominal Duration: 28 HOURS Credits: 2.8
General Learning Outcome: Apply management concepts
Specific Learning Outcomes: On completion of this unit the trainee will be able to:

19.1.1 Identify the nature of Management
19.1.2 Identify organisations in Management
19.1.3 Identify purpose and types of controls
19.1.4 Identify responsibility of management
19.1.5 Identify types of business organisation charts

Conditions: The trainee must have access to:

- library
- Handouts
- Lecture notes
- Journals
- Newspapers

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities:
The Trainer must ensure that the learning activities for the trainee include the following:

- **Identifying the nature of Management**
  - Concept of Management
  - Theories of Management
  - Functions of management
    - Planning
    - Organizing
    - Controlling
    - Directing

- **Identifying organisations in Management**
  - Organisation and systems
  - Developments in systems theories
  - Organisation structures
- **Identifying purpose and types of controls**
  - Management information systems
  - Personnel
  - Management accounting
  - Control methods
  - Budgets
  - Break-even Analysis
  - Ratios

  - Quality control
  - Techniques for control
  - Network analysis
  - Simulation
  - Inventory controls
- **Identifying responsibility of management**
  - Purchasing
  - Production
  - Research and Development
  - Marketing
  - Administration
  - Finance and Accounting
  - Management services
- **Identifying types of business organisation charts**
  - Functional organisation structure
  - Product based structure
  - Geographically based structure
  - Divisionalised structure
  - Flat organisational structure
  - Tall organisational structure
  - Line and Staff structure

Assessment Criteria

- nature of Management identified correctly
- Organisations in Management identified correctly
- purpose and types of controls identified correctly
- responsibility of management identified correctly
- types of business organisation charts identified correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Tests
- Assignments
- Group work
- Field visits

Recommended Readings/Texts

- Morgan, G (1996) Images and Organisations
- Turnner, P.H.& Turner, D.E (1994) Business Management Studies

MODULE: 412-19-C MANAGEMENT AND ORGANISATION
UNIT C19.2 IDENTIFYING SCIENTIFIC MANAGEMENTT ELEMENTS AND PRINCIPLES OF ECONOMICS
Nominal Duration: 26 HOURS Credits: 2.6
General Learning Outcome: IDENTIFY SCIENTIFIC MANAGEMENTT ELEMENTS AND PRINCIPLES OF ECONOMICS
Specific Learning Outcomes: On completion of this unit the trainee will be able to:

19.2.1 Apply the concepts of scientific management
19.2.2 Examine evolution of management thought
19.2.3 Identify elements of scientific management
19.2.4 Identify the functions of production planning

Conditions: The trainee must have access to:

- library
- Handouts
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- **Applying the concepts of scientific management**
  - Concept of scientific management
  - Characteristics of Scientific management
  - Advantages and disadvantages
- **Examining evolution of management thought**
  - Concept of management thought
  - Characteristics of management thought
- **Identifying elements of scientific management**
  - Elements of Scientific management
  - Advantages and disadvantages
- **Identifying the functions of production planning**
  - Functions of production planning

Assessment Criteria

- concepts of scientific management applied correctly
- evolution of management thought examined correctly
- elements of scientific management identified correctly
- the functions of production planning identified correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Tests
- Assignments
- Practical

Recommended Readings/Texts

- Buchanan, D & Huczyniki, (1997) Organisation Behaviuor: Introductly tests, 3rd edition. London. Practice Hall Inter

MODULE: 412-19-C MANAGEMENT AND ORGANISATION
UNIT C19.3 EMPLOYING VARIOUS TECHNIQUES OF MANAGEMENT
Nominal Duration: 22 HOURS Credits: 2.2
General Learning Outcome: EMPLOY VARIOUS TECHNIQUES OF MANAGEMENT
Specific Learning Outcomes: On completion of this unit the trainee will be able to:

19.3.1 Apply financial management skills
19.3.2 Apply Time management skills
19.3.3 Apply Risk management skills

Conditions: The trainee must have access to:

- library
- Handouts
- Lecture notes
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities:
The Trainer must ensure that the learning activities for the trainee include the following:

- **Applying Financial management skills**
  - Managing Finances
- **Applying Time management skills**
  - Concept of time management
  - Importance of time management
- **Applying Risk management skills**
  - Concept of risk management
  - Identification of risks
  - Management of risks

Assessment Criteria

- Financial management skills applied correctly
- Time management skills applied correctly
- Risk management skills correctly applied correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Tests
- Assignments
- Practical

Recommended Readings/Texts

- Buchanan, D & Huczyniki, (1997) Organisation Behaviuor: Introductly tests, 3rd edition. London. Practice Hall Inter

MODULE: 412-19-C MANAGEMENT AND ORGANISATION

UNIT C19.4 APPLYING MANAGEMENT TECHNIQUES

Nominal Duration: 22 HOURS Credits: 2.2

General Learning Objective: APPLY MANAGEMENT TECHNIQUES

Specific Learning Outcomes: On completion of this unit the trainee should be able to:

19.4.1 Motivate others
19.4.2 Train others
19.4.3 Manage conflicts
19.4.4 Provide good leadership styles
19.4.5. Apply public relations skills
19.4.6 Observe health and Safety procedures

Conditions: The trainee must have access to:

- Library
- Literature
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities:
The Trainer must ensure that the learning activities for the trainee include the following:

- **Motivating others**
  - Motivational theories
- **Training others**
  - Training methods
- **Managing conflicts**
  - Conflict management theories
- **Providing good leadership styles**
  - Types of leadership styles
- **Applying public relations skills**
  - Workshop reception
  - Customer complaints
  - Forecourt services

- **Observing health and Safety procedures**
  - Health and safety
  - Housekeeping
  - Work environment

Assessment Criteria:

- Motivate as required
- Train correctly
- Manage conflicts applied correctly
- Good leadership styles provided as per set standards
- Public relations skills applied correctly
- health and Safety procedures observed correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Class exercise
- Write theory tests
- Assignments
- Practical

Recommended Texts/Books:

- T.Lucey : Costing
- Morgan, G (1996) Images and Organisations,

MODULE: 412-19-C MANAGEMENT AND ORGANISATION

UNIT C19.5 APPLYING LAW

Nominal Duration: 22 HOURS Credits: 2.2

General Learning Outcomes: APPLY LAW

Specific Learning Outcomes: On completion of this unit the trainee should be able to:

19.5.1 Apply the terms and conditions of a valid contract
19.5.2 Apply principles of agreement
19.5.3 Apply vitiating factors
19.5.4 Identify the factors for breach of documents
19.5.5 Use petroleum regulations

Conditions: The trainee must have access to:

- Library
- Literature
- Internet

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities: The Trainer must ensure that the learning activities for the trainee include the following:

- **Applying the terms and conditions of a valid contract**
  - Full definition of a contract:
    - Offer and offeree'
    - Counter offer
    - Acceptance
    - Capacity
    - Consideration
    - Consent
    - Legal formalities
    - Intention to create legal obligation
    - Certainty and possibility of performance

- **Applying principles of agreement**
    - Full definition of conditions of a valid contract
    - Agreement
    - Consideration

- **Applying vitiating factors**
    - Misrepresentation
    - Mistake
    - Undue influence

- **Identifying factors for breach of contracts**
    - Liquidated damages

Assessment Criteria:

- Terms and conditions of a valid contract applied correctly
- Principles of agreement applied correctly
- Vitiating factors applied correctly
- Factors for breach of documents identified correctly

Assessment Methods: To achieve the above assessment the trainee will be given

- Class exercise
- Practical
- Write theory tests
- Assignments

Recommended Texts/Books:

- “Business Law” Painter A and Lawson R. Hienemann
- “English Law” Smith and Keenan, ELBS, Pitmad

MODULE: 412-20-C LOCKSMITH OPERATION
UNIT B 20.1 EXPLAINING LOCKSMITH OPERATION

Nominal Duration: 40 Hours **Credits: 4**

General Learning Outcomes：EXPLAIN LOCKSMITH OPERATION

Specific Learning Outcomes：On completion of this unit the trainee will be able to：

20.1.1 Apply safety in locksmith operation
20.1.2 Use tools and equipment
20.1.3 Analyse fitter process

Conditions：The trainee must have access to：

- Library
- Training rooms
- Handouts
- Tutorials
- Tool
- Workshops
- Laboratories

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- **Applying safety in locksmith operation**
- **Using tools and equipment**
  - Bench table
  - Vice
- **Analysing fitter process**
  - The range of workmanship of locksmith
  - Fitter's application

Assessment Criteria：

- Safety in locksmith operation applied correctly
- Tools and equipment used correctly
- Fitter process analysed correctly

Assessment Methods：To achieve the above assessment the trainee will be given:

- Exercises
- Assignments
- Theory test
- Practice test

Recommended Books：

XuTingguo,*Locksmith operation*

MODULE: 412-20-C LOCKSMITH OPERATION

UNIT C 20.2 USING TOOLS AND EQUIPMENT

Nominal Duration: 40 Hours **Credits: 4**

General Learning Outcomes：USE TOOLS AND EQUIPMENT

Learning Outcome：On completion of this unit the trainee will be able to：

20.2.1 Use Marking tool
20.2.2 Use Crossed benchmark
20.2.3 Use Marking method
20.2.4 Use Practice operation

Conditions：The trainee must have access to：

- Library
- Training rooms
- Handouts
- Tutorials
- Tool
- Consumables
- Workshops

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- Using Marking tool
  - Benchmarking tool
  - Measuring tool
  - Direct line tool
- Using Crossed benchmark
  - The concept of marking datum
  - The selection principle of marking datum
  - Type of underline datum
- Using Marking method
  - Plane crossed
  - Stereo crossed
- Using Practice operation
  - Draw a simple figure on the sheet plane

- D line marking of simple parts

Assessment Criteria:

- Marking tool used correctly
- Crossed benchmark used correctly
- Marking method used correctly
- Practice operation used correctly

Assessment Methods：To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- The practice test

Recommended Books：

XuTingguo, *Locksmith operation*

MODULE: 412-20-C LOCKSMITH OPERATION

UNIT C 20.3 APPLYING CUTTING TECHNIQUES

Nominal Duration: 40 Hours **Credits: 4**

General Learning Outcomes：APPLY CUTTING TECHNIQUES

Learning Outcome：On completion of this unit the trainee will be able to：

20.3.1 Use chisel
20.3.2 Use saws
20.3.3 Use Files
20.3.4 Employ operation method of cutting
20.3.5 Carryout locksmith operation

Conditions：The trainee must have access to：

- Library
- Training rooms
- Handouts
- Tutorials
- Tool
- Consumables
- Workshop

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- **Using chisel**
  - Types and uses of chisels
  - The geometric Angle of chisels and the cutting edge grinding method
  - Basic operation of hammer and burin
- **Using saws**
  - Types of Saws
  - Specification and installation of saw blade
- **Using Files**
  - Type and structure of file

    - Purpose and maintenance of file
    - How to use file
- **Employing operation method of cutting**
    - The method of cutting plane
    - The method of filing plane and circular surface
    - The method of burring and chamfering parts
    - Testing of cut faces
- **Carrying out locksmith operation**
    - Chisel to cut the surface of steel
    - Saw cutting metal rods or specified parts
    - File plane (20*90mm)
    - Parts burr, chamfer

Assessment Criteria：

- Chisel used correctly
- Saws used correctly
- Files used correctly
- Operation method of cutting employed correctly
- Locksmith operation carried out correctly

Assessment Methods：To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- The practice test

Recommended Books：

XuTingguo*, Locksmith operation*

MODULE: 412-20-C LOCKSMITH OPERATION

UNIT C 20.4 CARRYING OUT HOLE AND THREAD PROCESSING

Nominal Duration: 40 hours **Credits: 4**

General Learning Outcomes：CARRYOUT HOLE AND THREAD PROCESSING

Specific Learning Outcome：On completion of this unit the trainee will be able to：

20.4.1 Use hole tool
20.4.2 Use tapping tool and tool kit
20.4.3 Carryout operation method of hole and thread machining
20.4.4 Carryout locksmith operation

Conditions：The trainee must have access to：

- Library
- Training rooms
- Handouts
- Tutorials
- Tool
- Consumables
- Workshops

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- **Using hole tool**
  - The structure, Angle and cutting edge of the twist drill
  - Type, structure, operation and maintenance of drill press
  - The way the drill is clamped

- Type and usage of reamer

- **Using tapping tool and tool kit**
  - Type and specification of tap and die
  - Selection and usage of tap and die
- **Carrying out hole and thread machining**
  - Boring method
  - Reaming method
  - Reaming method
  - Calculation of tapping base hole diameter
  - Determine the diameter of the wire rod
  - Method of tapping and threading
- **Carrying out locksmith operation**
  - Drill, ream and ream holes on the steel pieces
  - Drill and tap the steel
  - Round wire rod set

Assessment Criteria：

- Hole tool used correctly
- Tapping tool and tool kit used correctly
- Hole and thread machining carried out correctly
- Locksmith operation carried out correctly

Assessment Methods：**To achieve the above assessment the trainee will be given**

- Exercises
- Assignments
- The practice test

Recommended Books：

XuTingguo,*Locksmith operation*
Yang, H.Y., 2014. Recent fluid power research activities in China. 9th International Fluid Power Conference, Aa-chen, Germany.
Liu, R., Alleyne, A., 2000. Nonlinear force/pressure tracking of an electro-hydraulic actuator. Journal of Dynamic Systems, Measurement, and Control, 122 (1):232-236

MODULE: 412-20-C LOCKSMITH OPERATION

UNIT C 20.5 APPLYING HAMMERING TECHNIQUES

Nominal Duration: 40 Hours **Credits: 4**

General Learning Outcomes：APPLY HAMMERING TECHNIQUES

Learning Outcome：On completion of this unit the trainee will be able to：

20.5.1 Apply Hand hammer processing technology
20.5.2 Carryout locksmith operation

Conditions：The trainee must have access to：

- Library
- Training rooms
- Handouts
- Tutorials
- Tool
- Consumables

Values/ Attitudes:

- Sober minded
- Honest
- Hardworking
- Committed
- Smart
- Bold
- Driven
- Time conscious

Learning Activities：The trainer must ensure that the learning activities for the trainee include the following：

- Applying Hand hammer processing technology
  - Hand hammer part map reading
  - Operating procedure of making hammer
  - Matters needing attention
- Carrying out locksmith operation
  - Finish the hammer processing
  - Finish the hardness test after heat treatment

Assessment Criteria：

- Hand hammer processing technology applied
- locksmith operation carried out correctly

Assessment Methods：To achieve the above assessment the trainee will be given

- Exercises
- Assignments
- The practice test

Recommended Books：

XuTingguo,*Locksmith operation*
Yang, H.Y., 2014. Recent fluid power research activities in China. 9th International Fluid Power Conference, Aa-chen, Germany.
Liu, R., Alleyne, A., 2000. Nonlinear force/pressure tracking of an electro-hydraulic actuator. Journal of Dynamic Systems, Measurement, and Control, 122 (1):232-236